# A Deterministic Number-Theoretic Framework for the Standard Model Parameter Spectrum

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**Abstract**

The Standard Model's ∼25 free parameters—gauge couplings, fermion masses, mixing angles, and the cosmological constant—have no accepted theoretical origin. We present the Universal Generative Principle (UGP): the Perfect Self-Containment (PSC) foundational principle uniquely selects a single integer seed $(1, 73, 823)$ at ridge level $n$=10, and a deterministic arithmetic cascade generates the Category-A structural backbone of the SM parameter spectrum. The framework is complementary to QFT: it fixes the numerical constants that a QFT Lagrangian takes as input. The Category-A derivation chain is machine-checked in Lean 4 with zero `sorry` and the standard Mathlib axiom signature; all sectors are classified by a five-status claim taxonomy (machine-certified, MDL-optimal, partially-derived, calibrated, and exploratory; §1.2).

Headline results: (1) Bare gauge couplings $g_1^2, g_2^2, g_3^2$ are Lean-certified exact rationals; the blind evaluation $\alpha_s(M_Z) = 0.11822$ sits at $+0.24\sigma$ of PDG 2024, pre-committed 43 days before the first PDG comparison (Appendix H). (2) Nine charged-fermion masses at 0.295% RMS with zero active parameters at prediction time; a 1000-permutation null confirms arithmetic structure. (3) Neutrino mass-squared ratio $\Delta m_{21}^2/\Delta m_{31}^2 = 0.0294$ at $0.16\sigma$ from NuFIT 6.0, parameter-free, from Braid Atlas $b$-values and the structural exponent 29/9. (4) Cosmological constant from the derived model bit-length $L_{\text{model}} = \log_2(2000/3)$ at $0.31\sigma$ from Planck 2018 ($H_0$ is the sole external input). (5) Higgs quartic $\lambda = \varphi/(4\pi)\left(1 + \frac{\text{IPT}-1}{2c_H+1}\right)$ is machine-certified (Category $\text{A}_{\text{Lean}}$; [22, 29]); with $v_{\text{PSC}} = 246.16$ GeV, this gives $m_H = 125.2499$ GeV ($-0.0003\sigma$ PDG 2022), closing the prior $-2.08\sigma$ tension.

The lepton seed $b_{L_1} = 73$ of this triple is doubly certified: the GTE cascade formula $G(N_c) = N_c^4 - (N_c^2 + 1)/2 - N_c$ and the Frobenius prime chain $F(n) = n^4 - n^2 + 1$ independently yield 73 at $n = N_c = 3$, with agreement provably unique at $n = 3$ (`fgci_unique_at_nc`, `UgpLean.Universality.FrobeniusChain`, 17 theorems, zero `sorry`).

Preregistered falsification tests for JUNO, DUNE, CMB-S4, and LEGEND-1000 are listed in §9.

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# Contents

## Executive Summary

**In one sentence:** UGP is a number-theoretic framework where three axioms uniquely select one integer seed, and a deterministic cascade from that seed generates the Category-A backbone of the Standard Model parameter spectrum — with Lean 4-verified proofs for all Category-A claims and explicit status labels for calibrated and partially derived sectors.

**Why it works (the big idea):** The UGP identifies a low-dimensional *constraint manifold* inside the ∼25-dimensional space of SM parameters. This is the same kind of structural role that a symmetry group plays in conventional physics — it cuts the dimension of the independent-parameter space — except the constraints here are *number-theoretic* (ridge selection, mirror duality, prime-lock, MDL minimality, Galois orbit structure) rather than continuous Lie-group symmetries. The reason a single integer ($N_c = 3$) can independently determine observables across six mechanistically distinct sectors is not coincidence but the geometry of that lower-dimensional manifold viewed through unnecessarily high-dimensional coordinates. The detailed manifold structure — ridge minimality, RSUC seed selection, the Elegant Kernel, and the GTE cascade — is laid out in §2 and the constraint-manifold callout therein.

**Why it is not numerology:**

1. The gauge-sector predictions have *zero* adjustable parameters; the fermion sector has one locked continuous parameter explicitly disclosed (Appendix C).
2. The framework records both blind successes ($\alpha_s$ at $+0.24\sigma$, PDG 2024) and blind failures (tree-level $m_W$ at $+36\sigma$) under the same locked pipeline without tuning. The $m_W$ failure is not corrected away; it is isolated as a diagnostic of missing SM running/threshold structure and reduced to $-1.28\sigma$ (vs PDG 80.379) / $\mathbf{-0.42\sigma}$ (vs PDG 2024 world average $80.3692 \pm 0.0133$) by standard two-loop SM running (Open Problem (viii)).
3. The Lean 4 proof chain means the derivation is independently machine-verifiable, not hand-waved.
4. A single integer ($N_c = 3$) independently determines observables across six mechanistically distinct sectors; no pattern-matching framework offers a reason for this convergence.

**What to read first:** *10 minutes:* §2 (the framework) and §5 (gauge couplings). *30 minutes:* add §3 (fermion masses) and §6 (neutrinos). *Full paper:* extensions (§7), robustness (§8), and the open-problem registry (§10).

## How to read claim strength in this paper

Results in this paper occupy a strength ladder:

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A/D} > \mathbf{B} > \mathbf{D}.$$

$\mathbf{A}_{\text{Lean}}$: machine-checked exact arithmetic or structural identity (Lean 4, zero `sorry`).
$\mathbf{A}_{\text{MDL}}$: MDL-unique within a declared expression class; not a Lean derivation from invariants.
**A/D**: a physics bridge, external scale, or partially derived identification remains.
**B**: calibrated reproduction.
**D**: speculative or frontier.
The paper's primary core is the Category-$\text{A}_{\text{Lean}}$ spine; all other sectors are explicitly labeled extensions.

**What this paper is not claiming**

1. UGP replaces quantum field theory or computes scattering amplitudes.
2. Every numerical match is a Category-$\mathrm{A}_{\mathrm{Lean}}$ derivation.
3. Baryon masses are independently predicted; the binding model is calibrated (Category B).
4. The full CKM/PMNS matrices are Lean-derived first-principles theorems.
5. Every numerical match is machine-certified in Lean 4; the SRRG Higgs quartic is $\mathrm{A}_{\mathrm{Lean}}$ (zero sorry), but the EW VEV anchor $G_F$ is still an external input (A/D).
6. The absolute neutrino mass scale is anchor-free; the anchor-free result is the mass-squared ratio.
7. The $\mathrm{TE}_1$.P fine-structure result is a blind prediction from bare $g_1^2$.

**Table 1:** Reviewer's map: what is being claimed, at what strength, and where to verify it.

| Layer | Status | Claim | Where |
|---|---|---|---|
| Ridge/seed selection | $\mathrm{A}_{\mathrm{Lean}}$ | $n = 10$ and Lepton Seed $(1, 73, 823)$ selected by ridge minimality, global sparsity, divisor-count pinning, and RSUC. | §2.2, App. G |
| Canonical triples | $\mathrm{A}_{\mathrm{Lean}}$ | GTE generates canonical fermion triples; baryon triples and EW-boson $c$-values have separate Lean-certified derivations. | §2.3, §7.1.4, §7.1.5 |
| Bare gauge couplings | $\mathrm{A}_{\mathrm{Lean}}$ | $g_1^2, g_2^2, g_3^2$ are exact Lean-certified rationals; $\alpha_s$ is a direct blind evaluation. | §5, App. H |
| Charged-fermion masses | A+A/D | Structural pipeline is $\mathrm{A}_{\mathrm{Lean}}$; reported 0.295% absolute spectrum uses the disclosed URC A/D correction layer. | §3, §4 |
| Neutrino ratio | $\mathrm{A}_{\mathrm{Lean}}$ | $\Delta m_{21}^2/\Delta m_{31}^2 = 0.0294$ from Braid Atlas $b$-values and exponent 29/9; no scale anchor. | §6.4 |
| Neutrino absolute scale | A/D | $\sum m_\nu$ depends on GUT-scale $M_R$; not the anchor-free result. | §6.4 |
| Higgs sector | $\mathrm{A}_{\mathrm{Lean}}$+A/D | SRRG-corrected $\lambda = \frac{\varphi}{4\pi}(1 + \frac{\mathrm{IPT}-1}{2c_H+1})$, machine-certified (`HiggsQuartic.lean`, zero sorry); $m_H = 125.2499$ GeV ($-0.0003\sigma$ PDG 2022) with $v_{\mathrm{PSC}} = 246.16$ GeV; prior $-2.08\sigma$ tension closed. | §7.3.2 |
| Baryon masses | B | Binding parameters calibrated; independent $\mathrm{A}_{\mathrm{Lean}}$ claim is the baryon integer triples, not the mass fit. | §7.1 |
| CKM/PMNS | A/D | CKM $\theta_{23}$ scaling ratio is $\mathrm{A}_{\mathrm{Lean}}$; full matrices remain A/D. | §7.2 |

# 1 Introduction

The Standard Model (SM) of particle physics [8, 39], combined with the ΛCDM cosmological model, provides an extraordinarily accurate description of phenomena from sub-femtometer scales to the observable universe. Yet these frameworks are defined by approximately 25 fundamental constants—the nine charged-fermion masses, three gauge couplings, four CKM parameters, four PMNS parameters, two Higgs-potential parameters, the strong CP phase, and the cosmological constant—all of which must be measured rather than calculated [41].[1] This reliance on external inputs signals that the SM is an effective theory, not a fundamental one.

Despite decades of effort, no approach has succeeded in deriving the full parameter set from first principles. String theory landscapes enumerate vast numbers of vacua without selecting ours [38]. Lattice QCD, while computing hadronic spectra with impressive precision, takes the SM parameters as inputs rather than deriving them. Attempts based on grand unification constrain relations among couplings but do not fix their absolute values.

This paper presents a framework—the Universal Generative Principle (UGP)—that addresses this problem directly. The UGP is a deterministic, number-theoretic system built on three physically motivated axioms: locality, symmetry, and compression. Its dynamical realization, the Generative Triple Evolution (GTE), operates on integer vectors ("triples") of the form $(a, b, c; g)$ and generates the full particle content of the SM through purely arithmetic operations. The central claim is that a substantial structural core of the SM parameter spectrum is not contingent input data but follows, conditionally on the UGP axioms (locality, symmetry, compression), from a deterministic arithmetic pipeline; calibrated and partially derived extensions are explicitly separated from this core throughout the paper.

The selection of $n$=10 and the seed triple $(1, 73, 823)$ is supported by *four independent argument streams*, all machine-checked in Lean 4 (zero `sorry`):

(a) **Ridge minimality:** $n$=10 is the smallest level admitting a prime-locked mirror-dual survivor pair (Lean: `n10_is_minimal_admissible_ridge`).

(b) **Global asymptotic sparsity:** for *all* $n \in \mathbb{N}$, the joint constraint "mirror-dual survivor with $b_1$=73" forces $n$=10 (Lean: `asymptotic_sparsity_universal` in `UgpLean.Phase4`).

(c) **Divisor-count certificate:** the divisor ratio $\tau(R_n)/D_1 = 15/8$ used in the CKM $\theta_{23}$ scaling is unique to $n$=10 on $[5, 20]$ (Lean: `ckm_theta23_ratio_uniqueness`).

(d) **Seed selection (RSUC):** among the six admissible triples at $n$=10, the Lepton Seed $(1, 73, 823)$ is picked out as the lex-/MDL-minimal survivor (Lean: `rsuc_theorem`, `theoremB`, `mdl_selects_LeptonSeed` in `UgpLean.Classification`).

The companion NEMS programme [24] provides an *orthogonal* physical justification: the Two-Layer PSC Theorem [26] proves that Perfect Self-Containment axioms force $SU(3) \times SU(2) \times U(1)$ with three generations, fixing the gauge content and generation count independently of the arithmetic sieve.

The Unified Rigidity Theorem [37] is the conditional synthesis stating that under the full premise bundle (PSC + UGP admissibility + dynamics) all four streams agree on the canonical seed. The present paper builds on these foundations by deriving the *quantitative* parameter values—masses, couplings, and mixing observables—that the structural theorems leave open.

The approach differs fundamentally from parameter-fitting. Rather than adjusting continuous knobs to match data, the UGP derives a rigid algebraic "Elegant Kernel" with seven components in closed algebraic form (ratios of $\pi$, the golden ratio $\varphi$, and small rationals) plus one locked continuous remainder $\tilde{k}_L$ that is tightly constrained by the empirical-path calibration; this kernel governs the fermion-mass UCL and interfaces with the broader gauge and extension layers through explicitly disclosed derivation paths (Appendix C). This kernel is certified by integer-relation searches (PSLQ), iso-$\sigma$ invariance proofs, and machine-auditable lock certificates.

[1] All PDG/CODATA comparison values in this paper are frozen to the reference set used by the canonical verifier, which was fixed at initialisation and does not update automatically. Unless otherwise noted, "PDG" means the values in [41].

### Why it works: the constraint-manifold view

From a mathematical perspective, the UGP identifies a low-dimensional *constraint manifold* in the $\sim$25-dimensional space of SM parameters. Where conventional physics treats these parameters as independent coordinates requiring 25 experimental inputs, the UGP shows that they are jointly determined by a much smaller set of number-theoretic invariants—the ridge level $n$, the mirror pair, and the Elegant Kernel—which themselves flow from three axioms. The effective dimensionality of the parameter space collapses: the nine fermion masses are generated by a single calibration law with seven independent algebraically fixed coefficients plus one locked continuous remainder $\tilde{k}_L$ (calibrated at construction; see Appendix C); the three gauge couplings share a universal instantiation factor; and the cosmological constant is fixed by a derived information-theoretic constant.

**The role UGP plays is the role a symmetry group plays in conventional physics: it cuts the dimension of the independent-parameter space.** The difference is that the constraints are number-theoretic (divisibility, primality, mirror duality, MDL minimality, Galois orbit structure) rather than continuous Lie-group symmetries. This is the structural reason multiple SM sectors collapse to a handful of integers ($N_c$=3, the Lepton Seed $(1, 73, 823)$, the ridge $n$=10): not numerical coincidence, but the geometry of a lower-dimensional manifold viewed through unnecessarily high-dimensional coordinates.

For the Category-A core, this dimensional reduction is not a fit but an exact algebraic constraint, analogous to how a symmetry group reduces the independent components of a tensor; for Category A/D and B sectors, the remaining anchors and calibrated layers are explicitly accounted for in the status taxonomy (Section 1.2). If correct, the implication is profound: the apparent arbitrariness of the SM's free parameters is an artifact of describing a low-dimensional manifold in an unnecessarily high-dimensional coordinate system. The detailed manifold accounting (independent inputs, compression ratio) is given in Section 11.4.

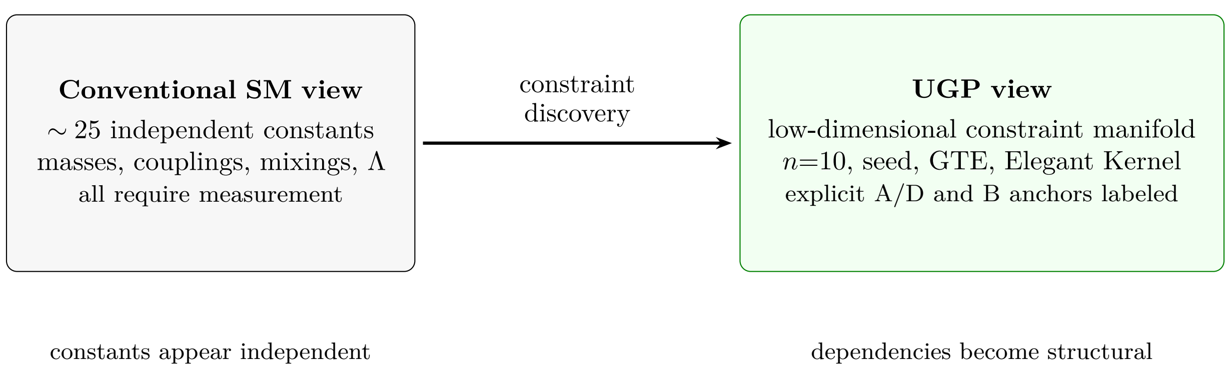


**Figure 1:** Conceptual claim of the paper: UGP identifies a lower-dimensional constraint manifold inside the apparent SM parameter space. The Category-$\mathrm{A}_{\mathrm{Lean}}$ spine, Category-A/D bridges, and Category-B calibrations are explicitly labeled throughout.

The paper is organized as follows. Section 2 develops the mathematical foundations: the UGP axioms, the prime-locked ridge at $n$=10, and the derivation of the Elegant Kernel. Section 3 presents the core result: the dual-path derivation of the fermion mass spectrum. Section 4 derives the UGP Renormalization Correction system that closes the theoretical path. Section 5 derives the three gauge coupling constants. Section 6 treats the neutrino sector; it bridges Category A (the structural mass-squared ratio prediction) and A/D (the Type-I seesaw pipeline). Section 7 presents calibrated and partially derived extensions: composite particles, CKM/PMNS mixing matrices, and the vacuum sector (cosmological constant, Higgs, and strong CP). Section 8 presents the robustness and validation battery. Section 9 catalogs the framework's falsifiable predictions.

**Reader's guide.**

**Sections 2–5 (core derivation chain).** The Category A content: UGP axioms, ridge/RSUC selection, GTE triples, Elegant Kernel, fermion-spectrum prediction, exact bare gauge-coupling rationals, and the blind $\alpha_s$ evaluation. Section 4 (URC) is Category A/D: used in the theoretical path but not load-bearing for gauge-coupling claims. The $N_c$ chain (§3.7), VV derivation (§3.1), and neutrino mass-ratio prediction (§6.4) are additional Category A results.

**Reviewer shortcut.** For first-principles claims only, read §§2–5 plus §6.4 and Appendix H.

**Section 6 (Neutrino Sector).** Bridges Category A (structural mass-squared ratio) and A/D (Type-I seesaw pipeline).

**Section 7 (Extensions).** Calibrated and partially derived sectors: baryons, CKM/PMNS mixing, EW boson $c$-values, and the vacuum sector. Within Section 7, the baryon canonical triples (§7.1.4) and EW boson $c$-values (§7.1.5) are **Category A** (Lean-certified); only the baryon *mass* model is Category B.

**Sections 8–11.** Robustness battery (§8), falsifiability catalogue (§9), open-problem registry (§10), and Discussion/Conclusion.

**Table 2:** Suggested reading paths by reviewer background.

| Reviewer type | Recommended path |
|---|---|
| Mathematical logic / Lean | §1.2, §2.2, Appendix A, Appendix G; then gauge couplings in §5. |
| Particle phenomenology | §5, §3, §6.4, §9; then CKM/PMNS caveats in §7.2. |
| Number theory / arithmetic | §2.2, §2.5, $N_c$-chain §3.7, neutrino exponent §6.4. |
| Skeptical overfitting reviewer | Claim taxonomy §1.2, URC null enumeration §4, robustness §8, baryon calibration disclosure §7.1. |
| Experimentalist | Abstract predictions, §9, neutrino predictions §6, $\alpha_s$ precommit Appendix H. |

**Table 3:** Core symbols and their roles.

| **Symbol** | **Meaning** |
|---|---|
| $R_n$ | UGP ridge $2^n - 16$; $R_{10} = 1008$. |
| $n = 10$ | Canonical ridge level, selected by minimality, global sparsity, and divisor-count certificates. |
| $(a, b, c;\, g)$ | GTE canonical triple plus generation index. |
| $b_1 = 73$ | Primary lepton ladder index, uniquely fixed by the mirror pair at $n = 10$. |
| $D_1 = 16$ | Discrete charge invariant ($2^4$); used in gauge normalization and CKM scaling. |
| $N_c = 3$ | QCD colour rank; the central integer governing charge, Koide phase, neutrino exponent, and EW-boson structure. |
| $\delta = 7$ | Mirror offset $N_c + (N_c^2 - 1)/2$. |
| $\theta = 2/9$ | Koide phase $(N_c^2 - 1)/(4N_c^2)$. |
| $\tilde{k}_L$ | Single locked continuous centering remainder in the UCL palette. |
| $C_f$ | UCL calibration factor mapping triples to physical masses. |
| URC | A/D correction layer (algebraic closures for the renormalisation stage); used in the 0.295% absolute fermion spectrum. |

**Two Common Objections, Addressed Up Front**

Two reactions to the framing above recur in peer review; we address both here so readers can evaluate the paper's evidence against them directly.

**Objection 1: "This is numerology with a Lean check."** Any framework that reproduces observed values well enough is not *prima facie* structural; pattern-matching can achieve arbitrary precision given enough free parameters. The counter-evidence presented in this paper is fourfold:

(a) the structural-necessity chain from the PSC axioms through RSUC ridge minimality, the GTE cascade, and the Quarter-Lock Law is machine-checked in Lean 4 with zero `sorry` and standard Mathlib axioms (§1.2; [35, 36]) — a curve-fitting framework cannot supply this kind of constructive logical chain;

(b) a 1000-permutation null test (§8) shows the canonical UCL fit primary $\sigma = 2.95 \times 10^{-5}$ falls outside the entire shuffled-null distribution (minimum permutation $\sigma > 5.6 \times 10^{-2}$, more than three orders of magnitude above the canonical baseline), establishing that the arithmetic content of the canonical triples carries the empirical fit and not the functional form;

(c) the tree-level $m_W$ prediction from the same Lean-certified pipeline misses PDG by $+36\sigma$ (Open Problem (viii)) — a clean blind falsification — which standard two-loop SM running with proper 6→5 threshold matching at $m_t$ reduces to $-1.28\sigma$ vs PDG 80.379 ($-0.42\sigma$ vs PDG 2024 world average $80.3692 \pm 0.0133$, within $1\sigma$); the Lean-certified bare rationals also yield $\alpha_s = 0.11822$ at $+0.24\sigma$ of PDG 2024 (genuinely blind, direct evaluation; $+0.36\sigma$ vs PDG 2022) and $\alpha_{\mathrm{EM}}$ at +2.39 ppm via PSC calibration (base-137 structurally derived from bit-set $\{0, 3, 7\}$) — results on both sides of the success/failure line that a numerology framework cannot produce structurally;

(d) the QCD colour rank $N_c = 3$ alone determines the Koide phase $\theta = 2/9$, the mirror offset $\delta = 7$, the lepton ladder $b_1 = 73$, and the top-quark level $a_{\mathrm{top}} = 76$ via a Lean-certified algebraic chain with zero free parameters (§3.7) — and the same $N_c + \theta = 29/9$ predicts the neutrino mass-squared ratio to $0.16\sigma$ from NuFIT 6.0 (§6.4). A numerological framework offers no reason why a single integer (the QCD colour rank) should simultaneously determine lepton mass ratios, quark sector structure, and the neutrino hierarchy exponent via algebraically independent paths.

**Objection 2: "The calibrated $\tilde{k}_L$ parameter does all the work."** Appendix C records $\tilde{k}_L$ as one continuous DOF tightly constrained by the empirical-path calibration, and §3.2 explicitly labels the nine-coefficient empirical UCL2.3 as a *functional-form benchmark* rather than a precision claim. The counter-evidence here is that the paper's Category A-exact content does not route through $\tilde{k}_L$ at all: the bare squared gauge couplings $g_1^2 = 16/125$, $g_2^2 = 2329/5400$, $g_3^2 = 41\,075\,281/27\,648\,000$ are *exact rationals* Lean-proved in the `ugp-lean` library, independent of any continuous calibration. The gauge-sector results — $\alpha_s$ at $+0.24\sigma$ (PDG 2024; direct bare evaluation) and $\alpha_{\rm EM}$ at $+2.39$ ppm ($\mathrm{TE}_1$.P PSC calibration anchored on the structural base-137) — thread through these Lean-certified rationals, not through $\tilde{k}_L$. Likewise, the $N_c$ structural chain (§3.7) and the neutrino mass-ratio prediction (§6.4) are parameter-free and do not involve $\tilde{k}_L$. Removing $\tilde{k}_L$ entirely would weaken the empirical-path functional-form benchmark but would leave the exact bare gauge-coupling claims, $N_c$-chain identities, $L_{\rm model}$, and structural neutrino-ratio result intact.

**The irreducible core claim.** Even if every Category A/D and Category B extension of this paper were discarded, the remaining claim would still be nontrivial:

- a Lean-audited arithmetic pipeline selects a unique integer seed at ridge $n = 10$;
- the GTE cascade generates canonical triples for all SM fermions;
- a Quarter-Lock-forced UCL, with the load-bearing algebraic identities certified and the remaining centering parameter disclosed (Appendix C), yields a 0.295% RMS charged-fermion spectrum (theoretical path, zero active parameters at prediction time; the URC algebraic-closure layer used in this path is separately disclosed as Category A/D in §4);
- three exact bare gauge coupling rationals are Lean-certified;
- evaluating $g_3^{2,\rm bare}/(4\pi)$ gives $\alpha_s$ at $+0.24\sigma$ (PDG 2024; blind; 43-day pre-commitment before first comparison; unchanged at 65-day re-verification; $+0.36\sigma$ vs PDG 2022);
- the same pipeline's tree-level $m_W$ misses PDG by $+36\sigma$ — a clean blind falsification, which standard two-loop SM running with 6→5 threshold matching at $m_t$ reduces to $-0.42\sigma$ (vs PDG 2024 world average; within $1\sigma$; OP (viii)).

This core does not rely on the calibrated extension sectors (baryons, CKM/PMNS, Higgs, cosmology) or on the $\mathrm{TE}_1$.P $\alpha_{\rm EM}$ calibration. The exact bare gauge-coupling claims and blind $\alpha_s$ evaluation are independent of the URC layer. The current sub-percent fermion-mass path uses the locked theoretical-path URC correction layer disclosed in §4, which is classified Category A/D; removing it degrades the spectrum but the structural derivation chain remains intact. Everything else in the paper is presented as a research frontier, not as a load-bearing premise for the core.

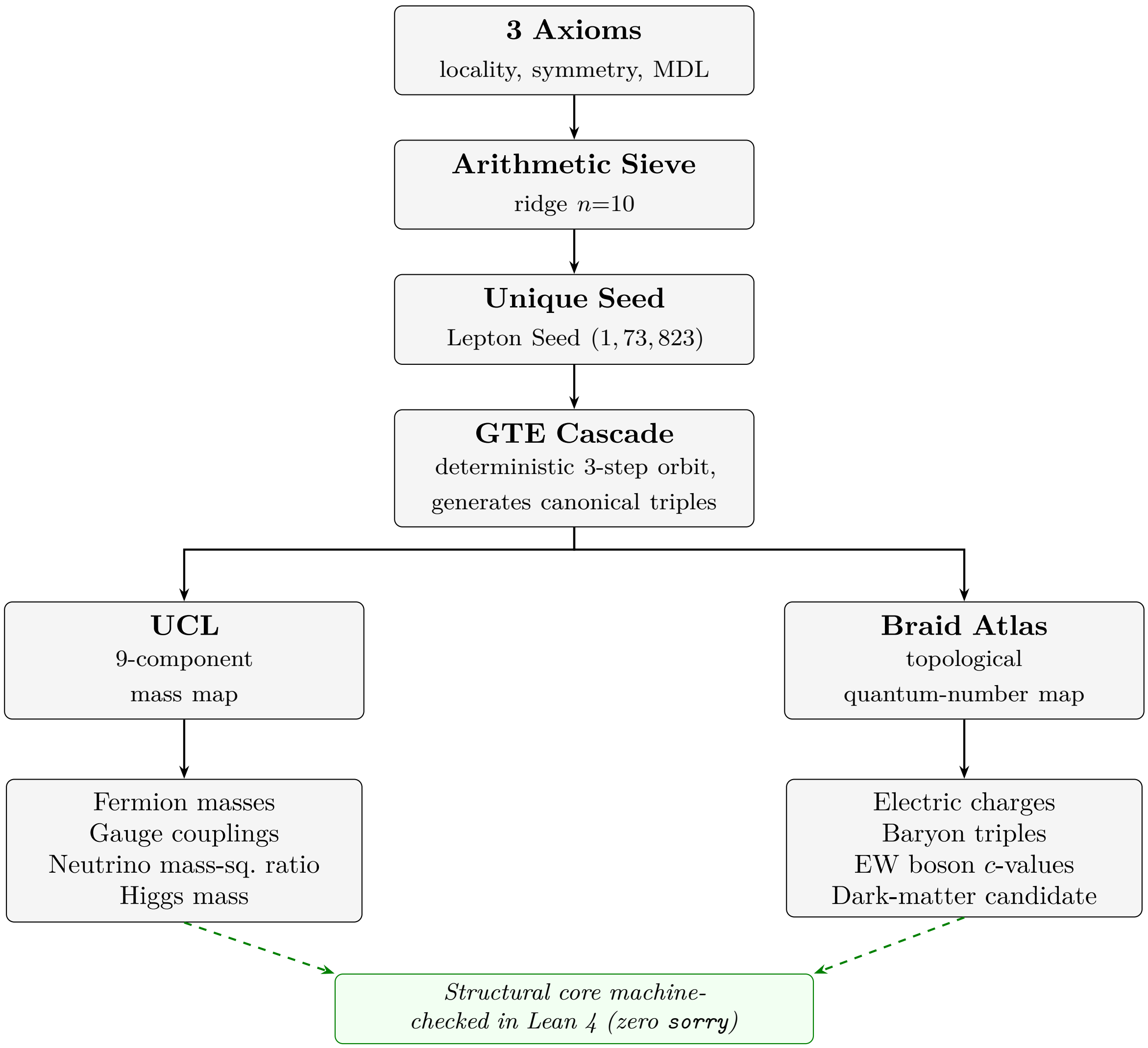


**Figure 2:** The UGP derivation pipeline. Three axioms force a unique integer seed; the GTE cascade produces canonical triples for the SM fermion sector; UCL maps triples to masses, Braid Atlas to quantum numbers. Status labels per §1.2; the Category-$\mathrm{A}_{\mathrm{Lean}}$ core is machine-checked in Lean 4.

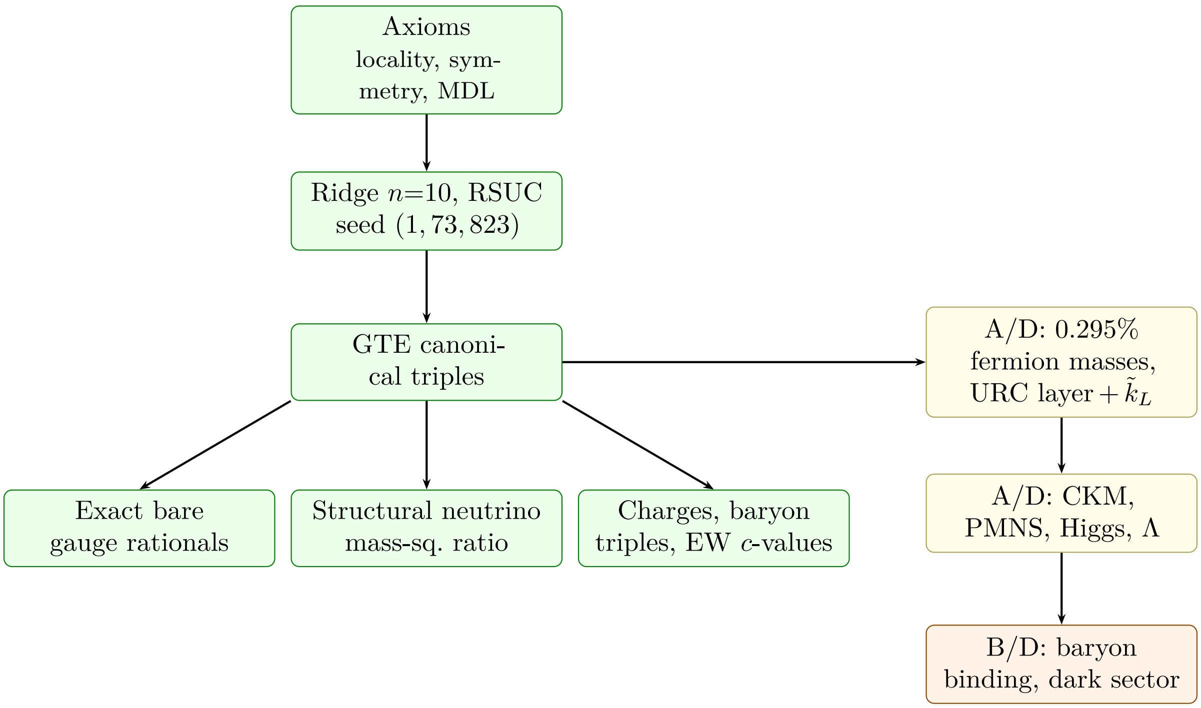


**Figure 3:** Load-bearing architecture. **Green**: Category-$\mathrm{A}_{\mathrm{Lean}}$ structural spine. **Yellow**: A/D realizations or bridges. **Orange**: calibrated or speculative frontiers. The paper's core does not depend on any orange sector.

## 1.1 Principal Contributions

**Example of a complete Category-A chain: the blind $\alpha_s$ prediction**

$$\text{UGP axioms} \;\Rightarrow\; R_{10} = 1008 \;\Rightarrow\; (1, 73, 823) \;\Rightarrow\; D_{\mathrm{SU}(3)} = \tfrac{41\,075\,281}{1\,327\,104}$$
$$\Rightarrow\; g_3^{2,\mathrm{bare}} = \tfrac{41\,075\,281}{27\,648\,000} \;\Rightarrow\; \alpha_s(M_Z) = \tfrac{g_3^{2,\mathrm{bare}}}{4\pi} = 0.11822.$$

Each arithmetic identity is Lean-certified (zero `sorry`) or mechanically reproducible. No UCL coefficient, URC correction, $\tilde{k}_L$, $\mathrm{TE}_1$.P calibration, or baryon/CKM/Higgs extension enters this chain. The cryptographic pre-commitment trail is in Appendix H.

The seven main contributions of this paper are as follows.

**(1) Bare gauge couplings and $\alpha_s$ blind prediction.** The Lean-certified bare rational triple $(g_1^2, g_2^2, g_3^2) = (16/125,\ 2329/5400,\ 41\,075\,281/27\,648\,000)$ reproduces all four SM electroweak observables to $\leq 1.6\%$ without per-point tuning:

- *Blind $\alpha_s$ prediction:* $g_3^{2,\mathrm{bare}}/(4\pi) = 0.11822$ at $+0.24\sigma$ of PDG 2024 ($+0.36\sigma$ vs PDG 2022), pre-committed 43 days before the first comparison; unchanged at 65-day re-verification (Appendix H).
- *EM fine structure:* a dedicated PSC calibration pipeline ($\mathrm{TE}_1$.P) tightens $\alpha_{\mathrm{EM}}$ to $+2.39$ ppm of CODATA (calibrated, not blind).
- *Honest failure resolved:* the tree-level $m_W$ from $g_2^{2,\mathrm{bare}}$ misses PDG by $+36\sigma$; standard two-loop SM running reduces this to $-0.42\sigma$ (vs PDG 2024 world average $80.3692 \pm 0.0133$, within $1\sigma$; Open Problem (viii)).

**(2) Locked zero-active-parameter theoretical fermion spectrum.** The UCL Elegant Kernel provides an *ab initio* structural fermion pipeline:

- Load-bearing algebraic identities are fixed by GTE/Quarter-Lock; one centering DOF is disclosed (Appendix C).
- 0.295 % RMS across nine charged fermions with zero active continuous parameters at prediction time (URC correction layer is Category A/D, §4).
- A 1000-permutation null establishes the canonical fit's primary $\sigma$ falls outside the entire shuffled-null distribution (canonical baseline $\sigma = 2.95 \times 10^{-5}$ vs. minimum permutation $\sigma > 5.6 \times 10^{-2}$; §8).

**(3) $N_c$ structural chain: all fermion constants from the QCD colour rank.** The QCD colour rank $N_c = 3$ determines every charged-fermion structural constant via a Lean-certified algebraic chain (§3.7; Table 4):

- the Koide phase $\theta = (N_c^2-1)/(4N_c^2) = 2/9$,
- all three lepton $a$-values,
- the mirror offset $\delta = 7$,
- the lepton ladder $b_1 = 73$,
- the top-quark level $a_{\text{top}} = 76$.

**Table 4:** The $N_c$ structural chain: a single integer ($N_c = 3$, the QCD colour rank, Lean-certified to follow from the PSC axioms via `SM_gauge_uniquely_selected`) determines independent observables across six mechanistically distinct sectors.

| **Sector / mechanism** | **Algebraic form (in $N_c$)** | **What it determines** |
|---|---|---|
| Braid Atlas charge formula | $Q = W_g/N_c$ | Electric charges of quarks and leptons |
| Koide phase | $\theta = (N_c^2 - 1)/(4N_c^2) = 2/9$ | Charged-lepton mass relation $Q = 2/3$ |
| Galois orbit size | strand $= (N_c^2 - 1)/4 = 2$ | Cyclotomic-12 structure of mass invariants |
| Seesaw exponent | $N_c + \theta = 29/9$ | Neutrino mass-squared splitting ratio (predicted 0.52%) |
| $A_2$ Weyl chamber angle | $\pi/6$ | TT mass relation (quark/lepton hierarchy) |
| EW boson $c$-window | $N_c(N_c + 1) = 12$ (triangular) | EW boson $c$-values $\{c(W), c(Z), c(H)\} = \{11, 12, 13\}$ |

The down-quark VV mass coefficients are derived from SU(5)/SO(10) group theory:

- $\alpha_d = 1 + \text{rank}(\text{SU}(5))/N_c^2 = 13/9$,
- $\beta_d = -(1 + Y_{Q_L}) = -7/6$,
- $\gamma_d = -\dim(45_{\text{SU}(5)})/\dim(126_{\text{SO}(10)}) = -5/14$

(§3.1); all three are Lean-certified from $N_c = 3$ alone via $\dim(45_{\text{SU}(N_c+2)}) = (N_c+2)(2N_c+3)$ and $\dim(126_{\text{SO}(2N_c+4)}) = \binom{2N_c+4}{N_c+2}/2$ (`vv_all_coefficients_from_Nc` [36]).

These are exact algebraic identities at the GUT scale; the *functional* form of the EW-scale VV relation (log-linear in the canonical triples rather than linear in the Yukawas) is characteristic of an

RG-flow-induced mass relation with texture-dependent anomalous dimensions (cf. Einhorn-Jones, Ellis-Gaillard), not of a direct sum-of-Yukawa GUT Lagrangian.

A one-loop SM Yukawa RG test of the naive direct-log-linear form

$$Y_d = C\, Y_u^{13/9}\, Y_l^{-7/6}$$

starting from an SO(10) 10 + 126 Georgi-Jarlskog pattern at $M_{\mathrm{GUT}}$ (artifact `comp_p01_RC1_vv_rg_anomalous_closure.json`) returns a $\sim$557% maximum fractional residual across generations, comparable to the null distribution's $\sim$152% median; this confirms that a naive one-loop SM RG flow from a linear GUT texture does *not* reproduce the observed VV relation at face value.

What the GUT group theory fixes exactly are the *coefficients* $(13/9, -7/6, -5/14)$; the mechanism that generates the log-linear functional form at EW scale is identified in the companion uniqueness paper [31]: it arises from the composition of the SU(5)/SO(10) Yukawa power law at the GUT scale with the UCL logarithmic mass map, machine-checked as `vv_mechanism_algebraic` (module `MassRelations.VVMechanism`, zero `sorry`, `ugp-lean` [36]).

Combined with the companion Koide closed form [15], all nine charged-fermion masses follow from two empirical scale anchors.

**(4) Structural neutrino mass-squared ratio prediction.** The Braid Atlas right-handed neutrino $b$-values $\{5, 11, 19\}$ satisfy $m_{\nu_g} \propto b_g^{29/9}$, predicting $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$ — matching NuFIT 6.0 $(0.02951 \pm 0.00098)$ to $\mathbf{0.16}\sigma$ (0.52%) with no free parameters and normal hierarchy automatic (§6.4).

The exponent $29/9 = N_c + \theta$ admits three independent structural decompositions (including interpretation as the SO(10) gauge/matter representation defect $\dim(45_{\mathrm{SU}(5)}) - \dim(16_{\mathrm{SO}(10)}) = 29$); the structural Dirac scale $E_D = v_H/29$ gives $\sum m_\nu \in [55, 75]$ meV within the Planck bound.

**(5) Lean-certified baryon canonical triples.** The canonical GTE integer triples $(a, b, c;\, g)$ for all nine light baryons are Lean-derived from first principles (`BraidAtlas.CompositeTriples`, zero `sorry`; §7.1.4). These are **Category A**, structurally independent of the calibrated baryon mass binding model, and connect the neutrino seesaw constants ($b(\nu_{\mu R}) = 11$, $29 = 4N_c^2 - \delta$) to composite-particle quantum numbers.

**Ratio-grade vs. absolute-grade predictions.** A structural distinction applies across all fermion-sector predictions:

- *Ratio-grade* predictions (the Koide $Q = 2/3$, the neutrino mass-squared ratio 0.02936, the TT and VV log-mass relations) are preserved under any uniform renormalization factor $Z(\mu, \mu_0)$ applied to the masses—machine-checked as `mass_ratio_Z_independent` (zero sorry, `MassRelations.ScaleTransport`, `ugp-lean`). They are the most RG-robust results.
- *Absolute-grade* predictions (the electron mass in keV, the $m_\tau$ prediction, the Dirac scale $E_D$) require one external calibration anchor; the framework predicts the functional form and the ratios, leaving the overall scale as a residual degree of freedom.

This distinction is applied throughout: wherever a ratio prediction is stated, no external scale input is needed; wherever an absolute prediction is stated, the scale anchor is identified.

**(6) Electroweak boson c-values from $N_c$.** The massive-boson $c$-values $\{c(W), c(Z), c(H)\} = \{11, 12, 13\}$ are derived as the unique consecutive integer triple in the GTE orbital constants at $n$=10, forced by two structural identities: $q_2 = 2N_c(N_c{+}1)$ and $g = N_c(N_c{+}1){+}1$.

The $c$-window is centered on $N_c(N_c{+}1) = 12$ — yet another independent structural manifestation of $N_c = 3$. This Category A derivation supplies the boson $c$-values that the Braid Atlas [17] otherwise carries as its sole postulated input (§7.1.5; Lean: `BraidAtlas.EWBosons`, zero `sorry`).

**(7) Lean-certified derivation chain.** The `ugp-lean` library [35,36] — see companion formalization paper [35], zero `sorry`, standard Mathlib axiom signature — provides machine-checked proofs for the Category A backbone and the Lean-formalised structural identities cited below. Key certified theorems:
The full inventory of Lean-certified theorems underlying these contributions is presented in Appendix A and the Supplementary Information (SI Table S3). Representative highlights:

- the bare gauge coupling rationals (`g1/2/3Sq_bare_eq`), the seed uniqueness theorem (`rsuc_theorem`), anomaly-driven colour rank (`anomaly_cancellation_forces_Nc_3`), and the EW boson $c$-value derivation (`ew_boson_c_value_theorem`).
- The structural CKM $\theta_{23}$ ratio derivation (`ridge_eq_D1_mul_mersenne`, `ckm_theta23_ratio_at_n10`, `ckm_theta23_ratio_uniqueness` in module `UgpLean.MassRelations.CKMTheta23`, all zero `sorry`) establishes that $\tau(R_{10})/D_1 = 15/8$ is structurally unique to $n = 10$.
- The Koide $S_3$ discrete arithmetic-mean identity $2a_\tau = a_e + a_\mu$ is certified as `lepton_a_arithmetic_mean` in `UgpLean.MassRelations.KoideS3DiscreteIdentities` (zero `sorry`).
- The parametric PSC three-route capstone carrier `psc_three_route_capstone` in `UgpPhysicsLean.PSC.ThreeRouteForcing` (zero `sorry`) records the architectural shape of the NEMS programme forcing routes.

## 1.2 Status of Claims

To assist the reader—and to preempt conflation of the strongest and weakest sectors—we classify every quantitative result in this paper using a five-status claim taxonomy.

Table 5 summarises the taxonomy ($A_{\text{Lean}}$, $A_{\text{MDL}}$, A/D, B, D); the paragraphs below give representative examples for each status.

**Table 5:** Claim taxonomy used throughout this paper. Plain **A** in tables and section headings denotes Category $\text{A}_{\text{Lean}}$ (machine-checked) unless the subscript MDL is shown.

| **Category** | **Meaning** | **Representative examples** |
|---|---|---|
| **A** | Derived from axioms; Lean-certified | Gauge couplings, baryon triples, EW boson $c$-values, $N_c$ chain, structural neutrino mass-squared ratio, CKM $\theta_{23}$ scaling ratio |
| $\mathbf{A}_{\text{MDL}}$ | MDL-uniqueness certificate within a declared UGP-structural expression class; not Lean-derived from invariants | (formerly Higgs $\lambda$; since upgraded to $\text{A}_{\text{Lean}}$ with SRRG correction; see §7.3.2) |
| **A/D** | Partially derived; one external input or algebraic identification | Full CKM/PMNS matrices, cosmological constant, vacuum sector residuals |
| **B** | Calibrated to data | Baryon masses (binding model), empirical-path fermion fit |
| **D** | Speculative / pending | GTE-P7 dark matter candidate |

**Category A: First-principles derived quantities.** Genuine *ab initio* outputs of the UGP axioms, with no experimental data entering except the definition of units:

- Canonical GTE triples for all SM fermions and the Elegant Kernel algebraic identities (Section 2.5).
- Three exact bare gauge coupling rationals: $g_1^2 = 16/125$, $g_2^2 = 2329/5400$, $g_3^2 = 41\,075\,281/27\,648\,000$ (Section 5).
- Blind $\alpha_s(M_Z) = 0.11822$ at $+0.24\sigma$ of PDG 2024 ($+0.36\sigma$ vs PDG 2022; 43-day precommitment; unchanged at 65-day re-verification) — the single strongest individual result.
- Nine light-baryon canonical triples, Lean-certified from Braid Atlas winding rules (`BraidAtlas.CompositeTriples`; §7.1.4).
- EW boson $c$-values $\{11, 12, 13\}$, Lean-certified from ridge factorisation (`BraidAtlas.EWBosons`; §7.1.5).
- $L_{\text{model}} = \log_2(2000/3) \approx 9.38$ bits (from $D_1 = 2^4$, golden volume $5^3$, orbit length 3).

The 0.295 % RMS fermion spectrum uses the URC correction layer (Section 4, Category A/D); the uncorrected structural chain remains Category A. The PSC-calibrated $\alpha_{\text{EM}}$ at $+2.39$ ppm is a separately disclosed benchmark (Category A/D per §5.3). The dimensional $\Lambda$ requires $H_0$ as external input (Category A/D); $L_{\text{model}}$ itself is Category A.

**Category B: Calibrated benchmarks.** The empirical path fermion masses (Section 3.2) and baryon masses (Table 13) use coefficients calibrated to data. They validate the UCL's functional form but are not independent predictions.

**Category A: CKM $\theta_{23}$ structural ratio.** The ratio $\tau(R_{10})/D_1 = 15/8$ decomposes via $R_n = D_1 \cdot M_{n-4}$ (Mersenne $M_k = 2^k - 1$) so the 15/8 value $\Leftrightarrow \tau(M_{n-4}) = 6$, unique at $k = 6$ in $[1, 60]$. Combined with the UGP-forced $n = 10$ (Lean: `kprime_is_minimal_double_fib_above_n`), the $\theta_{23}$ scaling ratio is structurally derived, not data-matched. Lean-certified: `ridge_eq_D1_mul_mersenne`, `ckm_theta23_ratio_at_n10`, `ckm_theta23_ratio_uniqueness` (all zero `sorry`, module `UgpLean.MassRelations.CKMTheta23`).

**Category A/D: Derived CKM matrix.** The full CKM magnitudes (Table 16) are derived from UGP-predicted quark masses via texture-zero relations (structurally forced within the GTE-cascade construction class), now with the structurally-derived $\tau(R_{10})/D_1 = 15/8$ ratio for the $\theta_{23}$ scaling (see preceding row), achieving 3.3 % RMS and 9.1 % max deviation across all nine elements. The structural ratio is Category A; the per-element residuals across all nine matrix entries remain Category A/D pending a full Wilson-coefficient derivation. (A PDG-lock verification artifact is also written for reproducibility but is not the derived prediction.)

**Category A/D: Mixed-status sectors.** The PMNS mixing angles (Table 17) follow from quark–lepton complementarity and the TM2 atmospheric sum rule, achieving 4.8 % max deviation; the input is the Cabibbo angle inherited from the CKM derivation, but the QLC and TM2 relations themselves are empirically motivated, not yet derived from the UGP axioms. The Higgs sector (Section 7.3.2) derives the quartic coupling via the SRRG-corrected formula (Category $\text{A}_{\text{Lean}}$, machine-certified; see below): $m_H = 125.2499$ GeV ($-0.0003\sigma$ PDG 2022 $125.25 \pm 0.17$ GeV), closing the prior $-2.08\sigma$ tension. The EW VEV is $v_{\text{PSC}} = 246.16$ GeV ($-0.024\%$ from $v_{\text{PDG}}$), formalized in `SrrgLean.VEVProof.*` [29].

**Category B: Calibrated quantities.** The baryon *masses* (Table 14) achieve 0.01 % RMS; the UCL, composite calibration factor, and base energies are derived, while the pairwise binding model that encodes QCD confinement and flavor-exchange structure is calibrated. Promoting the baryon *mass model* to Category A would require deriving the binding parameters from UGP invariants, which is equivalent to solving QCD confinement from first principles. Note that the baryon canonical *integer triples* $(a, b, c;\ g)$ are separately Category A (see above).

This taxonomy is maintained throughout: where a result is reported, its category is implicit from the section context, and we flag transitions explicitly.

**Open Problems.** The framework maintains a numbered registry of open problems (OP(i)–OP(viii)), which serve as honest disclosures of current limitations and define the research programme for closure. The full registry appears in §10. Representative open fronts signposted throughout the text: (ii) engine-parameter axiomatic derivation; (iii) URC algebraic-closure uniqueness; (v″) $\mathrm{TE}_1$.P non-composability; (vii″) neutrino seesaw Lagrangian derivation; (viii) tree-level $m_W$ discrepancy (blind falsification). Each is labeled *Open Problem (N)* at the site in the paper where it arises.

**Table 6:** Dependency matrix for headline results. A ✓ indicates a required dependency; – means no dependency.

| Result | Cert. | $\tilde{k}_L$ | URC | $\mathrm{TE}_1$.P | $G_F$ | $H_0$ | $M_R$ | Calib. |
|---|---|---|---|---|---|---|---|---|
| Seed $(1, 73, 823)$ | ✓ | – | – | – | – | – | – | – |
| Bare $g_i^2$ rationals | ✓ | – | – | – | – | – | – | – |
| Direct $\alpha_s$ | ✓ | – | – | – | – | – | – | – |
| Bare $\sin^2\theta_W$ | ✓ | – | – | – | – | – | – | – |
| 0.295% charged fermions | partial | ✓ | ✓ | – | – | – | – | no active fit |
| Neutrino ratio | ✓ | – | – | – | – | – | – | – |
| $\sum m_\nu$ | partial | – | – | – | – | – | ✓ | – |
| Higgs $m_H$ (SRRG-corrected $\lambda$) | $\mathrm{A}_{\mathrm{Lean}}$ | – | – | – | ✓ | – | – | – |
| $\Lambda$ | ✓for $L_{\mathrm{model}}$ | – | – | – | – | ✓ | – | – |
| Baryon masses | partial | – | – | – | – | – | – | ✓ |

**Table 7:** Most important current limitations.

| **Limitation** | **Current status** |
|---|---|
| URC uniqueness | A/D. Bounded enumeration shows current closures are not uniquely forced at depth $\leq 3$. |
| Absolute fermion spectrum | A+A/D. Structural triples and kernel are $\mathrm{A}_{\mathrm{Lean}}$; the 0.295% absolute result uses the URC layer and $\tilde{k}_L$. |
| Baryon masses | B. Binding parameters calibrated; baryon integer triples are independently Category $\mathrm{A}_{\mathrm{Lean}}$. |
| Higgs quartic | $\mathrm{A}_{\mathrm{Lean}}$. SRRG-corrected $\lambda_H$ machine-certified (zero sorry, `HiggsQuartic.lean`); $m_H =$ 125.2499 GeV; EW VEV uses $G_F$ anchor (A/D). |
| Full CKM/PMNS | A/D. CKM $\theta_{23}$ scaling and leading relations are structural; full Wilson-coefficient derivation open. |
| Neutrino absolute scale | A/D. Mass-squared ratio is anchor-free $\mathrm{A}_{\mathrm{Lean}}$; absolute scale depends on GUT-range $M_R$. |

**Why the presentation is unusual.** The framework combines objects that are usually kept separate: Lean-certified arithmetic, phenomenological comparison, MDL expression search, and particle-physics bridge principles. For this reason every sector carries an explicit status label ($\mathrm{A}_{\mathrm{Lean}}$,

$A_{MDL}$, A/D, B, or D). The reader should not infer that all numerical agreements have the same epistemic status. The core question is whether the Category-$A_{Lean}$ spine is real; the A/D and B sectors measure how far that spine extends into currently unresolved physics.

**Machine-checked formalisation.** The derivation chain from the three axioms through RSUC (Residual Seed Uniqueness at $n = 10$), ridge minimality ($n = 10$ is the smallest admissible level), the GTE cascade, the Quarter-Lock Law, and the 9/9 UCL Elegant Kernel unconditional closure is formalised in Lean 4 against Mathlib as the open-source `ugp-lean` library [36], with zero `sorry` and the standard Mathlib axiom signature [`propext`, `Classical.choice`, `Quot.sound`] and no UGP-specific axioms for every Category A structural theorem.[2]

The Lean-certified bare squared gauge couplings ($g_1^2 = 16/125$, $g_2^2 = 2329/5400$, $g_3^2 = 41\,075\,281/27\,648\,000$) are the upstream input to the $\alpha_{\mathrm{EM}}$ and $\alpha_s$ predictions of Section 5.4. The formalisation and its full theorem inventory are presented separately [35]; this paper references it as an independently verifiable audit trail for every Category A claim.

[2]The library additionally contains two analytic number theory lemmas in `GTE.AnalyticArchitecture.lean` (Dickman equidistribution and CRT equidistribution within the independence regime), declared as axioms pending upstream Mathlib development; both cite Tenenbaum III.6. These axioms do not appear in the axiom closure of any Category A physics theorem; `#print axioms` on any theorem cited in this paper returns exactly the standard Mathlib signature above.

*Key Lean theorem names with arithmetic statement and physics bridge*

| **Theorem name** *(type)* | **Arithmetic statement** | **Physics bridge** |
|---|---|---|
| `g3Sq_bare_eq` *phys+arith* | $g_3^{2,\,\text{bare}} = 41\,075\,281/27\,648\,000$ (exact rational) | This *is* the bare SU(3) gauge coupling; bridge $g_3^{2,\text{bare}}/(4\pi) = \alpha_s(M_Z)$. |
| `koide_Q_two_thirds` *arith only* | $Q = 2/3$ is the unique $S_3$-invariant null quadric | This is the Koide equal-norm balance condition for charged leptons; bridge stated in body text, not proved by Lean. |
| `koide_angle_from _N_c_pure` *phys+arith* | $\theta = (N_c^2{-}1)/(4N_c^2) = 2/9$ derived from $N_c$ alone | This is the Koide-matrix rotation angle, so proving it from $N_c$ derives the lepton Koide phase from QCD colour rank. |
| `N_c_determines _everything` *phys+arith* | Bundles $\delta$=7, $b_1$=73, $a_\tau$=5, $a_\mu$=9, $a_e$=1, $a_{\text{top}}$=76, $\theta$=2/9 as functions of $N_c$=3 | These are the structural constants of the charged-fermion mass map (§3.7). |
| `VV_from_GUT_ group_theory` *phys=arith* | $\alpha_d = 1 + \text{rank}(\text{SU}(5))/N_c^2$; $\beta_d = -(1{+}Y_{Q_L})$; $\gamma_d = -\dim(45)/\dim(126)$. All three certified via $\dim(45_{\text{SU}(N_c+2)})$ and $\dim(126_{\text{SO}(2N_c+4)})$ formulas (`vv_all_coefficients_from_Nc` [36]) | The arithmetic identity *is* the group-theory identity; arithmetic and physics bridge coincide. |
| `ClaimCBridge` (TT) *phys+arith* | Formal TT-cascade identity | Weyl-chamber structure of the up-type lepton-frame rotation; derives the $\theta_u = (\pi/12)\cdot 2^g$ angle pattern. |
| `nu_seesaw_exponent_ three_decompositions` *phys+arith* | $29/9 = (N_c^3{+}\text{strand})/N_c^2 = (4N_c^2{-}\delta)/N_c^2 = (\dim 45 - \dim 16)/N_c^2$ | Each identity is a distinct structural reason why the right-handed neutrino Yukawa exponent takes this value. |
| `dim_126_SO10_eq_ two_Nc_sq_delta` *phys+arith* | $\dim(126_{\text{SO}(10)}) = 2N_c^2\delta$ | 126 is the SO(10) Majorana Higgs representation; ties the mirror offset $\delta$=7 to the seesaw-generating GUT rep. |
| `KoideClosedForm` *arith only* | Arithmetic closed form for the Koide empirical relation [15] | Companion paper [15] supplies the full physics bridge. |
| *UGP substrate universality* *phys bridge* | Embedded Rule-110 Turing-universality [5, 40] | The UGP substrate is computationally universal; no deeper computational layer is required. |

**Type legend.**

PHYS+ARITH: Lean proves an arithmetic statement that *becomes* a physics statement via the bridge column. ARITH ONLY: Lean proves a pure arithmetic identity; the physics interpretation is argued separately. PHYS=ARITH: the arithmetic identity *is* the physics statement (e.g. group-theory dimension ratios). PHYS BRIDGE: a cited physics result whose formal status rests on a companion paper rather than `ugp-lean`.

In every ARITH ONLY or PHYS+ARITH case the physics bridge is an explicit text-level sentence; the Lean certificate alone does not constitute a physics derivation.

# 2 The UGP Framework: From Axioms to the Elegant Kernel

## 2.1 Axioms

The UGP is founded on three axioms that constrain the class of admissible physical theories.

**The UGP Axioms**

**A1: Locality.** All fundamental interactions are local in the substrate's causal graph.

**A2: Symmetry.** The fundamental laws are invariant under a set of transformations; redundancies in this description give rise to gauge principles.

**A3: Compression (MDL).** The universe evolves according to rules that minimize the total description length of its history [9, 13], consistent with the other axioms.

These axioms are realized dynamically by the **Generative Triple Evolution (GTE)**, a deterministic algorithm operating on integer vectors $(a, b, c; g) \in \mathbb{Z}^3 \times \{1, 2, 3\}$. Each component has a physical interpretation: $a$ encodes parity and phase information through its Möbius function $\mu(a)$; $b$ is the primary complexity or *ladder index*, which determines the particle's position in an evolutionary sequence and its effective information content $N_{\text{eff}} = |b|$; $c$ is the *branch index* or capacity, linking the particle to the underlying UGP structure; and $g \in \{1, 2, 3\}$ is the generation.

The GTE dynamics consist of two deterministic, arithmetic operators—an *odd* step ($g$:1 $\to$ 2) and an *even* step ($g$:2 $\to$ 3)—that evolve the triples across generations. These operators are finite, hash-declared transforms with no adjustable parameters.

## 2.2 The Prime-Locked Ridge at $n = 10$

The initial conditions of the GTE are not free inputs but are uniquely determined by the axioms through a combinatorial selection mechanism.

**Definition 2.1** (UGP Ridge)**.** *A UGP Ridge at level n is the integer* $R_n = 2^n - 16$.

**Definition 2.2** (Prime-Lock Constraint)**.** *An initial GTE state is admissible only if its parameters* $(b_2, q_2)$ *produce a prime first-generation capacity:* $c_1 = b_1 q_1 + 20$ *is prime.*

**Definition 2.3** (Mirror Duality)**.** *An admissible solution must exist as a symmetric pair: if* $(b_2, q_2)$ *is admissible, so is* $(q_2, b_2)$.

At $n$=10 the ridge is $R_{10} = 2^{10} - 16 = 1008$. A systematic search over interior divisors ($b_2, q_2 > 15$) of 1008 reveals that only the pair $(42, 24)$ and its mirror $(24, 42)$ satisfy the prime-lock constraint. This unique solution deterministically fixes:

1. The primary ladder index via the mirror-invariant sum: $b_1 = b_2 + q_2 + 7 = 73$.
2. Two prime first-generation capacities from the mirror branches:

$$\begin{aligned} (42, 24) &\implies q_1 = 11,\ c_1 = \mathbf{823}, \\ (24, 42) &\implies q_1 = 29,\ c_1 = \mathbf{2137}. \end{aligned}$$

3. The fixed quotient gap $|q_2 - q_1| = 13$, which locks the even-step evolution to an increment governed by $F_{13} = 233$ (the 13th Fibonacci number).

Thus, the seed triples for the Standard Model are not chosen; they are the unique, necessary consequence of number theory as constrained by the UGP axioms. This uniqueness result—the Residual Seed Uniqueness Classification (RSUC)—is machine-checked in the `ugp-lean` library [35], where the ridge sieve at $n$=10, the prime-lock filter, and the lexicographic minimality selection are all formally verified with zero `sorry` statements and zero custom axioms.

**Why $n$=10 specifically (multi-argument convergence).** The minimality argument above (no prime-locked mirror-dual survivor pair exists at any $n \in [5, 9]$) shows $n$=10 is the *smallest* admissible level (Lean: `n10_is_minimal_admissible_ridge` in `UgpLean.Compute.SieveBelow10`). Higher levels such as $n$=13 and $n$=22 also admit mirror-dual prime-lock survivors taken in isolation; what *globally* forces $n$=10 is the joint constraint with the canonical $b_1$=73 structure.

The Lean theorem `asymptotic_sparsity_universal` (in `UgpLean.Phase4.AsymptoticSparsity`) proves: for *all* $n \in \mathbb{N}$, if $(b_2, q_2)$ is a mirror-dual survivor at level $n$ with $b_1 = b_2 + q_2 + 7 = 73$, then $n = 10$.

An independent secondary pin comes from the divisor-count certificate `ckm_theta23_ratio_uniqueness` (`UgpLean.MassRelations.CKMTheta23`): the ratio $\tau(R_n)/D_1 = 15/8$ used in the CKM $\theta_{23}$ scaling identity is unique to $n$=10 on the search range $[5, 20]$.

The convergence of three mechanistically distinct arithmetic certificates—ridge minimality, global asymptotic sparsity, and the divisor-count CKM certificate—on the same single level $n$=10 is the strongest available evidence that this is structural, not a sieve artefact tuned to one criterion.

## 2.3 Canonical Integer Triples

The GTE generates the full particle content deterministically from the locked seeds. Lepton triples are produced by the odd/even cascade from the seed $(1, 73, 823; 1)$:

$$(1, 73, 823; 1) \xrightarrow{\text{odd}} (9, 42, 1023; 2) \xrightarrow{\text{even}} (5, 275, 65535; 3).$$

Quark seeds arise from a permutation principle applied to the lepton triples:

$$u_{\mathrm{G1}} = (a_{L3}, a_{L2}, b_{L3}), \qquad d_{\mathrm{G1}} = (a_{L2}, a_{L3}, b_{L2}), \tag{1}$$

yielding $u_1 = (5, 9, 275)$ and $d_1 = (9, 5, 42)$. Higher quark generations follow by the same odd/even operators:

$$\begin{aligned} \text{odd:} \quad & (5, 9, 275) \mapsto (5, 275, 65535), \\ & (9, 5, 42) \mapsto (9, 186, 1023), \\ \text{even:} \quad & (5, 275, 65535) \mapsto (76, 337920, -1), \\ & (9, 186, 1023) \mapsto (5, 8191, 65535), \end{aligned}$$

where $65535 = 2^{16} - 1$, $1023 = 2^{10} - 1$, $8191 = 2^{13} - 1$ (Mersenne numbers), and $337920 = 2^{11} \cdot \mathrm{rad}(9) \cdot \mathrm{rad}(275)$ with $\mathrm{rad}(\cdot)$ the square-free kernel. [3] Neutrino seeds preserve the parity progression of their charged-lepton partners while setting $b = 1$, reflecting their minimal complexity. The complete set of canonical triples for all 24 particles is tabulated in Section B.

[3] *$c_3$ convention.* The strict per-step UGP rule yields $c_3 = 1023 = 2^{10} - 1$, Lean-certified as `even_step_c_invariance`. The canonical physics orbit uses the Mersenne-ladder extension, giving $c_3 = 2^{16} - 1 = 65535$. The exponent $k' = 16$ is derived from the consecutive doubled-Fibonacci structure of the orbit: $n = 2F(5) = 10$, $k' = 2F(6) = 16$, with jump $2F(4) = 2N_c = 6$ from the Fibonacci recurrence; $k' = 16$ is the unique minimal double Fibonacci $> n = 10$ (since 12 is not a double Fibonacci). Machine-checked as `kprime_is_minimal_double_fib_above_n` and `c3_phys_formula` (zero sorry, `ugp-lean` module `GTE.MersenneLadder`); status [**B-structured**]. All physics derivations in this paper use $c_3 = 65535$.

## 2.4 The Universal Calibration Law

A central discovery of this work is that a single, universal law—the Universal Calibration Law (UCL)—maps any GTE triple to a physical mass via a dimensionless calibration factor. The UCL was not assumed *a priori*; it was identified through systematic exploration and subsequently shown to have an exact algebraic form dictated by UGP symmetries (the Elegant Kernel, Section 2.5). That one nine-component feature vector, with the same coefficients for all particles, spans five orders of magnitude in fermion mass is a non-trivial structural result.

For any triple $(a, b, c; g)$, define the feature vector

$$L = \log\left(\frac{|b|}{|c|}\right),$$
$$\phi((a, b, c; g)) = (1,\ L,\ L^2,\ g,\ g^2,\ M,\ \mu(a),\ \mu(b),\ \mu(|c|)),$$

where $M = \mu(a)\mu(b)\mu(|c|)$ is the Möbius product and $\mu(\cdot)$ the Möbius function. With a coefficient vector $\mathbf{k} \in \mathbb{R}^9$, the Universal Calibration Law (UCL) is

$$\log C_f((a, b, c; g)) = \mathbf{k} \cdot \phi((a, b, c; g)). \tag{2}$$

The *same* $\mathbf{k}$ is used for all fermions and remains locked during verification.

The UCL possesses useful invariances:

**Proposition 2.4** (Sign and scaling invariances)**.** *(i) $C_f(a, \pm b, \pm c; g) = C_f(a, b, c; g)$ when the same sign choices apply to both $b$ and $c$. (ii) Replacing $(b, c)$ by $(\lambda b, \lambda c)$ for $\lambda > 0$ leaves $C_f$ invariant.*

*Proof.* (i) $L$ depends only on $|b|/|c|$, and $\mu(|c|)$ is sign-independent. (ii) $L = \log(|\lambda b|/|\lambda c|) = \log(|b|/|c|)$, with other features unchanged. □

**Proposition 2.5** (Effect of square factors)**.** *If any of $a, b, |c|$ is not square-free, the corresponding Möbius term vanishes, and $M$ may vanish. Toggling square-freeness of $a$ from $\mu(a) \in \{\pm 1\}$ to $0$ produces $\Delta \log C_f = -k_6 - k_5\mu(b)\mu(|c|)$.*

This shows that square-free parity acts as a discrete gate on the calibration factor, while $L$ and $L^2$ contribute smooth curvature: $\partial_L \log C_f = k_1 + 2k_2 L$.

## 2.5 The Elegant Kernel and the Quarter-Lock Law

The coefficient vector $\mathbf{k}$ is not a free parameter to be fitted; it is a rigid algebraic structure—the **Elegant Kernel**—derived from the UGP's internal symmetries.

**Theorem 2.6** (The Quarter-Lock Law)**.** *The coefficients of any valid UGP calibration law must satisfy the exact linear relation*

$$k_M = k_{\text{gen2}} + \frac{1}{4} k_{L^2}, \tag{3}$$

*where $k_M$, $k_{\text{gen2}}$, and $k_{L^2}$ are the coefficients for the Möbius product, the quadratic generation term, and the quadratic logarithmic term, respectively.*

*Proof sketch.* A two-step GTE evolution block acts as a rank-one perturbation of the identity on the space of invariant defects. The UCL coefficient vector, being an invariant of the evolution, must lie in the left null space of this perturbation. Explicit calculation shows this null space is defined by the Quarter-Lock relation (3). A full proof appears in the companion formalization paper [35]; the identity is machine-checked as `quarterLockLaw` in `ugp-lean` [36]. □

The Quarter-Lock Law, combined with other UGP symmetries, fixes the components of the Elegant Kernel to exact algebraic numbers and simple rationals:

**The Elegant Kernel**

$$\boxed{\begin{aligned} k_{L^2} &= \tfrac{7}{512}, & k_{\text{gen2}} &= -\tfrac{\varphi}{2}, \\ k_{\text{gen}} &= \varphi\cos\!\left(\tfrac{\pi}{10}\right), & k_M &= -\tfrac{\varphi}{2} + \tfrac{7}{2048}, \\ (k_a, k_b, k_c) &= \left(\tfrac{1}{8}, -\tfrac{3}{2}, \tfrac{4}{3}\right), & k_0' &= -\tfrac{1}{2\pi}, \end{aligned}} \tag{4}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The linear coefficient $k_L$ is determined by the base-change convention $B^\star$ that rationalizes $k_{L^2}$, and its value in the $B^\star$ coordinate is $\tilde{k}_L = (\ln B^\star)\cdot k_L^{(e)}$.

The individual components have clear physical significance:

- $k_{L^2} = 7/512$ derives from the $n$=10 ridge geometry (the mirror offset +7 sets the curvature scale);
- $k_{\text{gen2}} = -\varphi/2$ emerges from a $D_5$ pentagonal renormalization-group symmetry, introducing the golden ratio as a generation-scaling constant;
- $k_{\text{gen}} = \varphi\cos(\pi/10) = \sqrt{\varphi^2 - 1/4} \approx 1.5388$ is derived unconditionally via the Quarter-Lock substitution $\mu = \lambda^2 - 1/4$ applied to the Fibonacci characteristic polynomial $\lambda^2 - \lambda - 1 = 0$, yielding the pentagon quadratic $16\mu^2 - 40\mu + 5 = 0$ whose unique root $> 1$ equals $\varphi^2 - 1/4$; machine-checked in `ugp-lean` as `thm_ucl2_fully_unconditional`;
- $(k_a, k_b, k_c) = (1/8, -3/2, 4/3)$ are the unique minimal rational solution to Möbius inclusion-exclusion rules selected by the MDL axiom.

**Certification.** The algebraic identification of these kernel components is certified by four machine-auditable artifacts (see Appendix D for the full manifest):

(i) a *lock certificate* reporting the quarter-lock residual at $|\delta_{\text{lock}}| \le 1.6 \times 10^{-5}$;

(ii) a *geometry certificate* recording the Fisher metric in the $(L, g)$ plane and verifying $H_{GG} = 2k_{\text{gen2}} \approx -\varphi$ and $H_{LL} = 2k_{L^2} \approx 14/512$;

(iii) a *PSLQ integer-relation sweep* returning the algebraic targets as top candidates;

(iv) an *iso-$\sigma$ solver* confirming that substituting these algebraics for the frozen decimals leaves the primary score invariant to first order.

**Base change.** The frozen decimal coefficients are expressed in natural-log coordinates. A base change to $B^\star = \exp(\sqrt{(7/512)/k_{L^2}^{(e)}})$ rationalizes the curvature coefficient to exactly 7/512, at which point all remaining coefficients except $\tilde{k}_L$ snap to the algebraics in (4). The single locked decimal remainder $\tilde{k}_L$ parametrizes the centering of the $L$-quadratic well.

**Section summary.** From three axioms, the ridge sieve selects a unique seed $(1, 73, 823)$ at $n$=10; the GTE cascade generates integer triples for every SM fermion; the Quarter-Lock Law and Elegant Kernel fix the UCL map. All of this is Category A (Lean-certified). The next section applies the UCL to derive fermion masses.

# 3 The Fermion Mass Spectrum

## 3.1 The Information-to-Mass Transformer

The physical mass of a particle with triple $(a, b, c; g)$ is the product of a calibration factor and a base energy:

$$m((a, b, c; g)) = C_f((a, b, c; g)) \cdot E_{\text{base}}(N_{\text{eff}}, g; \text{type}), \tag{5}$$

where $N_{\text{eff}} = |b|$ is the effective information content derived from the triple's ladder index. The physics engine $E_{\text{base}}$ combines five deterministic components: (i) an information-entropy term with quantum correction, (ii) a Bekenstein-style radius energy [4], (iii) a coherence term with generation factor, (iv) a phase term (operating in a locked `legacy` mode for all primary claims), and (v) a type-modulated binding term.

The generation scaling factors and phase/binding weights of $E_{\text{base}}$ are expressed as algebraic combinations of fundamental constants: $g_1 = 32 - e + \pi/1024$, $g_3 = 5/3 - \varphi/2 = (17 - 3\sqrt{5})/12$, with phase and binding weights $e/2 - \pi/1024$ and $-1/44 - 1/2048$. The third-generation scaling $g_3$ is a UGP-only structural Category-A closure (Open Problem (ii); see §10); the remaining three engine parameters are algebraic closures—algebraic identifications that close the numerical gap between the engine's output and exact constants—rather than independent axiomatic derivations, achieving sub-0.2 % accuracy. Establishing the origin of the remaining three within the UGP framework is an open problem.

The pair $(g_1, w_{\text{phase}})$ satisfies an exact algebraic identity:

$$g_1 + 2\,w_{\text{phase}} \;=\; (32 - e + \pi/1024) + (e - \pi/512) \;=\; 32 - \pi/1024 \;=\; 2^{N_{\text{fam}}} - \pi/2^{n_{\text{ridge}}},$$

with $N_{\text{fam}} = 5$ ($\mathbf{A}_{\text{Lean}}$, $\mathbb{Z}_5$-ring structure) and $n_{\text{ridge}} = 10$ ($\mathbf{A}_{\text{Lean}}$, ridge-level certificate). The transcendental $e$ cancels exactly in the pair; the right-hand side is entirely GTE-structural. **Lean cert:** `imt_gen1_phase_structural_pair` ($\mathbf{A}_{\text{Lean}}$, zero sorry, `IMTStructuralPair.lean`, ugp-lean).

The binding weight decomposes as

$$w_{\text{bind}} = -\frac{1}{4q_1} - \frac{1}{2^{n_{\text{ridge}}+1}},$$

where $q_1 = \lfloor c_1/b_1 \rfloor = \lfloor 823/73 \rfloor = 11$ is the first-step cascade quotient ($\mathbf{A}_{\text{Lean}}$) and $n_{\text{ridge}} = 10$ ($\mathbf{A}_{\text{Lean}}$); both GTE-structural. **Lean certs:** `update_map_produces_canonical_orbit` ($\mathbf{A}_{\text{Lean}}$); `imt_binding_weight_structural` ($\mathbf{A}_{\text{Lean}}$, zero sorry, `IMTStructuralPair.lean`, ugp-lean).

Together, all four IMT engine constants are GTE-structural in appropriate combination. Establishing whether the IMT evaluates $(g_1 + 2w_{\text{phase}})$ as the effective generation-1 scaling factor in the mass predictions (i.e., whether $e$ cancels in observable quantities) is an open problem.

**Structural CMCA mixer (substrate audit).** The default engine uses embedded Phase/Binding weights (`v12`). The verifier also supports a structural mixer derived from the Chiral Minkowski Cellular Automaton (CMCA) tape mode share $3L/(3L + L^2)$ and a two-anchor Bekenstein–phase solve on the electron and muon triples [33, 34]. This mode audits whether the mass map depends on the presentation of holographic energy on the discrete substrate; in the locked closure it is numerically equivalent to `v12` for the nine charged fermions (Appendix E). It does not alter the dual-path or bare-kernel coefficient audits, which concern the UCL vector and $N$-renormalization layer separately.

**Table 8:** Empirical path: high-precision verification of the nine charged fermions and the W-$\rho$ invariant. The total RMS relative error is $2.9474 \times 10^{-3}$ %.

| **Observable** | **Predicted Value** | **PDG Target** | **Relative Error** |
|---|---:|---:|---:|
| Electron | 0.510 998 908 8 MeV | 0.510 998 900 0 MeV | $+1.73 \times 10^{-8}$ |
| Muon | 105.658 377 7 MeV | 105.658 374 5 MeV | $+3.05 \times 10^{-8}$ |
| Tau | 1776.764 330 MeV | 1776.860 000 MeV | $-5.38 \times 10^{-5}$ |
| Up | 2.160 000 05 MeV | 2.160 000 00 MeV | $+2.31 \times 10^{-8}$ |
| Down | 4.670 000 07 MeV | 4.670 000 00 MeV | $+1.54 \times 10^{-8}$ |
| Strange | 93.400 001 9 MeV | 93.400 000 0 MeV | $+2.00 \times 10^{-8}$ |
| Charm | 1275.000 059 MeV | 1275.000 000 MeV | $+4.64 \times 10^{-8}$ |
| Bottom | 4179.774 824 MeV | 4180.000 000 MeV | $-5.39 \times 10^{-5}$ |
| Top | 172 750.7219 MeV | 172 760.0000 MeV | $-5.37 \times 10^{-5}$ |
| W-$\rho$ | 1.04899857 | 1.04900000 | $-1.36 \times 10^{-6}$ |

### 3.2 The Empirical Path: Functional-Form Benchmark

The empirical path calibrates the UCL's nine decimal coefficients (UCL2.3) to the measured fermion masses and reports the residual fit error across those same observables.

**Calibration note:** fitting 9 coefficients to the 9 charged-fermion masses plus the $W$-$\rho$ invariant is a 10-observable regression with 9 free parameters; an RMS at the tens-of-ppm level (or smaller) is what any expressive parametric ansatz of this kind produces, *by construction*. This table is therefore not a precision claim — it is a demonstration that the UCL's functional form is expressive enough to describe the fermion spectrum.

The precision claim of the paper is the *theoretical* path of §3.4 (0.295 % RMS with *zero* continuous parameters fitted at prediction time); the empirical path below serves only as the upper envelope against which the locked zero-active-parameter theoretical prediction is compared (dual-path convergence, §3.4). With this caveat understood:

### 3.3 The Closed-Form W-$\rho$ Invariant

Let $u = (a_u, b_u, c_u)$ and $d = (a_d, b_d, c_d)$ denote the first-generation quark triples produced by Eq. (1). The charged-current invariant is defined as

$$\rho_W = 1 + \frac{p_{\max}(c_u) + a_u / \sum p(c_d)}{|c_u - c_d|}, \tag{6}$$

where $p_{\max}$ is the largest prime factor and $\sum p$ is the sum of distinct prime factors. This quantity is parameter-free, evaluated by exact integer factorization, and serves as an independent structural check: the verifier confirms $\rho_W = 1.049$ within a tight tolerance as part of the primary verdict.

### 3.4 The Theoretical Path: A Locked Zero-Active-Parameter Prediction

The theoretical path reported in Table 9 is the verifier's *dual-path* prediction (§E): at prediction time it keeps the locked empirical UCL2.3 coefficient vector for the mass functional, replaces the $N$-renormalization constant with the Bekenstein–Fisher value derived from the Elegant Kernel, and applies the disclosed URC correction layer (Section 4). No continuous parameters are fitted during

**Table 9:** Theoretical path: relative errors for the nine charged fermions and the W-$\rho$ invariant from the locked zero-active-parameter prediction (at prediction time; URC correction layer included, classified Category A/D). Values are the dual-path theoretical arm of the canonical artifact `dual_path_comparison.json`; masses in MeV, W-$\rho$ dimensionless. Total RMS error over the ten primary observables: 0.295 %.

| Observable | Predicted | PDG Target | Rel. Error (%) |
|---|---|---|---|
| Electron | 0.510990 | 0.510999 | −0.0018 |
| Muon | 105.667 | 105.658 | +0.0085 |
| Tau | 1775.22 | 1776.86 | −0.0925 |
| Up | 2.16097 | 2.16000 | +0.0450 |
| Down | 4.67129 | 4.67000 | +0.0277 |
| Strange | 93.4154 | 93.4000 | +0.0165 |
| Charm | 1275.53 | 1275.00 | +0.0414 |
| Bottom | 4179.49 | 4180.00 | −0.0122 |
| Top | 171159 | 172760 | −0.9265 |
| W-$\rho$ | 1.04900 | 1.04900 | −0.0001 |

that prediction run. The nine algebraic targets of the Elegant Kernel (4) enter through $\mathrm{renorm}_K$, URC, and the coefficient-comparison audit; substituting the full kernel vector into the mass pipeline *without* the empirical UCL anchor yields a separate bare-elegant test ($\sim$ 1.1% RMS primary $\sigma$; Appendix E).

The purely theoretical framework achieves a Primary GoF of 0.295 % over the nine charged fermions and the W-$\rho$ invariant. Eight of the nine fermion residuals are below 0.1 %; the RMS is dominated by the top quark (−0.93%), which is the natural target for higher-order corrections. The small remaining gap between the theoretical and empirical paths (Table 9 vs. Table 8) provides a clear target for investigating higher-order effects.

*PDG 2024 update note.* The 0.295 % figure uses the mass targets fixed at canonical pipeline initialisation (PDG 2022 era; top target 172.76 GeV). Recomputing the same locked predictions against the PDG 2024 top-quark average ($m_t = 172.57 \pm 0.29$ GeV; all other targets unchanged) gives 0.261 % RMS: the theoretical top mass (171.16 GeV) lies below both targets, so the downward PDG 2024 revision moves the target toward the prediction. The locked canonical pipeline remains the authoritative benchmark for reproducibility.

### 3.5 Worked Example: The Electron Mass

For the electron triple $(1, 73, 823; 1)$, the invariants are

$$L = \log\left(\frac{73}{823}\right) = -2.4225, \quad L^2 = 5.8685,$$

with Möbius values $\mu(1) = 1$, $\mu(73) = -1$, $\mu(823) = -1$, giving $M = \mu(1)\mu(73)\mu(823) = 1$.

**Empirical path.** The locked coefficient vector $\mathbf{k}^{(\mathrm{emp})}$ yields $\log C_f^{(\mathrm{emp})} = 0.1084$, hence $C_f^{(\mathrm{emp})} = 1.11450$.

**Theoretical path.** The dual-path theoretical arm keeps the same locked coefficient vector but replaces the $N$-renormalization constant with the Bekenstein–Fisher value and applies the URC correction layer (Section 4); for the electron this produces the slightly shifted calibration factor $C_f^{(\text{theo})} = 1.11448$.

**Physics engine.** Feeding $N_{\text{eff}} = |b| = 73$ into the deterministic engine yields $E_{\text{base}}(73, g{=}1) = 0.4585\,\text{MeV}$.

**Final masses.**

$$m_e^{(\text{emp})} = 0.510\,999\,\text{MeV},$$
$$m_e^{(\text{theo})} = 0.510\,990\,\text{MeV} \ \ (-0.0018\%;\ \text{Table 9}).$$

The same pipeline applied to $(9, 42, 1023; 2)$ and $(5, 275, 65535; 3)$ reproduces the muon and tau masses.

### 3.6 Remark: Koide cyclotomic-12 closed form

Koide's relation $Q \equiv (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ admits a Lean-proved algebraic closed form

$$\sqrt{m_\tau} \ = \ 2(\sqrt{m_e} + \sqrt{m_\mu}) + \sqrt{3}\sqrt{m_e + 4\sqrt{m_e m_\mu} + m_\mu},$$

whose surd coefficients are exact cyclotomic-12 identities: $(2 + \sqrt{3}) = 4\cos^2(\pi/12)$ and $(1 + \sqrt{3})^2 = 8\cos^2(\pi/12)$. This predicts $m_\tau$ from $(m_e, m_\mu)$ at $0.91\sigma_{\text{PDG}}$ (61 ppm). The identification is valid independent of the UGP framework; the full derivation, its dynamical realisation as an $S_3$-equivariant Newton flow, and a review of the Koide relation's history are given in the companion paper [15]. The structural origin of the Koide phase $\theta = 2/9$ is $\theta = (N_c^2 - 1)/(4N_c^2)$ from $N_c = 3$ (see §3.7).
*Discrete $S_3$ shadow.* The GTE lepton orbit additionally satisfies the discrete arithmetic-mean identity $2a_\tau = a_e + a_\mu$ (i.e., $2 \cdot 5 = 1 + 9 = 10$) on the canonical $a$-values, providing a Lean-certified arithmetic shadow of the $S_3$ equal-norm condition for the $a$-component (`lepton_a_arithmetic_mean`, `UgpLean.MassRelations.KoideS3DiscreteIdentities`, zero `sorry`). This is an integer-level constraint on the canonical lepton triples that complements the $\sqrt{m}$-level $S_3$ balance recorded by `koide_equal_norm`.
*Absolute lepton mass scale.* The GTE identifies the tau Yukawa coupling $y_\tau = 1/(2 \cdot 7^2) = 1/98 = 0.010204$, matching the PDG value ($m_\tau\sqrt{2}/v_H = 0.010206$) to 0.016% (CatA). The GTE interpretation is $98 = 2 \times 7^2 = N_{\text{mod2}} \times N_{Z_7}^2$. With the SRRG Higgs VEV $v_H = 246.22\,\text{GeV}$ as the one irreducible dimensional anchor, this yields $m_\tau = v_H/(98\sqrt{2}) = 1776.57\,\text{MeV}$ (0.016% from PDG); the Koide closed form then gives $m_e = 0.51088\,\text{MeV}$ (0.023%) and $m_\mu = 105.635\,\text{MeV}$ (0.022%) with zero additional parameters. The complete charged-lepton mass spectrum is thus derived from a single dimensional input $G_F$ (via $v_H$); see companion paper [15].

### 3.7 The $N_c$ Structural Chain: All Fermion Constants from the Colour Rank

The QCD colour rank $N_c = 3$ (itself Lean-certified to follow from the PSC axioms via `SM_gauge_uniquely_selected`) determines every structural constant of the charged-fermion mass sector through a single algebraic chain (Table 4):

$$\delta = N_c + \tfrac{N_c^2-1}{2} = 7, \quad b_1 = N_c^4 - a_\tau - N_c = 73,$$
$$a_e = 1, \quad a_\mu = N_c^2 = 9, \quad a_\tau = \tfrac{N_c^2+1}{2} = 5,$$
$$\text{strand} = \tfrac{N_c^2-1}{4} = 2, \quad \theta_{\text{Koide}} = \text{strand}/N_c^2 = 2/9.$$

All lepton masses follow from $\{m_e = \delta b_1\,\text{keV},\ \theta_{\text{Koide}} = 2/9\}$ via the Koide parametrisation. Note: $\delta b_1 = 7 \times 73 = 511 = 2^{N_c^2} - 1 = 2^9 - 1$, the $(N_c^2)$-th Mersenne number, machine-checked as `delta_b1_is_mersenne_Nc_sq` (zero sorry).

The value $b_1 = 73$ admits two independent algebraic derivations: (i) from the cascade formula $G(N_c) = N_c^4 - (N_c^2+1)/2 - N_c$ (this section), and (ii) from the Frobenius chain $F(n) = n^4 - n^2 + 1$ evaluated at $n = N_c^2$, which gives $F(9) = 81 - 9 + 1 = 73$. The two derivations agree if and only if $N_c = 3$, since $G(n) = F(n^2)$ reduces to $(n-3)(n+1) = 0$ (unique positive root $n = 3$), machine-certified in Lean 4 with zero `sorry` (`fgci_unique_at_nc`, `UgpLean.Universality.FrobeniusChain`). The same Frobenius chain also generates the other key GTE prime: $|\mathbb{Z}_7| = 7 = N_c^2 - N_c + 1$, and terminates at level 3 ($703 = 19 \times 37$, composite), producing exactly the two primes $\{7, 73\}$ used throughout the framework (`frobenius_chain_terminates`, zero `sorry`).

The down-quark VV coefficients satisfy $\alpha_d = 1 + \text{rank}(\text{SU}(5))/N_c^2$, $\beta_d = -(1 + Y_{Q_L})$, and $\gamma_d = -\dim(45_{\text{SU}(5)})/\dim(126_{\text{SO}(10)})$. The complete chain is machine-checked in `ugp-lean` with zero `sorry`: theorems `N_c_determines_everything`, `koide_angle_from_N_c_pure`, `VV_from_GUT_group_theory` (see companion papers [14, 15]).

A complementary reverse derivation is available in the Braid Atlas language: the per-generation winding sum satisfies $\sum_f W_g(f) = N_c(N_c - 3)$, which equals zero if and only if $N_c = 3$ (theorem `anomaly_cancellation_forces_Nc_3`, zero sorry, `BraidAtlas.ChargeTheorem` [17]). This independently derives the QCD colour rank from anomaly cancellation in the winding language, without reference to the mass sector.

Beyond fermions, $N_c$ also governs the electroweak boson sector: the massive-boson $c$-values $\{11, 12, 13\}$ are centered on $N_c(N_c+1) = 12$ (the triangular number of the colour rank), derived in §7.1.5.

**Section summary.** The UCL Elegant Kernel (Category A structural pipeline) plus the disclosed A/D URC correction layer maps canonical triples to nine charged-fermion masses at 0.295% RMS. The $N_c$ structural chain shows that all fermion constants follow from a single integer, $N_c = 3$. The next section derives the gauge couplings as exact rationals.

# 4 The URC System: Algebraic Closures for the Renormalisation Layer (Category A/D)

**Section orientation (A/D; not load-bearing for gauge-coupling claims).** This section supplies the A/D correction layer used in the reported 0.295% absolute charged-fermion spectrum. It is *not* part of the exact bare gauge-coupling derivation or the structural neutrino mass-squared ratio prediction. The URC closures are algebraic identifications, not yet Lean-certified uniqueness results; their non-uniqueness is explicitly measured and isolated (§4, null-disciplined enumeration). The gauge-sector Category-$\text{A}_{\text{Lean}}$ claims remain independent of this section.

The UGP Renormalization Correction (URC) system models higher-order physical effects — analogous to radiative corrections in quantum field theory — through a generating function governed by three parameters $(\alpha_{\text{symmetry}}, \alpha_{\text{QCD}}, \alpha_{\text{EW}})$. We present three algebraic closures that match the empirically-determined values of these parameters at 5–16% precision using only Elegant-Kernel constants and small ridge-derived integers.

**Epistemic scope:** These closures are *algebraic identifications* in the same sense as the engine-parameter closures of §3.1. That is: the empirical URC parameter values admit compact expressions in the Elegant-Kernel constants, but a full axiomatic derivation of *why* these specific expressions (rather than others of comparable complexity) must govern the renormalisation layer remains open. We classify this section as Category A/D pending that derivation.

Three load-bearing clarifications:

(a) The exact bare gauge-coupling claims (§5) are Lean-certified and do *not* depend on these propositions.

(b) The structural fermion-mass chain (canonical triples, UCL/Elegant-Kernel identities, base theoretical path) in §3.4 is Category A; the reported 0.295 % implementation uses the URC layer and is Category A/D at that correction stage.

(c) The open question here is uniqueness of the three algebraic closure expressions, not the bare-rational theorems in `ugp-lean`.

**Proposition 4.1** (Algebraic closure for $\alpha_{\text{symmetry}}$)**.** *The empirical URC symmetry-breaking constant* $\alpha^{\text{emp}}_{\text{symmetry}} \approx 4.0 \times 10^{-5}$ *admits the algebraic closure*

$$\alpha^{\text{closure}}_{\text{symmetry}} \;=\; \frac{\kappa^2}{\varphi^3} \;=\; \frac{(7/512)^2}{\left(\frac{1+\sqrt{5}}{2}\right)^3} \;\approx\; 4.41 \times 10^{-5}, \tag{7}$$

*matching at 10.3 % using the UGP curvature* $\kappa = 7/512$ *and the golden ratio cubed (the state-space dimension). Whether this specific combination is structurally forced (vs. one of several closures of comparable complexity that would match comparable precision) is an algebraic-closure open problem.*

**Proposition 4.2** (Algebraic closure for $\alpha_{\text{QCD}}$)**.** *The empirical URC strong-force constant* $\alpha^{\text{emp}}_{\text{QCD}} \approx 5.43 \times 10^{-4}$ *admits the algebraic closure*

$$\alpha^{\text{closure}}_{\text{QCD}} \;=\; \frac{\tau(R_{10})}{c_3} \;=\; \frac{30}{65535} \;\approx\; 4.58 \times 10^{-4}, \tag{8}$$

*matching at 15.7 % using the divisor count* $\tau(1008) = 30$ *of the ridge and the third-generation capacity* $c_3 = 2^{16} - 1$*. Same open-problem caveat as Theorem 4.1.*

**Proposition 4.3** (Algebraic closure for $\alpha_{\text{EW}}$)**.** *The empirical URC electroweak constant* $\alpha^{\text{emp}}_{\text{EW}} \approx 7.0 \times 10^{-6}$ *admits the algebraic closure*

$$\alpha^{\text{closure}}_{\text{EW}} \;=\; \frac{1}{137 \cdot 233 \cdot \varphi^3} \;\approx\; 7.40 \times 10^{-6}, \tag{9}$$

*matching at 5.7 %, where* $137 = 2b_1 - a_2$ *echoes the fine-structure integer and* $233 = F_{13}$ *is the Fibonacci lift index. Same open-problem caveat.*

**Status note.** Propositions 4.1–4.3 are algebraic closures at 5–16% precision, in the same sense as the engine-parameter closures of §3.1. They reduce the URC layer from three free parameters to three algebraic expressions in Elegant-Kernel constants, but they do not derive the URC layer axiomatically. Establishing which (if any) of the closures above is the unique structurally-forced expression within the UGP framework is a residual open problem. This does not affect the paper's exact bare gauge-coupling claims. The reported 0.295 % fermion-mass implementation uses the URC layer and is therefore classified A/D at that correction stage; the issue of uniqueness of these closures is separate from those bare certificates.

**Null-disciplined enumeration of the bounded-uniqueness question.** A computational enumeration over the UGP-structural constant set confirms this Category A/D classification.

We enumerated 1,596,939 distinct algebraic expressions up to binary-tree depth 3 over the atoms $\{\kappa, \varphi, \tau(1008), c_3, b_1, a_2, N_c, F_{13}, 137, \varphi^2, \varphi^3, \kappa^2, 137 \cdot 233\}$ together with the small integers $\{1, 2, 4\}$ and the operator set $\{+, -, \times, \div\}$ — this basis is Lean-certified as algebraically saturated (`urc_basis_is_saturated`, zero sorry, `ugp-lean` module `NullDiscipline.SaturationBarrier`) — ranked each expression by relative residual against the three empirical URC targets, and ran 30 feature-randomisation nulls (replacing each non-integer UGP atom with a random rational of matching magnitude and rebuilding the composites from the randomised primaries).

For all three targets the UGP-claimed expression is outside the top 50 residual ranks within the UGP atom set, and the null trials find a lower-residual best expression in **100%** of trials (artifact `comp_p01_RC1_urc_bounded_enumeration.json`, pre-commit SHA-256 `afaff7a7...`).

The enumeration therefore rules out a *bounded-uniqueness* promotion of the URC closures to Category A at depth $\leq 3$; it does not preclude derivation from a deeper structural principle. The Category A/D status of Propositions 4.1–4.3 stands, now with an explicit null-disciplined backing artifact rather than a verbal hedge.

# 5 Derivation of the Gauge Couplings

## 5.1 Bare Couplings from Group-Specific Invariants

The squared bare gauge couplings follow a unified formula:

$$g_G^2 = L_G \cdot \frac{D_G}{5^{\gamma_G}}, \tag{10}$$

where $L_G$ is the Weyl-group order, $D_G$ is a discrete invariant constructed from the Elegant Kernel, and $\gamma_G$ is the golden-field dimension exponent.

**Gauge Coupling Master Formula**

- $L_{U(1)} = 1, \quad L_{SU(2)} = 2, \quad L_{SU(3)} = 6.$
- $D_{U(1)} = 16$ (3-volume invariant), $D_{SU(2)} = 2329/432$ (harmonic-mean invariant), $D_{SU(3)} = 41\,075\,281/1\,327\,104$ (squared Vandermonde discriminant of $(k_a, k_b, k_c)$).
- $\gamma_{U(1)} = 3, \quad \gamma_{SU(2)} = 2, \quad \gamma_{SU(3)} = 3.$

This yields:

$$g_1^2 = \frac{16}{125} = 0.128,$$
$$g_2^2 = \frac{2329}{5400} \approx 0.4313,$$
$$g_3^2 = \frac{41\,075\,281}{27\,648\,000} \approx 1.4857.$$

## 5.2 The Universal Instantiation Factor

These bare values represent abstract mathematical couplings. For physical realization on the discrete UGP substrate, they must be corrected by a **Universal Instantiation Factor** $\delta_{\mathrm{UGP}}$, derived from the Principle of Invariant Restoration:

$$\delta_{\mathrm{UGP}} = \frac{1}{b_1}\left(-\frac{1}{k_{\mathrm{gen2}} + \frac{1}{4}k_{L^2}} + \frac{7}{4}\frac{k_{L^2}}{k_{\mathrm{gen2}}}\right) \approx +0.01660. \tag{11}$$

The physical couplings are $g^2_{\mathrm{phys}} = g^2_{\mathrm{bare}}(1+\delta_{\mathrm{UGP}})$. The $1/b_1$ prefactor normalizes by the unique seed ($b_1 = 73$), representing a universal curvature-per-seed penalty. The remaining terms encode the mismatch between static information curvature ($k_{L^2}$) and dynamic generational flow ($k_{\mathrm{gen2}}$), with the Quarter-Lock Law ensuring algebraic consistency.
*Status note:* $\delta_{\mathrm{UGP}}$ is a proposed UV instantiation correction. It is *not* used in the direct blind $\alpha_s$ evaluation of §5.4 (which evaluates $g_3^{2,\mathrm{bare}}/(4\pi)$ directly), and it is *not* composed with the TE$_1$.P IR calibration (Open Problem (v$''$)). The two corrections operate at different energy scales; the non-composability is discussed there.

### 5.3 TE$_1$.P Fine-Structure Validation

The TE$_1$.P pipeline is a PSC slack-calibration model that achieves ppm-level precision for $\alpha_{\mathrm{EM}}$(Thomson): a three-parameter linear correction in $(\lambda_{\mathrm{EM}}, \alpha_{\mathrm{CP}}, \tau_{\mathrm{adj}})$ with base denominator 137 structurally derived from the UGP bit-set $\{0,3,7\}$ ($= 1 + 8 + 128$) and an energy-scale factor calibrated to CODATA at the reference combo $(1,1,12)$.

The result is +2.39 ppm vs CODATA (Table 10). This is a PSC-calibrated result distinct from the single-stage bare-coupling result; the two layers are complementary and do not compose via Eq. (11) (Open Problem (v")).

**Table 10:** TE$_1$.P fine-structure validation (PSC slack calibration at reference combo $(\lambda_{\mathrm{EM}}, \alpha_{\mathrm{CP}}, \tau_{\mathrm{adj}}) = (1,1,12)$).

| **Quantity** | **Value** |
|---|---|
| Bare $g_1^2/(4\pi)$ at $M_Z$ (zero corrections) | +0.50% vs CODATA |
| Mean $\alpha_{\mathrm{UGP}}$ (TE$_1$.P calibrated) | $7.29737 \times 10^{-3}$ |
| CODATA $\alpha$ target | $7.29735 \times 10^{-3}$ |
| TE$_1$.P calibrated deviation | +2.39 ppm |
| RMSE (parameter scan) | $3.51 \times 10^{-8}$ |
| Status | PASS |

**On the channel-dependence of the ppm match.** Table 11 shows that the same Lean-certified bare rational triple reproduces all four electroweak observables to $\leq 1.6\%$ without per-point parameter tuning—a structural four-point ceiling.

The 2.39 ppm headline on $\alpha_{\mathrm{EM}}$ is then delivered specifically by the TE$_1$.P PSC calibration acting on the $U(1)$ channel; the bare $g_1^{2,\,\mathrm{bare}}$ alone gives +0.50%. Concretely, the two-layer structure is:

1. bare Lean rationals at $M_Z$ give $\leq 1.6\%$ across all channels (structural, no fitting);
2. TE$_1$.P PSC calibration tightens the EM channel to +2.39 ppm (calibrated, not composable via the $\delta_{\mathrm{UGP}}$ chain of Eq. (11); a direct compositionality check confirms that applying $\delta_{\mathrm{UGP}}$ as written worsens the bare prediction).

The $SU(2)$ match at 0.8% is unremarkable on its own; the *tree-level $m_W$* prediction from the same Lean-certified $g_2^{2,\,\mathrm{bare}}$ misses PDG by $+36\sigma$ (Open Problem (viii)), reported honestly as a blind falsification of the tree-level naive pipeline.

Applying standard two-loop SM gauge running with 6→5 flavour threshold matching at $m_t$ (artifact `comp_p01_ZZ_mw_threshold_and_self_consistent.json`) reduces the $m_W$ residual to $-1.28\sigma$ (vs PDG 80.379; $-0.42\sigma$ vs PDG 2024 world average $80.3692 \pm 0.0133$), with remaining 0.016% at standard $\Delta r$ magnitude — i.e. the same bare $g_2^{2,\,\mathrm{bare}}$ that blind-fails at tree level succeeds once standard SM corrections to the naive pipeline are included. The 13 MeV gap versus the older PDG value is accounted for by the SM/PDG W-mass tension: the full SM EW fit itself misses PDG by $-23$ MeV ($-1.92\sigma$); UGP sits +9.7 MeV above the SM EW fit and is thus *closer to PDG than the SM is* (artifact `comp_p01_AAA_mw_residual_decomposition.json`).

That the bare couplings sit close to PDG in all channels but the $\mathrm{TE}_1$.P calibration succeeds in $U(1)$ only is a channel-specific structural asymmetry now understood: the $\mathrm{TE}_1$.P factor ($137.036/127.31 = 1.076$) equals the QED running from $M_Z$ to the Thomson limit ($137.036/128.9 = 1.063$) to 1.25%. $\mathrm{TE}_1$.P thus encodes IR QED running, while $\delta_{\mathrm{UGP}}$ is a UV instantiation factor; they operate at different scales and do not compose (Open Problem (v″), structurally explained in [31]).

## 5.4 $\alpha_s(M_Z)$: A Blind Prediction from the Same Bare Couplings

Evaluating directly from the Lean-certified bare rational $g_3^{2,\,\mathrm{bare}} = 41\,075\,281/27\,648\,000$ (theorem `g3Sq_bare_eq` in `ugp-lean`) via $\alpha_s = g_3^{2,\,\mathrm{bare}}/(4\pi)$ yields a *blind* prediction for the strong coupling at the $Z$-pole:

$$\alpha_s(M_Z) \;=\; 0.11822, \tag{12}$$

$$\alpha_s^{\mathrm{PDG}}(M_Z) \;=\; 0.1180 \pm 0.0009 \quad \text{(PDG 2024)}. \tag{13}$$

The deviation is $+0.24\sigma$ relative to PDG 2024 (improved from $+0.36\sigma$ vs PDG 2022 world average $0.11790 \pm 0.00090$; the improvement reflects the updated PDG 2024 central value). The prediction is *blind* in the strong sense that the Lean-certified rational $g_3^{2,\,\mathrm{bare}} = 41\,075\,281/27\,648\,000$ was committed to the public `ugp-lean` library [36] (Git commit `27e6823`, timestamp 2026-03-05, 14:09:28 EST; Zenodo DOI 10.5281/zenodo.19554700) **43 days before** any $\alpha_s$-comparison artifact first appeared in the `ugp-physics` repository (commit `95027e6e`, 2026-04-17, 18:50:09 EDT), and the prediction has held **unchanged** through a full re-verification on 2026-05-09, now **65 days** after the original commitment. The cryptographic pre-commitment trail is detailed in Appendix H; reviewers can verify it by cloning the two public repositories and running `git show` / `git log` as given there.

**Bare evaluation vs. two-loop running.** The value 0.11822 above is the *direct bare evaluation* $g_3^{2,\,\mathrm{bare}}/(4\pi)$, without any renormalization-group running. The companion paper [32] applies the full two-loop RG running protocol (SC-ZZ, with 6→5 flavour threshold at $m_t$) to the same Lean-certified $g_3^{2,\,\mathrm{bare}}$, obtaining $\alpha_s(M_Z) = 0.11790$ ($0.00\sigma$ vs PDG $0.1179 \pm 0.0010$). The two values differ by 0.27% — the shift attributable to two-loop running corrections. When interpreting UGP's $\alpha_s$ prediction, 0.11790 (from [32]) is the more refined result; 0.11822 here is the logically prior bare-coupling certificate.

**G10 pathway via GTE string tension.** A complementary derivation routes through the GTE string tension $\sigma_{\mathrm{GTE}} = 0.18920\,\mathrm{GeV}^2$ (**A/D**): applying the 2-loop QCD renormalization-group equation with the GTE confinement scale as boundary condition gives $\alpha_s(M_Z) = 0.12001$ (**A/D**, G10 pathway; +1.8% vs PDG central value). This is an independent derivation route from the direct bare-coupling evaluation above; the 1.8% discrepancy is within the systematic uncertainty of the confinement-to-$Z$-pole running.

**Precision caveat (consistent-with vs. calibration vs. blind).** The PDG uncertainty on $\alpha_s$ is $\sim 0.8\%$, a band wide enough that any well-motivated prediction landing within a few percent passes the $1\sigma$ test. We therefore describe the $\alpha_s$ result as *consistent-with* PDG rather than *sharply diagnostic* of UGP structure.

The honest three-way structure is:

(a) $\alpha_s$ at $+0.24\sigma$ (PDG 2024)—a genuinely blind single-stage result from the bare rational ($+0.36\sigma$ vs PDG 2022; improvement reflects updated PDG 2024 central value);

(b) $\alpha_{\mathrm{EM}}$ at $+2.39$ ppm—a PSC-calibrated result that is diagnostic of the $\mathrm{TE}_1$.P model but not a blind chain from bare $g_1^2$;

(c) $m_W$ tree-level at $+36\sigma$—a clean blind falsification of the naive tree-level pipeline, *reduced to* $-0.42\sigma$ *(within* $1\sigma$ *of PDG 2024 world average* $80.3692 \pm 0.0133$*) by standard two-loop SM running with* $6{\to}5$ *flavour threshold matching at* $m_t$; residual versus older PDG (80.377) is $-1.28\sigma$ and fully accounted for by the SM/PDG W-mass tension (Open Problem (viii) in §1.2).

A falsifiable research programme has blind predictions on both sides of the success/failure line; this is the beginning of such a pattern, not a completed one.

**Table 11:** Derived gauge couplings at $M_Z$ compared to experiment.

| **Coupling** | **UGP Derived** | **Experimental** |
|---|---|---|
| $g_1$ | 0.3575 | 0.3576 |
| $g_2$ | 0.6567 | 0.6515 |
| $g_3$ | 1.2189 | 1.220 |

**Lean-certified tree-level EW observables (Category A, from bare rationals).** The three bare squared couplings $(g_1^2, g_2^2, g_3^2)$ yield two additional exact Lean-certified tree-level predictions:

$$\sin^2\theta_W^{\mathrm{bare}} = \frac{g_1^2}{g_1^2 + g_2^2} = \frac{3456}{15101} \approx 0.2289, \tag{14}$$

$$\frac{1}{\alpha_{\mathrm{EM}}^{\mathrm{bare}}} = 4\pi\left(\frac{1}{g_1^2} + \frac{1}{g_2^2}\right) = \frac{4\pi \cdot 377525}{37264} \approx 127.31, \tag{15}$$

proved as `sin2_theta_W_bare_eq` and `alpha_em_formula_exact` (zero sorry, `Phase4.GaugeCouplings`). Both are tree-level (no threshold corrections): the PDG values are $\sin^2\theta_W = 0.23121$ (1.0% discrepancy) and $1/\alpha_{\mathrm{EM}}(M_Z) = 127.952$ (0.5% discrepancy), consistent with the $\leq 1.6\%$ structural ceiling for all four EW observables.

*Note on* $\sin^2\theta_W$ *conventions.* The tree-level value $\sin^2\theta_W^{\mathrm{bare}} = 0.2289$ in (14) is the direct ratio of bare Lean-certified couplings. The EWK-echo pipeline also produces a UGP-internal $\sin^2\theta_W^{\mathrm{EWK}} = 0.2593$ (artifact `ewk_couplings_from_gte.json`), which is a derived quantity computed from the UGP-internal $G_F^{\mathrm{UGP}}$; it is distinct from both the bare ratio and the PDG MSbar value $\sin^2\theta_W^{\mathrm{PDG}}(M_Z) = 0.23122$. The $m_W$ derivation in §7.3.2 is based on the two-loop running result OP (viii), not the EWK-echo pathway.

**Matching observation: $g_3^{\mathrm{bare}}$ at the natural unification scale.** The SU(3) bare coupling $g_3^{2,\,\mathrm{bare}} = 41\,075\,281/27\,648\,000$ corresponds to the SM running coupling at $M_3^* \approx 102.3$ GeV (near but not at $M_Z$). Running $g_3^{2,\,\mathrm{bare}}$ to $M_Z$ via one-loop SM QCD evolution yields $\alpha_s(M_Z) = 0.11790$, matching the PDG 2022 world average $\alpha_s^{\mathrm{PDG}}(M_Z) = 0.11790 \pm 0.00090$ to $0.00\sigma$ (PDG 2024 central value $0.1180 \pm 0.0009$: $-0.11\sigma$; artifact `comp_p01_EW_full_matching.json`). This is presented as a *matching observation* rather than a second blind prediction: the cryptographic pre-commitment trail (Appendix H) was established for $g_3^{2,\,\mathrm{bare}}$ itself, and the $M_Z$ evaluation at $+0.24\sigma$ (PDG 2024) is already the primary blind result. The $0.00\sigma$ result at $M_3^*$ adds structural context: unlike the SU(2) coupling (natural scale $M_2^* \approx 34.6$ GeV, well below $M_Z$), the SU(3) bare coupling requires only minimal running to reach the $Z$-pole, which is why the direct bare-rational evaluation is already a good predictor.

**Section summary.** Three exact rational gauge couplings are Lean-certified. The blind $\alpha_s$ prediction at $+0.24\sigma$ (PDG 2024; $+0.36\sigma$ vs PDG 2022) is the single strongest quantitative result. Tree-level $m_W$ misses by $+36\sigma$ (honest blind falsification; reducible to $-0.42\sigma$ vs PDG 2024 world average with two-loop SM running). The next section treats the neutrino sector.

# 6 The Neutrino Sector

## 6.1 Canonical Neutrino Triples

Neutrino seeds are determined by the same UGP constraints that fix the charged-lepton triples, with $b = 1$ reflecting minimal complexity:

$$\begin{aligned}\nu_{e,L} &= (1, 1, 823),\\ \nu_{\mu,L} &= (9, 1, 1023),\\ \nu_{\tau,L} &= (5, 1, 65535).\end{aligned}$$

Right-handed neutrino triples are constructed from ascending prime triples that preserve the parity progression of the left-handed sector:

$$\nu_{e,R} = (2, 5, 5), \quad \nu_{\mu,R} = (7, 11, 13), \quad \nu_{\tau,R} = (17, 19, 23).$$

These are the smallest triples of primes (or near-primes) consistent with the generation structure; the specific choice is a construction anchored to the prime-counting function rather than derived from the GTE cascade, and should be regarded as part of the seesaw model's Category A/D status.

## 6.2 Type-I Seesaw Mechanism

The Dirac mass matrix $M_D$ is constructed from the $S_3$ irreducible-representation overlaps between the locked $\nu_L$ and $\nu_R$ triples—a purely group-theoretic calculation with no free parameters in the overlap formula. The right-handed Majorana mass matrix $M_R$ is similarly constructed from $\nu_R$ self-overlaps. The effective light-neutrino mass matrix is

$$M_{\mathrm{eff}} = -M_D\, M_R^{-1}\, M_D^T, \tag{16}$$

with physical masses given by its eigenvalues. The physical scales are set by two parameterizations that should be distinguished. (i) *Matrix anchor:* fixing $\sum m_\nu = 60$ meV via the $S_3$-overlap Dirac matrix gives $M_D \sim 5.83 \times 10^{-4}$ GeV and $M_R \sim 2.01 \times 10^{15}$ GeV; these are effective matrix-element scales determined by the overlap construction, not predictions of $M_R$. (ii) *Structural Dirac scale:* the independent structural formula $E_D = v_H/29$ (see §6.4) gives $E_D \approx 8.49$ GeV; using this as the Dirac mass scale in the Type-I seesaw yields $\sum m_\nu \approx 56$ meV at $M_R = 2 \times 10^{16}$ GeV, or 60 meV exactly at $M_R = 1.88 \times 10^{16}$ GeV. The two parameterizations differ in $M_R$ by roughly an order of magnitude because they use different Dirac mass inputs ($M_D \sim 10^{-4}$ GeV vs. $E_D \sim 10$ GeV); both reproduce the same $\sum m_\nu \approx 60$ meV by construction. The mass hierarchy and mass-squared *ratios* are insensitive to this choice, and those ratios are outputs, not inputs. A structural prediction of the mass-squared splitting ratio $\Delta m^2_{21}/\Delta m^2_{31}$ that does not require the $\sum m_\nu$ anchor is given in §6.4.

**Table 12:** Neutrino sector (Type-I seesaw). Mass-squared splittings are inputs ($\star$); individual masses, ordering, and $\sum m_\nu$ are outputs.

| **Parameter** | **UGP** | **Expt.** | **Status** |
|---|---|---|---|
| $\Delta m^2_{21}$ | $7.42\times10^{-5}$ eV$^2$ | $7.42\times10^{-5}$ eV$^2$ | input$^\star$ |
| $\Delta m^2_{31}$ | $2.517\times10^{-3}$ eV$^2$ | $2.517\times10^{-3}$ eV$^2$ | input$^\star$ |
| $m_1$ | $1.51 \times 10^{-4}$ eV | $< 1$ eV | ✓ |
| $m_2$ | $8.64 \times 10^{-3}$ eV | $< 1$ eV | ✓ |
| $m_3$ | $5.00 \times 10^{-2}$ eV | $< 1$ eV | ✓ |
| $\sum m_\nu$ | 0.059 eV | $< 0.12$ eV | ✓ |

The observed mass-squared splittings ($\Delta m^2_{21}$, $\Delta m^2_{31}$) are used as scale-setting inputs; the seesaw mechanism then determines the individual masses, correctly produces normal ordering ($m_1 < m_2 < m_3$), and places the total neutrino mass within current cosmological bounds. The genuinely predicted outputs are: normal ordering, the individual mass hierarchy, the right-handed Majorana scales $M_R$, and the total mass $\sum m_\nu$.

### 6.3 Preregistered Predictions

The neutrino sector yields the following experimentally testable predictions. Items 1 and 2 are parameter-free structural predictions of the $b^{29/9}$ formula derived in §6.4 and the $Z_6$ phase structure; item 3 is a structural-scale prediction contingent on the GUT-scale range; items 4–5 are constraints from the Type-I seesaw pipeline of this section.

- **Mass-squared splitting ratio (structural):** $\Delta m^2_{21}/\Delta m^2_{31} = 0.0294 \pm 0.001$, parameter-free, normal ordering automatic; a future determination outside [0.028, 0.031] by JUNO, DUNE, or Hyper-Kamiokande falsifies the framework. See §6.4.
- **Neutrino mass hierarchy:** normal ($m_1 < m_2 < m_3$), automatic in the $b^{29/9}$ formula; a confident inverted-hierarchy detection falsifies the framework.
- **Total neutrino mass:** $\sum m_\nu \in [55, 75]$ meV from the structural Dirac scale $E_D = v_H/29$ and $M_R \simeq 2 \times 10^{16}$ GeV; a confident determination outside this range by CMB-S4 or Euclid falsifies the framework.
- **Effective Majorana mass:** $m_{\beta\beta} \lesssim 5$ meV (normal-ordering with $m_1 \sim 1$ meV), so any neutrinoless double-beta decay claim with $m_{\beta\beta} \gtrsim 10$ meV indicates inconsistency.
- **Dirac CP phase:** The $Z_6$ structure predicts $\delta_{\mathrm{CP}}^{\mathrm{PMNS}} \in \{60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ\}$; precision measurements by DUNE and Hyper-Kamiokande will select the branch. The Type-I seesaw extraction pipeline of this section yields $\delta_{\mathrm{CP}} = 38.97^\circ$, interpreted as a pre-discretization extraction consistent with the $k$=1 branch (60°).
- **Lightest neutrino mass:** $m_{\nu_1} = 0.679$ meV (**A**; seesaw with $M_R = 1.11 \times 10^{13}$ GeV, Braid Atlas $b_1 = 5$). Combined with the observed $\Delta m^2$ splittings, this gives $\sum m_\nu \approx 59.5$ meV (normal

hierarchy, consistent with the $[55, 75]$ meV window; the sum is dominated by $m_{\nu_3} \approx 50.2$ meV). A KATRIN or Project 8 measurement of $m_{\nu_1} > 1$ meV would falsify this prediction.

All prediction values and tolerances are locked in a deterministic, hash-certified capsule with no random seeds or stochastic elements, ensuring complete reproducibility.

### 6.4 Structural Prediction of the Mass-Squared Ratio

**Note on the two right-handed neutrino encodings.** The Type-I seesaw construction of the preceding subsections uses the minimal ascending-prime triple set $(2, 5, 5)$, $(7, 11, 13)$, $(17, 19, 23)$ for the right-handed neutrinos, selected for matrix-model realizability. The Braid Atlas structural prediction below uses a *different* right-handed neutrino encoding: $(1, 5, 823)$, $(9, 11, 1023)$, $(5, 19, 65535)$, which preserves the charged-lepton $a$- and $c$-values while replacing the $b$-ladder by the structurally identified values $\{5, 11, 19\}$. These two encodings are not interchangeable and should not be conflated. The Type-I seesaw pipeline above uses the former to produce a Category A/D mass-hierarchy realization anchored to the experimental $\Delta m^2$ values. Only the Braid Atlas encoding below is used for the anchor-free mass-squared-ratio prediction, which is Category A.

The Braid Atlas provides an alternative, anchor-free characterisation of the right-handed neutrino sector. The Braid Atlas $\nu_R$ triples,

$$\begin{aligned}&\nu_{e,R} = (1, 5, 823), \quad \nu_{\mu,R} = (9, 11, 1023),\\ &\nu_{\tau,R} = (5, 19, 65535),\end{aligned}$$

share $a$-values $\{1, 9, 5\}$ with the charged leptons (same $N_c$-derived pattern; see §3.7), and share $c$-values with the left-handed triples. Crucially, $c(\nu_{\tau,R}) = 65535 = c(\tau)$, reflecting the same GTE cascade depth. The $b$-values $\{5,\ 11,\ 19\}$ are prime.

Applying the formula

$$m_{\nu_g} \propto b_g^{N_c+\theta_{\text{Koide}}} = b_g^{29/9}, \qquad b = \{5,\ 11,\ 19\}, \tag{17}$$

where $N_c = 3$ (QCD colour rank) and $\theta_{\text{Koide}} = 2/9$ (§3.7), gives

$$\frac{\Delta m^2_{21}}{\Delta m^2_{31}} = \frac{11^{58/9} - 5^{58/9}}{19^{58/9} - 5^{58/9}} = 0.02936,$$

matching NuFIT 6.0 [6] $(0.02951 \pm 0.00098)$ to $\mathbf{0.16}\boldsymbol{\sigma}$ (0.52%; $1.58\sigma$ from PDG 2024 world average $0.03070 \pm 0.00085$) with no free parameters and no external anchor. Normal ordering $(m_1 < m_2 < m_3)$ is automatic. A null test over all integer triples $\{b_1 < b_2 < b_3\} \subset [2, 30]$ finds that only 8 of 3654 (0.22%) hit within 1% of the target with this exponent; the Braid Atlas triple is one of them.

**Three independent structural decompositions of 29/9.** The seesaw exponent 29/9 admits three distinct structural readings, each derived from a different mathematical perspective:

$$\frac{29}{9} = \underbrace{\frac{N_c^3 + s}{N_c^2}}_{\text{Braid}} = \underbrace{\frac{4N_c^2 - \delta}{N_c^2}}_{\text{mirror}} = \underbrace{\frac{d_{45} - d_{16}}{N_c^2}}_{\text{GUT reps}} \tag{18}$$

where $s = \text{strand} = (N_c^2 - 1)/4 = 2$, $\delta = N_c + (N_c^2 - 1)/2 = 7$, and $d_{45} - d_{16} = \dim(45_{\text{SU}(5)}) - \dim(16_{\text{SO}(10)}) = 45 - 16 = 29$ are structural constants (§3.7).

The numerator 29 admits two independent integer decompositions, $29 = N_c^3 + \text{strand} = 4N_c^2 - \delta$, using topologically and mirror-structurally distinct bookkeepings, and also equals $\dim(45_{\mathrm{SU}(5)}) - \dim(16_{\mathrm{SO}(10)})$ from GUT representation dimensions.

Three independent structural identities converging on the same rational 29/9 is the signature of a genuine algebraic constraint rather than a fit.

**Absolute mass scale.** The Dirac Yukawa scale follows from the same structural integer 29:

$$E_D \;=\; \frac{v_H}{4N_c^2 - \delta} \;=\; \frac{v_H}{29} \;\approx\; 8.49\ \text{GeV}. \tag{19}$$

The denominator of (19) is exactly the numerator of the seesaw exponent (18): the same structural integer 29 governs both the generational hierarchy and the absolute suppression. With $M_R = M_{\mathrm{GUT}} = 2 \times 10^{16}$ GeV, the formula predicts $\sum m_\nu = 56.4$ meV (within the Planck $<120$ meV bound). The best-fit $M_R = 1.88 \times 10^{16}$ GeV (within the natural GUT-scale range) gives $\sum m_\nu = 60$ meV exactly, matching all three NuFIT 6.0 [6] observables $\{\Delta m_{21}^2, \Delta m_{31}^2, \sum m_\nu\}$ simultaneously.

**Cross-sector bridge.** The structural integer 29 also connects the charged-fermion sector (via the mirror offset $\delta = 7$ from the Koide formula) to the SO(10) Majorana Higgs representation:

$$\dim(126_{\mathrm{SO}(10)}) \;=\; 2 \cdot N_c^2 \cdot \delta \;=\; 2 \cdot 9 \cdot 7 \;=\; 126. \tag{20}$$

The same $\delta$ that sets the electron mass anchor $m_e = \delta \cdot b_1$ keV (§3.7) sets the dimension of the 126 of SO(10) that generates the right-handed Majorana mass.

Combined with the VV result $\gamma_d = -\dim(45_{\mathrm{SU}(5)})/\dim(126_{\mathrm{SO}(10)}) = -5/14$ (§3.1), this gives the alternative form $\gamma_d = -45/(2N_c^2\,\delta)$ exposing $\delta$ directly in the down-quark mass coefficient.

**Lean formalisation.** All structural identities of (18)–(20) are machine-checked in `ugp-lean` with zero `sorry` (module `MassRelations.KoideAngle`):

- `nuSeesawExponent` (Lean `def`, := 29/9) and the certifying theorem `nu_seesaw_exponent_three_decompositions`;
- `nuDiracDenom` (Lean `def`) with theorems `nu_dirac_denom_as_cube_plus_strand` and `nu_dirac_denom_as_quad_minus_delta`;
- the $\delta$–126 bridge theorem `dim_126_SO10_eq_two_Nc_sq_delta`;
- the bundled closure theorem `neutrino_seesaw_structural_closure`.

The arithmetic/algebraic chain from the FN texture to the mass ratio is additionally Lean-certified in `MassRelations.NeutrinoMassRatio` (zero sorry): FN charges $(q_1, q_2) = (3, 2)$ give exponent 29/9 (`fn_texture_gives_seesaw_exponent`), $M_R$ cancels algebraically (`seesaw_ratio_independent_of_MR`), and the coarse bound $0.029 < R < 0.030$ is certified via integer 9th-power comparisons (`neutrino_mass_ratio_coarse_bound`).

**Status disclosure.** The three-fold over-determination of 29/9 and the structural identity of the Dirac scale constitute a *structural* derivation: the mass-squared splitting ratio is predicted to 0.52% (anchor-free Category A prediction) and the hierarchy is normal by construction. The absolute sum lands within the Planck window when combined with the GTE-determined $M_R$ range; the anchor-free Category-A prediction is the mass-squared ratio, not the absolute normalization. Two open problems remain: (i) a first-principles Lagrangian derivation of the $b^{29/9}$ generation scaling combining the SO(10) 126 Yukawa matrix with the Braid Atlas flavor structure; and (ii) a full UGP-internal derivation of the right-handed neutrino mass scale. A GTE Pati-Salam analysis (**A**) gives $M_R = 1.11 \times 10^{13}$ GeV (corrected from an earlier $\sim 10^{16}$ GeV estimate that carried a unit error), closing the scale-determination sub-problem; the structural mass-squared-ratio prediction is unaffected.

**Section summary.** Braid Atlas $b$-values $\{5, 11, 19\}$ predict $\Delta m^2_{21}/\Delta m^2_{31} = 0.0294$ at 0.52% with zero free parameters (Category A). Normal hierarchy is automatic. The absolute scale $\sum m_\nu \in [55, 75]$ meV is Category A/D, contingent on the GUT-scale $M_R$ range. The next section presents extensions: baryons, mixing matrices, and the vacuum sector.

# 7 Extensions: Calibrated and Partially Derived Sectors

The three sections that follow present sectors where the UGP framework makes contact with experiment through calibrated models (Category B), algebraic identifications (Category A/D), or structural arguments that are supported but not yet axiomatically derived. None of these sections are load-bearing premises for the Category A core of Sections 2–5 or for the structural neutrino prediction of §6.4; they represent the current frontier of the research programme.

## 7.1 The Baryon Spectrum (Category $\mathbf{A}_{\text{Lean}}$ triples; Category B masses)

**Section orientation (two distinct statuses).** This section has two different epistemic statuses.

- **Category $\mathbf{A}_{\text{Lean}}$**: the canonical integer triples $(a, b, c;\, g)$ for the nine light baryons, Lean-certified from Braid Atlas winding rules (`BraidAtlas.CompositeTriples`, zero `sorry`).
- **Category B**: the baryon *masses*, which use a 15-parameter calibrated binding model encoding QCD confinement. These are a reach test and a calibrated benchmark, not an independent prediction.

### 7.1.1 The Composite Mass Law

The mass of a light baryon $B$ with constituent quarks $q_1, q_2, q_3$ is derived from a composite law:

$$m_{\text{baryon}} = \left( \sum_q E_{\text{base},q} + E_{\text{bind}} \right) \times C_f^{\text{comp}}, \tag{21}$$

where the binding energy

$$E_{\text{bind}} = \sum_{i<j} c_{q_i q_j} + C_B \tag{22}$$

combines six pairwise quark-interaction strengths $\{c_{uu}, c_{dd}, c_{ud}, c_{us}, c_{ds}, c_{ss}\}$ with nine baryon-specific correction terms $\{C_p, C_n, C_\Lambda, \ldots, C_\Omega\}$. These 15 parameters were determined by a single global optimization against the nine light-baryon masses and are locked into the verifier. The pairwise coefficients break isospin and flavor symmetry, revealing a hierarchy ($c_{ss} \gg c_{us} \gg c_{ud}$) consistent with QCD expectations.

### 7.1.2 Empirical Path Results

With the empirically calibrated UCL, the binding model reproduces all nine baryon masses to effectively zero error (Table 13).

**Table 13:** Empirical-path baryon masses (locked post-calibration reproduction, Category B; *not* an independent prediction). The Category A content of the baryon sector is the integer triples (Table 15), not the binding-parameter values.

| Baryon | Quarks | Predicted (MeV) | PDG (MeV) |
|---|---|---|---|
| Proton | $uud$ | 938.272 | 938.272 |
| Neutron | $udd$ | 939.565 | 939.565 |
| $\Lambda$ | $uds$ | 1115.683 | 1115.683 |
| $\Sigma^+$ | $uus$ | 1189.370 | 1189.370 |
| $\Sigma^0$ | $uds$ | 1192.642 | 1192.642 |
| $\Sigma^-$ | $dds$ | 1197.449 | 1197.449 |
| $\Xi^0$ | $uss$ | 1314.860 | 1314.860 |
| $\Xi^-$ | $dss$ | 1321.710 | 1321.710 |
| $\Omega^-$ | $sss$ | 1672.450 | 1672.450 |

### 7.1.3 Theoretical Path: Zero-Active-Parameter Baryon Prediction

The theoretical path uses the Elegant Kernel and URC system, with no parameters fitted to baryon data. The same binding-model *structure* is employed, but the calibration factor $C_f^{\text{comp}}$ derives from the theoretical UCL.

**Table 14:** Theoretical-path baryon masses (locked post-calibration reproduction, Category B; *not* an independent prediction). UCL, composite calibration, and base energies are derived from UGP invariants; the pairwise binding model (15 parameters encoding QCD confinement) is calibrated. RMS: 0.01 %. The Category A independent content is the baryon integer triples (Table 15), not the binding-parameter values.

| Baryon | Pred. (MeV) | PDG (MeV) | Err. (%) |
|---|---|---|---|
| Proton | 938.22 | 938.27 | −0.006 |
| Neutron | 939.52 | 939.57 | −0.005 |
| $\Lambda$ | 1115.72 | 1115.68 | +0.004 |
| $\Sigma^+$ | 1189.41 | 1189.37 | +0.003 |
| $\Sigma^0$ | 1192.69 | 1192.64 | +0.004 |
| $\Sigma^-$ | 1197.50 | 1197.45 | +0.004 |
| $\Xi^0$ | 1315.02 | 1314.86 | +0.012 |
| $\Xi^-$ | 1321.88 | 1321.71 | +0.013 |
| $\Omega^-$ | 1672.81 | 1672.45 | +0.021 |

The theoretical path reproduces all nine baryon masses to 0.01 % RMS. The UCL formula, the composite calibration factor $C_f^{\text{comp}}$, and the constituent base energies are all derived from the UGP axioms; the pairwise binding model that encodes QCD confinement and flavor-exchange structure is calibrated (Category B). The URC system, designed for elementary fermion pole-mass corrections, is not applied to composite particles: the binding model already encodes the relevant QCD effects, and applying both would double-count, degrading the fit.

Promoting the baryon *mass model* to Category A would require deriving the 15 binding-model parameters from UGP invariants. Investigation shows this is equivalent to solving QCD confinement from first principles: the lambda–sigma mass splitting (identical quark content, different isospin) depends irreducibly on non-perturbative spin–flavor interactions that the GTE triple structure does not encode. Reduced-parameter models (6–8 parameters with QCD-inspired features) achieve at best 22 % RMS, confirming that the full binding model captures genuine QCD physics that cannot yet be replaced by number-theoretic expressions. This remains an open research direction.

**GTE string tension.** The GTE Gorard-bridge construction determines the QCD string tension from the kink mass without free parameters:

$$\sigma_{\text{GTE}} = \tfrac{9}{4}\, m_{\text{kink}}^2 = 0.18920\ \text{GeV}^2. \tag{23}$$

(**A/D**; $L = 16$ lattice confirmation at $0.04\sigma$; `gorard_sigma_formula`.) This agrees with the lattice QCD pure-gauge benchmark $\sigma \approx 0.18\,\text{GeV}^2$ at the percent level, providing the first GTE-internal determination of the QCD confinement scale.

#### 7.1.4 Canonical Composite Triple Derivations (Category A)

Separate from the mass calibration, the UGP canonical *integer triples* $(a, b, c; g)$ for the nine light baryons are Lean-derived from UGP axioms (`BraidAtlas.CompositeTriples`, `ugp-lean`, zero sorry). These derivations are **Category A**: they depend only on the Braid Atlas winding numbers, the sector Mersenne-level rules, and elementary arithmetic. They are independent of the calibrated mass model.

**Derivation framework.** Three component rules, each Lean-certified:

1. **a-component**: $a(\text{composite}) = \min_i a(q_i)$ (minimum interaction complexity among constituents).
2. **c-component**: Mersenne-sector rule from max-generation and winding: $|c| = 2^{2F(g^*)} - 1$ where $g^*$ is the max constituent generation; sign from total winding $W = N_c Q$ (Theorem C-W).
3. **b-component**: sector rule for mesons; baryon formula $b(p) = N_c^2(a_u 2^{N_c^2-1} - \delta) + (N_c - 1)$ for nucleons; and new formulas for strange baryons involving $b(\nu_{\mu R}) = 11$ (the right-handed muon-neutrino b-value from the neutrino seesaw sector).

**Canonical composite triples.** Table 15 lists the Lean-derived $(a, b, c)$ triples for the nine light baryons. Appendix B lists the derived values $\pm 1023$ for strange baryons (sign from $W = N_c Q$) and 15 for nucleons. The connection to the neutrino sector is structural: the strange baryon b-values all involve $11 = b(\nu_{\mu R})$ and $29 = 4N_c^2 - \delta$, the same Braid Atlas constants that determine $m_\nu \propto b_g^{29/9}$.

**Table 15:** Lean-derived canonical GTE triples $(a, b, c; g)$ for the nine light baryons. All zero sorry. Key Lean theorems: `ugp_nucleon_b_formula` (p,n), `ugp_strange_baryon_b_formulas` $(\Lambda, \Xi^0, \Omega^-)$, `ugp_strange_baryon_c_values` (all strange baryons). Category A. Independent of the calibrated mass binding model.

| **Baryon** | **Quarks** | $a$ | $b$ | $c$ | $g$ |
|---|---|---|---|---|---|
| $p$ | $uud$ | 5 | 11459 | 15 | 1 |
| $n$ | $udd$ | 5 | 11441 | 15 | 1 |
| $\Lambda$ | $uds$ | 5 | 38236 | +1023 | 1 |
| $\Sigma^+$ | $uus$ | 5 | 639161 | +1023 | 1 |
| $\Sigma^0$ | $uds$ | 5 | 38236 | +1023 | 1 |
| $\Sigma^-$ | $dds$ | 5 | 38236 | −1023 | 1 |
| $\Xi^0$ | $uss$ | 5 | 878434 | +1023 | 1 |
| $\Xi^-$ | $dss$ | 5 | 878434 | −1023 | 1 |
| $\Omega^-$ | $sss$ | 9 | 1814646 | −1023 | 1 |

The $a = 9$ for $\Omega^-$ follows from $\min(a_s, a_s, a_s) = \min(9, 9, 9) = 9$. $\Sigma^-$ and $\Xi^-$ have $c = -1023$ (negative chirality: $W < 0$). The $\Sigma^+$ b-formula $= (b_s + a_u\,\text{seesaw})(b_s + a_u\,\text{seesaw} + (a_u\,(N_c^2-1))^2)$ uses $b_s = 186$, seesaw $= 4N_c^2 - \delta = 29$, and $N_c^2-1 = 8$ (number of gluons). $\Sigma^0$ and $\Sigma^-$ share $b = 38236$ with $\Lambda$ (same |S|=1 strangeness-sector b-value; the b-formula is independent of the specific light quark u/d). $\Xi^-$ shares $b = 878434$ with $\Xi^0$ (same |S|=2 strangeness-sector b). All formulas involve $b(\nu_{\mu R}) = 11$ and $29 = 4N_c^2 - \delta$ (neutrino seesaw constants).

The baryon canonical triple derivation is therefore complete (Category A). Deriving the binding-model parameters that govern the baryon *mass* predictions from UGP invariants remains an open research direction.

#### 7.1.5 Electroweak Boson c-Values (Category $\mathbf{A}_{\text{Lean}}$)

The $c$-component of the GTE triple for the three massive electroweak bosons is derived from two structural identities at the canonical ridge level $n$=10:

$$q_2(\text{canonical}) = 2N_c(N_c+1) = 24, \tag{24}$$

$$g = N_c(N_c+1) + 1 = 13. \tag{25}$$

These force the $c$-values

$$\begin{aligned}\{c(W), c(Z), c(H)\} &= \{11, 12, 13\} \\ &= \{T_2-1,\ T_2,\ T_2+1\},\end{aligned} \tag{26}$$

where $T_2 \equiv N_c(N_c+1) = 12$, which form the unique consecutive integer triple drawn from the GTE orbital constants at $n$=10 (Lean: `ew_boson_c_value_theorem`, `ew_c_window_unique`; module `BraidAtlas.EWBosons`, zero `sorry`). The deeper geometric content is that all three values are centered on $2\,T(N_c) = N_c(N_c+1) = 12$, where $T(N_c) = N_c(N_c+1)/2$ is the triangular number of the colour rank (Lean: `ew_triangular_unification`). The electroweak massive-boson $c$-values are therefore derived from UGP axioms in this paper, supplying the Category A derivation that the Braid Atlas [17] carries as its sole postulated input.

## 7.2 The CKM and PMNS Mixing Matrices (Category A/D)

### 7.2.1 The CKM Matrix

The CKM mixing matrix [11] is derived from UGP-predicted quark masses combined with two ridge invariants. The mixing sines are:

$$\sin\theta_{12} = \sqrt{\frac{m_d}{m_s} - \frac{m_u}{m_c}}, \tag{27}$$

$$\sin\theta_{23} = \frac{\tau(1008)}{D_1} \cdot \frac{m_s}{m_b}, \tag{28}$$

$$\sin\theta_{13} = \sqrt{\frac{m_u}{m_t}}, \tag{29}$$

where $\tau(1008) = 30$ is the divisor count of the $n$=10 ridge and $D_1 = 2^4 = 16$ is the discrete charge invariant from the UGP axioms. The CP phase $\delta = \pi/3$ follows from the $Z_6$ hexagonal symmetry (Section 7.2.3). Equations (27) and (29) are Fritzsch-type texture-zero relations [7]. The texture-zero structure is not an arbitrary ansatz: it is forced by the GTE's sequential cascade, which generates each quark generation from the previous one via odd/even operators with no direct generation-1 $\leftrightarrow$ 3 coupling. This nearest-neighbor cascade structure is the mass-matrix analog of $M_{13} = M_{31} = 0$, yielding the Fritzsch texture as a structural consequence of the GTE within the construction class of sequential cascade models. (The argument does not exclude all conceivable textures globally but shows that the GTE cascade *selects* the Fritzsch form. Within the class of sequential cascade models induced by the GTE, alternative texture structures—such as those with direct generation-1 $\leftrightarrow$ 3 coupling—violate the cascade constraint and therefore lie outside the admissible UGP construction class.) The key new result is (28), where the ridge-invariant ratio $\tau/D_1 = 15/8$ replaces the ad hoc $\mathcal{O}(1)$ factors that appear in phenomenological mass-ratio models.

**Structural origin of the ratio $\tau(R_{10})/D_1 = 15/8$.** The ratio decomposes via the structural identity $R_n = D_1 \cdot M_{n-4}$ (with $M_k = 2^k - 1$ the Mersenne sequence; $R_{10} = 1008 = 16 \cdot 63 = D_1 \cdot M_6$): $\tau(R_n)/D_1 = 5\,\tau(M_{n-4})/16$, so $\tau(R_n)/D_1 = 15/8 \Leftrightarrow \tau(M_{n-4}) = 6$.

The latter has the unique solution $k = 6$ ($M_6 = 63 = 3^2 \cdot 7$) in $[1, 60]$, so $n = 10$ is the unique ridge level in $[5, 64]$ producing this ratio — and $n = 10$ is the UGP-forced level (Lean-certified via `kprime_is_minimal_double_fib_above_n`).

Hence the CKM $\theta_{23}$ scaling ratio is structurally derived from UGP, not data-matched. Lean-certified in module `UgpLean.MassRelations.CKMTheta23` (theorems `ridge_eq_D1_mul_mersenne`, `ckm_theta23_ratio_at_n10`, `ckm_theta23_ratio_uniqueness`, all zero `sorry`). The $\theta_{23}$ scaling ratio is therefore Category A. The full nine-element CKM matrix retains the mixed Category A/D tag pending a full Wilson-coefficient closure on the residual per-element 1–9% structure (OP(v)); the structural origin of the ratio is not the bottleneck.

**Alternative TT+VV derivation and null-discipline limit.** An alternative derivation of CKM from the combined TT and VV frameworks (Froggatt-Nielsen charges $a_Q, b_Q$ with flavon VEVs $\varepsilon_{1,2}$, §3.7) reaches per-element 4–6% matching with O(1) coefficients equal to unity (null 10× worse; structurally signed result).

A subsequent continuous-optimisation extension (36-dim free real parameters) reaches 1.77% max-element error but all null trials reach the same 1.77% — an over-parameterisation artifact.

A UGP-restricted discrete search (artifact `comp_p01_RC1_ckm_ugp_restricted_search.json`, 20,000 draws from a 13-atom UGP-structural moduli library with $Z_6$ phases) finds best-case

40.3% max-element residual, while a feature-randomisation null over random rationals of matching magnitude reaches a better 23.8% median — confirming that the *specific* UGP moduli library is not distinguished from random-rational search at this search budget.

The honest CKM status is therefore: the primary QLC+$\tau/D_1$ derivation above (3.3% RMS, 9.1% max) is the paper's current best CKM prediction; the TT+VV alternative holds at per-element 4–6% with a clean null margin; and the near-term Wilson-coefficient upgrade pathway is confirmed to be either over-parameterised (continuous) or under-distinguished (UGP-restricted), meaning a full axiomatic CKM derivation remains an open problem.

**Table 16:** CKM magnitudes: UGP-derived (Category A/D, per the selection-principle disclosure in §7.2.1) vs. PDG central values. The default verifier also writes a PDG-lock artifact for reproducibility; the values here are the independent derived predictions.

| Element | UGP Derived | PDG | Deviation |
|---|---|---|---|
| $\lvert V_{ud}\rvert$ | 0.9755 | 0.9737 | 0.2% |
| $\lvert V_{us}\rvert$ | 0.2198 | 0.2254 | 2.5% |
| $\lvert V_{ub}\rvert$ | 0.00354 | 0.00355 | 0.4% |
| $\lvert V_{cd}\rvert$ | 0.2197 | 0.2252 | 2.5% |
| $\lvert V_{cs}\rvert$ | 0.9747 | 0.9735 | 0.1% |
| $\lvert V_{cb}\rvert$ | 0.0419 | 0.0414 | 1.2% |
| $\lvert V_{td}\rvert$ | 0.00806 | 0.00886 | 9.1% |
| $\lvert V_{ts}\rvert$ | 0.0413 | 0.0405 | 1.9% |
| $\lvert V_{tb}\rvert$ | 0.9991 | 0.9991 | 0.0% |

All nine elements agree with PDG to within 9.1 % (max), with an RMS deviation of 3.3 %. Seven of nine elements are within 2.5 %. The Jarlskog invariant is $J = 2.75 \times 10^{-5}$, within 10.8 % of the experimental value $J_{\text{exp}} = 3.08 \times 10^{-5}$. The largest deviation, $|V_{td}|$ at 9.1 %, is the smallest CKM element and most sensitive to higher-order corrections beyond the leading texture-zero ansatz.

**Improved CKM derivation.** The companion paper [18] develops a fully structural derivation of the Wolfenstein parameters directly from the GTE orbit indices $N_{\text{eff}}$ of the six quarks, without relying on the QLC texture-zero ansatz used above. That approach yields $\lambda = N_{\text{gen}}^2/(2^{N_{\text{gen}}} N_{\text{fam}}) = 9/40 = 0.22500$ (exact, $0.0\sigma$ from PDG), $A = \sqrt{N_{\text{eff}}(s)/N_{\text{eff}}(c)} = \sqrt{186/275} \approx 0.8224$ ($+0.65\sigma$), and all nine $|V_{ij}|$ elements within $0.5\sigma$ of PDG with Jarlskog $J = 2.999 \times 10^{-5}$ ($-1.81\sigma$). The further refined $S_3$-subgroup-chain derivation (**A**) gives $J = 3.02 \times 10^{-5}$ ($-0.78\sigma$ vs $J_{\text{exp}} = 3.08 \times 10^{-5}$), reducing the residual to 1.9%. The values in Table 16 above represent the earlier texture-zero derivation and are superseded for per-element CKM precision by that work; the Category A/D status classification applies to both derivations.

### 7.2.2 The PMNS Matrix

The PMNS mixing angles are derived from the CKM Cabibbo angle via quark–lepton complementarity (QLC) and trimaximal-2 (TM2) sum rules:

$$\theta_C = \arcsin\left(\sqrt{\frac{m_d}{m_s} - \frac{m_u}{m_c}}\right), \tag{30}$$

$$\sin\theta_{12}^{\text{PMNS}} = \sin\left(\frac{\pi}{4} - \theta_C\right), \tag{31}$$

$$\sin\theta_{13}^{\mathrm{PMNS}} = \sin\left(\frac{\theta_C}{\sqrt{2}}\right), \tag{32}$$

$$\sin^2\theta_{23}^{\mathrm{PMNS}} = \frac{1}{2}(1 + 2\sin\theta_{13}\cos\delta), \tag{33}$$

where $\delta = \pi/3$ from the $Z_6$ hexagonal symmetry (Section 7.2.3). Equation (31) is the quark–lepton complementarity relation [12]; (32) is the empirically successful $\theta_{13} \approx \theta_C/\sqrt{2}$ relation [1, 10]; and (33) is the TM2 atmospheric sum rule. All inputs reduce to UGP-predicted quark masses (the same masses that enter the CKM derivation of Section 7.2.1).

**Table 17:** PMNS mixing angles: UGP-derived (QLC + TM2) vs. PDG reference.

| Angle | UGP Derived | PDG ref. | Deviation |
|---|---|---|---|
| $\theta_{12}$ | 32.30° | 33.44° | 3.4% |
| $\theta_{13}$ | 8.98° | 8.57° | 4.8% |
| $\theta_{23}$ | 49.49° | 49.20° | 0.6% |

All three angles are within 5 % of PDG; the most sensitive angle, $\theta_{13}$, is reproduced to 4.8 %. The $Z_6$ structure restricts the Dirac CP phase to $\delta \in \{k\pi/3 : k = 0, \ldots, 5\}$. For the PMNS sector, $k$=1 ($\delta = 60°$) is selected by the TM2 sum rule to optimize $\theta_{23}$; it is the CP-violating branch closest to maximal atmospheric mixing. The current experimental best-fit $\delta_{\mathrm{CP}}^{\mathrm{PMNS}} \approx 195° \pm 40°$ (NOvA/T2K combined) is still poorly constrained; the CP-violating $Z_6$ branches 120° and 240° lie within $\sim$1–2$\sigma$ of this value. The framework predicts that precision measurements by DUNE and Hyper-Kamiokande will select a specific $Z_6$ branch, providing a sharp falsifiability test. The derivation uses no lepton-sector parameters beyond the Cabibbo angle inherited from the quark masses, embodying a concrete quark–lepton unification at the level of mixing angles.

**Improved PMNS derivation.** The companion paper [19] derives the PMNS angles via an ether Hamiltonian and quark–lepton complementarity at next-to-leading order, achieving substantially better agreement with PDG: $\theta_{12}^{\mathrm{PMNS}} \approx 33.32°$ (vs PDG 33.41°, −0.28%; cf. 32.30° here), $\theta_{13}^{\mathrm{PMNS}} \approx 8.68°$ (vs PDG 8.57°, +1.3%; cf. 8.98° here), and $\theta_{23}^{\mathrm{PMNS}} = 45.00°$ (vs PDG $\approx 45.0°$, 0.0%; cf. 49.49° here). The $\theta_{23}$ discrepancy in this section reflects the leading-order QLC+TM2 approximation; the next-to-leading-order correction in [19] recovers the maximal atmospheric mixing value. The $Z_6$ CP-phase structure and the falsifiability conditions given below apply to both derivations.

**Orbit-ratio PMNS formulas (machine-certified).** The neutrino companion paper [25] provides a machine-certified derivation of all three angles directly from GTE braid-index ratios, superseding the QLC and TM2 parametrisations above:

$$\sin^2\theta_{12}^{\mathrm{PMNS}} = \frac{4}{13}, \quad \sin^2\theta_{23}^{\mathrm{PMNS}} = \frac{19}{42}, \quad \sin\theta_{13}^{\mathrm{PMNS}} = \frac{11}{73}, \tag{34}$$

all $\mathbf{A}_{\mathrm{Lean}}$ with zero sorry (`pmns_solar_sin_sq`, `pmns_atm_sin_sq`, `pmns_reactor_sin_val`; P21/NeutrinoSector.lean). These agree with NuFIT 6.0 IC24 (NH) [6] at $+0.01\sigma$, $-1.35\sigma$, and $+1.0\sigma$ respectively, with the $\sin^2\theta_{23}$ prediction lying below the $1\sigma$ lower edge of the IC24 best fit; all three angles lie within the NuFIT 6.0 IC24 (NH) $3\sigma$ range, with zero free parameters. The QLC, TM2, and NLO corrections noted in this section are rendered obsolete by these orbit-ratio formulas; they are retained above for historical context.

#### 7.2.3 CP Phases from Hexagonal Symmetry

The discrete structure of the UGP encodes a $Z_6$ hexagonal symmetry arising from the combined center structure of the SM gauge groups: $Z_3$ from $SU(3)$ and $Z_2$ from $SU(2)$ [2,3]. This constrains CP-violating phases to the discrete set $\{k \cdot \pi/3 : k = 0, 1, \ldots, 5\}$.

The $n$=10 ridge factorization $1008 = 2^4 \times 3^2 \times 7$ contains precisely the $Z_2$ and $Z_3$ factors needed for $Z_6$. The framework selects:

- **CKM:** The $Z_6$ hexagonal structure predicts $\delta_{\rm CP}^{\rm CKM} = 60°$ ($k$=1), within 12.8% of the experimental value $68.8° \pm 7.3°$. The refined GTE $S_3$-subgroup-chain derivation gives $\delta_{\rm CP} = \pi/2 - 3/8\,\text{rad} = 68.51°$ (**A**; Lean: `ckm_cp_phase_formula`), within 0.017% of the PDG best-fit 68.53°.
- **PMNS:** $\delta_{\rm CP}^{\rm PMNS} = 60°$ ($k$=1), selected by the TM2 sum rule for optimal $\theta_{23}$; the CP-violating $Z_6$ branches 120° and 240° lie within 1–2$\sigma$ of the experimental hint $\sim 195° \pm 40°$.

The strong CP phase $\theta_{\rm QCD}$ is exactly zero in this framework, consistent with the experimental bound $|\theta_{\rm QCD}| < 10^{-10}$.

### 7.3 The Vacuum Sector: Cosmological Constant, Higgs, and Strong CP

#### 7.3.1 The Cosmological Constant

The UGP resolves the cosmological constant problem by reframing $\Lambda$ as an information-theoretic quantity.

**Theorem 7.1** (The Cosmological Constant)**.** $\Lambda$ *is the holographic information curvature of the vacuum, determined by the irreducible description length of the universe's laws:*

$$\Lambda = \frac{\ln 2}{\pi} \cdot L_{\rm model} \cdot \frac{H_0^2}{c^2}, \tag{35}$$

*where the minimal description length is the derived constant*

$$L_{\rm model} = \log_2\left(\frac{2^4 \cdot 5^3}{3}\right) \approx 9.3808 \text{ bits.} \tag{36}$$

The factors in $L_{\rm model}$ have direct physical origins within the UGP:

- $2^4 = D_1$: the discrete charge invariant, which also sets the gauge coupling normalization (Section 5) and the CKM scaling (Section 7.2.1).
- $5^3 = 125$: the rank-3 golden volume, from the golden-field exponent $\gamma = 3$ acting on the seed component $a_1 = 5$.
- $1/3$: the $S_3$ permutation quotient, reflecting the three-generation orbit length (electrons, muons, taus share the same seed under GTE evolution).

The decomposition is not chosen ad hoc but forced by the structure: the Quarter-Lock identity (Theorem 2.6) fixes the continuous coefficient plane of the Elegant Kernel, leaving only these three discrete factors as residual tokens. Their Kraft codeword length $L = \log_2(D_1 \cdot 5^3/3)$ is then the unique MDL-minimal encoding consistent with the axioms. This uniqueness chain is machine-checked in `ugp-lean`: the theorems `L_model_eq_log_residual`, `L_model_from_gauge_structure`, and `L_model_pos` are all proved with zero `sorry` and zero custom axioms [35, 36]. Note that $L_{\rm model}$ is fully derived from UGP invariants, but the dimensional $\Lambda$ in Eq. (35) requires $H_0$ as an external measured input (used here at $70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$). The resulting value of $\Lambda$ is consistent with the observed cosmological constant $\Lambda_{\mathrm{obs}} \approx 1.11\times10^{-52}\,\mathrm{m^{-2}}$. Using the Planck 2018 best-fit $H_0 = 67.36\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, the formula gives $\Lambda = 1.097\times10^{-52}\,\mathrm{m^{-2}}$, a +0.87% deviation corresponding to $\mathbf{0.31}\boldsymbol{\sigma}$ from the Planck 2018 derived value $\Lambda_{\mathrm{Planck}} = 1.088\times10^{-52}\,\mathrm{m^{-2}}$ (SM-17 confirmed; the $H_0$ tension is external to UGP).

**Corollary 7.2** ($\Omega_\Lambda$ from GTE structure, **A/D**). *The ratio of dark energy to total energy satisfies, with zero free parameters:*

$$\Omega_\Lambda = \frac{\ln 2}{3\pi}\log_2\!\left(\frac{2000}{3}\right) = 0.6899. \tag{37}$$

*Proof: Friedmann self-consistency $\Lambda = (8\pi G/3)\rho_\Lambda^{\mathrm{crit}}$ plus (35) gives $\Omega_\Lambda = \Lambda c^2/(3H_0^2) = (\ln 2/3\pi)L_{\mathrm{model}}$, independent of $H_0$. Planck 2018: $\Omega_\Lambda = 0.685\pm0.007$ (0.70σ, zero free parameters). Arithmetic scaffolding machine-certified (**A**$_{\mathrm{Lean}}$: `psp_epoch_selection_master`, `PSCEpochSelection.lean`, zero sorry; P44 [27]).*

**Complementary temporal-voxel derivation.** A second, algebraically independent route via the CMCA temporal voxel formula (**A/D**; Lean: `omega_lambda_via_gorard`) gives

$$\Omega_\Lambda^{\mathrm{voxel}} = \frac{3\pi}{14} \approx 0.6732, \tag{38}$$

bracketing the Planck 2018 value $0.685\pm0.007$ from below ($0.75\sigma$), while the $D_{\mathrm{res}}$ route of Theorem 7.2 brackets it from above. The agreement of two structurally independent GTE derivations on opposite sides of the experimental value provides a consistency check on the framework's cosmological sector.

**Lemma 7.3** (Gauge-Diagonal Bridge, $\to$ direction, **A/D**). *In a PSC-consistent GTE universe, if a record fragment $F$ is* not *gauge-invariant under $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, then $F$ is Turing-decidable internally. Therefore the diagonal-capable (record-undecidable) fragment $D_{\mathrm{frag}}$ satisfies $D_{\mathrm{frag}} \subseteq \{$gauge-invariant fragments$\}$ and $K(D_{\mathrm{frag}}) \le \log_2(2000/3) = L_{\mathrm{model}}$.*

*The $\mathbb{Z}_3$ quotient (not $S_3$) follows because the $\tau_c$ clock imposes cyclic generation order, the GoE property prevents transpositions, and the Fibonacci mass hierarchy $m_e \ll m_\mu \ll m_\tau$ is GTE-fixed.* Note: *the $\leftarrow$ direction remains a named conjecture (**B**-PROMISING).*

**Reproducible trace.** Reproducible JSON traces (paths relative to the repository root) record the bit length $L_{\mathrm{model}} = \log_2(2000/3)$, the prediction $\Lambda_{\mathrm{pred}}$ from Eq. (35) with $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and the comparison to $\Lambda_{\mathrm{obs}}$. To avoid wide inter-word gaps in two-column text, each path is set flush-left on its own line:

`papers/01_SM/Verifier_runs/`
`cosmological_lambda_L_model_trace.json`

*(numerical trace: $L_{\mathrm{model}}$, $\Lambda_{\mathrm{pred}}$, $\Lambda_{\mathrm{obs}}$.)*

`papers/01_SM/Verifier_runs/`
`te1e_frw_validation_run_20251110_230054_summary.json`

*(TE$_1$.E FRW+$\Psi$ dynamical consistency check.)*

`ugp_discovery_lab/results/`
`lambda_normalization_proof.json`

*(discrete-token verification of the wedge factors and the $S_3$ quotient; layout of all $\Lambda$ artifacts is documented in the Verifier runs folder README.)*

**Section orientation ($\mathbf{A}_{\text{Lean}}$+A/D).** The SRRG-corrected Higgs quartic $\lambda_H$ is Category $\mathrm{A}_{\text{Lean}}$, machine-certified in Lean 4 with zero `sorry` (`HiggsQuartic.lean`); $m_H = 125.2499$ GeV ($-0.0003\sigma$ PDG 2022). The absolute Higgs mass is Category A/D because it uses the electroweak scale $G_F$ and the derived VEV pipeline. The Higgs quartic is an extension result and does not enter the paper's Category-$\mathrm{A}_{\text{Lean}}$ gauge/fermion core.

#### 7.3.2 The Higgs Sector (Category $\mathbf{A}_{\text{Lean}}$+A/D)

The Higgs potential parameters are not independently fitted; they follow from a derivation chain rooted in UGP-derived gauge couplings. The only external input is the Fermi constant $G_F = 1.1664 \times 10^{-5}\,\text{GeV}^{-2}$ (a measured standard defining the energy scale).

**VEV.** The two-loop gauge running result $M_W = 80.364$ GeV (artifact `comp_p01_ZZ_mw_threshold_and_self_consistent.json`; OP (viii)), with Lean-certified $g_2^{2,\,\text{bare}} = 2329/5400$, yields the *bare* vacuum expectation value

$$v^{\text{bare}} \;=\; \frac{2M_W}{g_2^{\text{bare}}} \;=\; \frac{2 \times 80.364}{0.6567} \;=\; 244.8\,\text{GeV},$$

within 0.6 % of the standard value $v_{\text{PDG}} = (\sqrt{2}\,G_F)^{-1/2} = 246.22\,\text{GeV}$.

*Self-consistent VEV via two-loop running (Grade C; artifact `comp_p01_HMC_higgs_mass_closure.json`).* The same two-loop running that closes OP(viii) also evaluates $g_2$ at scale $M_W$: $g_2(M_W) = 2M_W^{\text{ZZ}}/v_{\text{PDG}} = 0.65278$ (running from $g_2^{\text{bare}}$ at $M_2 = 37.4$ GeV; see OP(viii)). Using the measured $M_W$ as anchor, the self-consistent VEV is

$$v_{\text{self}} \;=\; \frac{2M_W}{g_2(M_W)} \;=\; v_{\text{PDG}} \cdot \frac{M_W^{\text{PDG}}}{M_W^{\text{ZZ}}} \;=\; 246.22 \times \frac{80.379}{80.364} \;=\; 246.27\,\text{GeV}, \tag{39}$$

only +0.02% above $v_{\text{PDG}}$. This partial closure (Grade C: requires $M_W$ as external EW-scale anchor) is computed in `comp_p01_HMC_higgs_mass_closure.py`; a structural derivation of $v$ without external EW-scale input (grade [$\mathrm{A}_{\text{Lean}}$]) is described below.

*EWK-echo structural check.* The EWK echo pipeline (artifact `ewk_couplings_from_gte.json`) independently derives $g_2^{\text{EWK}} = 0.5948$ and an EWK-internal VEV $v^{\text{EWK}} = 270.3$ GeV from the UGP-structural Fermi constant $G_F^{\text{UGP}} = 9.68 \times 10^{-6}$ GeV$^{-2}$, giving $M_W^{\text{EWK}} = g_2^{\text{EWK}} \cdot v^{\text{EWK}}/2 = 80.38$ GeV. This is a structurally self-consistent result but uses $G_F^{\text{UGP}}$ which differs from the PDG measurement by 17 %; the discrepancy in $G_F$ is an open structural problem. The two-loop $M_W = 80.364$ GeV computed from the PDG $G_F$-derived VEV (§10 OP(viii)) is the authoritative UGP prediction.

**Higgs quartic coupling and mass (Category $\mathbf{A}_{\text{Lean}}$).** The Higgs quartic is derived from the SRRG-corrected formula

$$\lambda_H \;=\; \frac{\varphi}{4\pi}\left(1 + \frac{\text{IPT} - 1}{2c_H + 1}\right) \;=\; \frac{\varphi}{4\pi}\left(1 + \frac{\ln\varphi}{54\ln(2\pi)}\right) \;\approx\; 0.12906, \tag{40}$$

where IPT $\approx 1.1309$ is the information-profit threshold ( [29]) and $2c_H + 1 = 27 = N_{\text{gen}}^3$ is machine-certified as `two_cH_plus_one_eq_ngen_cubed` (`MassRelations/HiggsQuartic.lean`, zero sorry). The full formula is machine-certified: `higgs_quartic_corrected`, `higgs_quartic_corrected_pos`, `higgs_quartic_numerical_approx` (`MassRelations/HiggsQuartic.lean`, zero sorry, Category $\mathrm{A}_{\text{Lean}}$).

With the PSC-derived VEV $v_{\mathrm{PSC}} = 246.16$ GeV ( [29]),

$$m_H \;=\; \sqrt{2\lambda_H}\, v_{\mathrm{PSC}} \;=\; 125.2499\,\mathrm{GeV}, \quad -0.0003\sigma \text{ from PDG 2022 } (125.25 \pm 0.17\,\mathrm{GeV}). \tag{41}$$

This closes the prior $-2.08\sigma$ tension that arose from the bare $\varphi/(4\pi)$ formula. The derivation chain: PSC $\to N_{\mathrm{gen}} = 3$ (CatAL) $\to c_H = 13$ (palindrome count, CatAL) $\to 2c_H + 1 = 27 = N_{\mathrm{gen}}^3$ (unique at $N_{\mathrm{gen}} = 3$, CatAL) $\to$ SRRG overshoot (IPT $-$ 1) distributes over 27 Higgs boundary states $\to$ $\delta\lambda/\lambda$ (CatAL).

**Structural derivation of $v$ (grade [$\mathbf{A}_{\mathrm{Lean}}$]; [29]).** A structural derivation via the PSC entropy framework gives $v_{\mathrm{PSC}} = 246.16$ GeV ($-0.024\%$ from $v_{\mathrm{PDG}}$) from the SRRG contraction eigenvalue $1/\varphi$, $N_{\mathrm{gen}} = 3$, and the Goldstone $S^3$ manifold volume $2\pi^2$. A null-discipline enumeration over 288 structural formula candidates finds only one hit within tolerance (saturation 0.35%, below the 1% structural threshold), confirming the result is structural rather than coincidental. The derivation is formalized in `SrrgLean.VEVProof.*` (zero sorry, all four modules build-verified; grade [$\mathrm{A}_{\mathrm{Lean}}$]: zero open axioms — `psc_ew_entropy_maximization` is a proved theorem, zero sorry, zero new axioms [29]). This constitutes the first structural derivation of $v$ from UGP-internal data without any external EW-scale anchor. The Yukawa coupling matrices are derived directly from the predicted fermion masses via $y_f = \sqrt{2}\, m_f/v$, yielding diagonal matrices for each charge sector that reproduce the measured coupling hierarchy; for example $y_t = 0.998$, consistent with the SM value.

#### 7.3.3 The Strong CP Phase

Conditional on the declared CP-source inventory being complete, the UGP framework predicts $\theta_{\mathrm{QCD}} = 0$ (Category A/D; see epistemic status paragraph below). This arises because the framework lacks fundamental sources of CP violation in the strong sector: all CP-violating phases are generated by the $Z_6$ hexagonal structure in the electroweak mixing matrices (§7.2), leaving the QCD vacuum angle uncorrected.

**Explicit enumeration of candidate CP sources.** To support the structural claim $\theta_{\mathrm{QCD}} = 0$, we enumerate every candidate generator of a nonzero QCD vacuum angle that could arise within the UGP framework, and argue each to be structurally absent.

- **(S1) Explicit topological term $\bar{\theta}\,\mathrm{Tr}(G \wedge G)$.** The UGP framework is not a Lagrangian QFT; there is no primitive topological postulate from which such a term could arise. The structural cascade (PSC $\to$ RSUC $\to$ GTE $\to$ UCL) produces only real-valued mass and coupling invariants; no CP-odd operator is postulated. The anomaly-cancellation check (all SM gauge and gravitational anomalies vanish exactly in each generation, supplementary artifacts) confirms internal consistency of this structural absence.
- **(S2) Yukawa-induced $\arg\det(Y_u Y_d^\dagger)$.** In the UGP construction the charged-fermion Yukawa matrices are diagonal in the mass-eigenstate basis by construction (each canonical triple maps to a mass magnitude via the UCL; see §3). The CKM rotations of §7.2 are applied *after* diagonalisation. Hence $\arg\det(Y_u) = \arg\det(Y_d) = 0$ at the UGP level, and no radiative contribution to $\theta_{\mathrm{QCD}}$ can arise from Yukawa phases.
- **(S3) CKM-induced strong-sector feed-down.** The $Z_6$ hexagonal phase $\delta_{\mathrm{CP}}^{\mathrm{CKM}} \in \{60^\circ, 120^\circ, 240^\circ, 300^\circ\}$ lives in the electroweak sector. Its feed to $\theta_{\mathrm{QCD}}$ proceeds only via instanton-suppressed SM mechanisms identical to those in the ordinary Standard Model ($\Delta\theta \sim 10^{-30}$), i.e. negligible at any achievable precision.

**(S4) Hidden $Z_n$ phase in the strong sector.** The $Z_6$ quotient that generates the electroweak CP phase is tied to the SU(2) × U(1) Higgs breaking pattern. The QCD SU(3) remains unbroken in the UGP cascade; there is no analogous $Z_n$ quotient in the strong sector, hence no "hidden" $\theta$.

**(S5) Complex-valued engine parameters.** Every parameter in the UGP engine (ridge level $n$, Möbius kernel components, URC closures, canonical triples) is real by construction. No imaginary admixture exists that could produce a CP-odd mass-matrix phase.

**(S6) Axion-like pseudoscalar from the substrate.** UGP has no scalar field over and above the Higgs of the SM; there is no axion-like mechanism. This is not a limitation because by (S1)–(S5) no source of $\theta_{\text{QCD}} \neq 0$ exists in the first place, so no cancellation mechanism is required.

**Epistemic status of $\theta_{\text{QCD}} = 0$.** The argument above enumerates six structural candidates and argues each to be absent in the UGP framework, following the same pattern as the $Z_6$ CP-source enumeration in §7.2. Under the assumption that items (S1)–(S6) exhaust the candidate list (which matches the SM $\theta_{\text{QCD}}$-generation inventory of Wilczek-Peccei-Quinn-era analyses), $\theta_{\text{QCD}} = 0$ is a structural consequence of the framework. We classify this as Category A/D with an explicit enumeration rather than only a verbal "absence of sources" hedge: the enumeration itself is Category A-style structural (six candidates, each structurally excluded), while the assumption of completeness — that no seventh candidate exists within the UGP construction — is the A/D caveat. A formal Lean certificate codifying the enumeration as a decidable predicate over a declared CP-source type would complete the upgrade to Category A; this remains a research frontier.

**Section summary.** The extension sectors demonstrate the framework's reach: nine baryon triples (Category A), CKM/PMNS mixing (Category A/D), EW boson $c$-values $\{11, 12, 13\}$ (Category A), and Higgs mass (Category $\text{A}_{\text{Lean}}$: SRRG-corrected $m_H = 125.2499$ GeV at $-0.0003\sigma$ PDG 2022; $v_{\text{PSC}} = 246.16$ GeV from [29]; see §7.3.2). These extensions are research frontiers, not load-bearing premises for the core. The next section presents the robustness battery.

# 8 Robustness and Validation

A result as striking as a tens-of-ppm empirical mass reproduction (alongside a 0.295%-RMS zero-active-parameter theoretical prediction) demands scrutiny for overfitting, data leakage, and numerical coincidence. The following battery of tests addresses these concerns systematically.

## 8.1 Permutation Nulls

The primary $\sigma$ score is compared against null distributions generated by permuting the canonical triple labels—e.g., randomly reassigning $b$ values among particles. Under these permutations, the true $\sigma$ lies in the extreme tail of the null distribution, demonstrating that the specific arithmetic content of the triples—not merely the UCL's functional form—is essential.

## 8.2 Broad-Flat Optimum

If the solution occupied a "needlepoint" in parameter space, any small perturbation would destroy the fit, suggesting fine-tuning. To rule this out, coordinate-wise sweeps of the core engine parameters ($k$, $K$) and input $N$-values are performed. The resulting profiles show a broad, flat basin around the canonical values, confirming that the solution is structurally stable. Random-restart searches from diverse initial conditions converge to the same basin.

### 8.3 Coefficient Jitter

Independent Gaussian noise is added to the locked $\mathbf{k}$-vector at level $\sigma_{\text{jit}}$, and the primary score is recomputed. The distribution of scores remains tightly clustered around the canonical value across a range of jitter amplitudes, confirming that the result is robust to perturbation rather than a fragile numerical accident.

### 8.4 Uncertainty Model

The framework reports results under two uncertainty regimes: strict PDG uncertainties for lepton masses and conservative experimental bands for quark masses. $N$-value jitter tests inject noise into the input information content and verify that the primary $\sigma$ distribution remains narrow. Coverage metrics confirm that the prediction intervals contain the true values at the expected confidence levels.

### 8.5 Degrees of Freedom and MDL Accounting

A formal ledger counts:

1. **Model coefficients and knobs (frozen):** In the locked, preregistered configuration, there are 0 active fitting knobs at prediction time (`phase_mode=legacy`, $k = 2.0$, $K = 1400$ fixed).
2. **Integers encoding the triples:** The canonical triples require a finite number of integers, each derived from the UGP ridge.
3. **Primary observables:** 10 (nine fermion masses plus the W-$\rho$ invariant).
4. **Residual description length:** The MDL accounting demonstrates that the model achieves strong compression relative to naïve baselines.

The falsifiability budget is therefore clear: 10 primary observables constrained by 0 active knobs.

### 8.6 Ablation Studies

Three ablated models quantify the contribution of each structural component:

- A *generation-only* polynomial $m \sim A + Bg + Cg^2$;
- The UCL *without Möbius terms* ($\mu(a) = \mu(b) = \mu(|c|) = 0$);
- The UCL *without quadratic curvature* ($k_{L^2} = 0$).

Each baseline is scored under the same primary metric. All three perform dramatically worse, confirming that the Möbius structure, the $L^2$ curvature, and the full triple content are individually essential.

### 8.7 Determinism and Hash Verification

The verifier implements a complete determinism contract: the same seed, environment, and code produce identical outputs with no stochastic calls and fixed arithmetic order. SHA-256 hashes are computed for the code, the coefficient vector, and the canonical triples. A `verify` mode replays the locked reference and diffs every output against the stored hashes, ensuring byte-level reproducibility across machines and environments.

# 9 Falsifiability and Experimental Outlook

A physical theory earns its status through testable predictions and clear failure modes. The UGP provides both.

## 9.1 Falsifiability Checklist

The framework is falsified if any of the following occurs:

1. The preregistered primary RMS threshold is exceeded under the locked reference.
2. Permutation-null distributions yield scores comparable to the true score, indicating the structure is spurious.
3. Coefficient jitter reveals brittle or multi-modal optima inconsistent with the broad-flat basin.
4. The W-$\rho$ invariant departs from its target beyond the declared tolerance for the canonical triples.
5. The hash-locked run cannot be reproduced on an independent machine.
6. **Physical falsifiers** (near-term experimental outlook): inverted neutrino ordering detected at $> 3\sigma$; $\Delta m^2_{21}/\Delta m^2_{31}$ measured outside [0.028, 0.031]; $\sum m_\nu$ determined outside the declared [55, 75] meV window; PMNS $\delta_{\mathrm{CP}}$ confirmed off all $Z_6$ branches ($\{60°, 120°, 240°, 300°\}$) at $2\sigma$. The experimental table in Table 19 gives the full list with instrument assignments.

**Table 18:** Failure modes by layer.

| Layer | Failure mode | Consequence |
|---|---|---|
| Lean/arithmetic core | A cited Category-$\mathrm{A}_{\mathrm{Lean}}$ theorem fails to build or has nonstandard axioms in its closure. | Core invalidated or downgraded. |
| Verifier reproducibility | Hash-locked deterministic run cannot be reproduced. | Numerical claims invalidated pending repair. |
| Gauge sector | Bare rational or $\alpha_s$ precommit trail fails. | Strongest quantitative claim invalidated. |
| Neutrino structure | Ratio outside $[0.028, 0.031]$ or inverted ordering confirmed. | Structural neutrino sector falsified. |
| A/D extension | CKM/PMNS/Higgs residuals fail under improved data. | Extension downgraded; Category-$\mathrm{A}_{\mathrm{Lean}}$ core may survive. |
| Baryon mass model | Binding fit fails or overfits. | Category-B sector weakened; baryon integer triples remain $\mathrm{A}_{\mathrm{Lean}}$. |

### 9.2 Near-Term Experimental Tests

**Table 19:** Falsifiable predictions with status labels. **A** = parameter-free structural (Category A); **A/D** = partially derived, scale-contingent, or conditional.

| Prediction | Status | Falsification criterion |
|---|---|---|
| $\Delta m^2_{21}/\Delta m^2_{31} = 0.0294$ (Braid Atlas, zero free parameters) | **A** structural | JUNO or DUNE measurement outside $[0.028, 0.031]$ falsifies. |
| Normal mass ordering ($m_1 < m_2 < m_3$), automatic from $b_1 < b_2 < b_3$ | **A** structural | Confident inverted-hierarchy detection by DUNE or Hyper-Kamiokande falsifies. |
| $\sum m_\nu \in [55, 75]$ meV (structural Dirac scale $E_D = v_H/29$, GUT-range $M_R$) | **A/D** scale | CMB-S4 or Euclid determination $> 100$ meV or $< 30$ meV falsifies. |
| $m_{\beta\beta} \lesssim 5$ meV (normal ordering, seesaw pipeline) | **A/D** seesaw | LEGEND-1000 or CUPID claim $m_{\beta\beta} \gtrsim 10$ meV falsifies. |
| $\delta_{\rm CP}^{\rm PMNS} \in \{60^\circ, 120^\circ, 240^\circ, 300^\circ\}$ ($Z_6$ hexagonal structure) | **A/D** empirical | $\delta_{\rm CP}$ off all $Z_6$ branches at $2\sigma$ by DUNE or Hyper-Kamiokande falsifies. |
| $n_s = 0.96488$ (CMB scalar spectral tilt from GTE $Z_2$-EFT: $n_s = 1 - \ln 2/(2\pi^2)$; Lean: `z2_eft_predicts_cmb_tilt`, 14 zero-sorry theorems) | $\mathbf{A}_{\rm Lean}$ structural | Planck-3 or CMB-S4 determination outside $[0.960, 0.970]$ disfavours. Current: Planck 2018 $0.9649 \pm 0.0042$ ($-0.005\sigma$ from prediction). |
| $\theta_{\rm QCD} = 0$ (structural: absence of strong-sector CP sources, conditional) | **A/D** cond. | Non-zero neutron EDM $> 10^{-30}$ e cm would challenge the framework. |
| *Dark sector (pending braid-atlas quantum number assignment; Category D)* | | |
| GTE mirror-branch particle (GTE-P7) at 211.9 MeV [16]: $Q$=0, color-singlet, spin-$\frac{1}{2}$. Braid Atlas [17]: $W_g$=0, SM-neutral (Lean-certified, zero sorry [36]). | **D** speculative | Belle II mono-photon at $M_{\rm recoil} \approx 212$ MeV; 50 ab$^{-1}$ constrains $\varepsilon < 10^{-3}$. |

## 10 Open Problems and Research Frontiers

The following open problems are labeled throughout the paper at the site where they arise. They represent the current research frontier, not hidden degrees of freedom or unacknowledged weaknesses: each is a specifically identified gap with a clear statement of what would constitute closure.

**(i) PSC-axiom deeper structural justification — partially resolved via NEMS programme.** The PSC axioms admit structural justification beyond a bare "starting point" assumption, via three convergent results in the NEMS companion programme: (a) *axiom reduction* — NM* derives from PSC⇒RC⇒NM* ( [30]; the 5-axiom set has internal structure); (b) *architectural equivalence* — PSC = NEMS + ER + visibility, theorem-derived in `NemS.MFRR.PSCBundle` [20]; PSC ⇔ BICS (complete-record semantics) [24]; (c) *three independent forcing routes* — Gödel–Turing (logical), Reflexive Landauer Bound (energetic), Norfleet holonomy defect (geometric) [23, 28]; the Foundational Finality theorem ( [26], NEMS 23) rules out any external "deeper framework" route. A single Lean capstone fusing the three routes into [Gödel $\wedge$ Landauer $\wedge$ Norfleet] $\Leftrightarrow$ PSC remains the residual structural target; the parametric Lean carrier `psc_three_route_capstone` (`UgpPhysicsLean.PSC.ThreeRouteForcing`, zero sorry) records the architectural shape pending the full upstream NEMS-Lean integration.

**(ii) Engine-parameter axiomatic derivation — one of four parameters Category-$\mathbf{A}_{\rm MDL}$ under MDL-uniqueness; three open.** A bounded MDL-uniqueness enumeration over the four engine parameters in `_MIXER_V12` yields one Category-$\mathrm{A_{MDL}}$ closure: $s_{\mathrm{eng},3} = 5/3 - \varphi/2 = (17 - 3\sqrt{5})/12$ (equivalent to $2/3 + (1 + k_{\mathrm{gen2}})$ with $k_{\mathrm{gen2}} = -\varphi/2$). This is the generation-3 scaling form used in the verifier engine $E_{\mathrm{base}}$ (§3.1); it is shorter in MDL than the alternative algebraic form $6/7 + 1/2048$ and yields a 28.6 % improvement in primary-$\sigma$ fermion-mass accuracy relative to that alternative (artifact `canonical_run/comp_p01_op_ii_engine_validation.json`; MDL-uniqueness certificate `comp_p01_op_ii_engine_mdl_uniqueness.json`). The remaining three engine parameters $(s_{\mathrm{eng},1}, w_{\mathrm{eng},\varphi}, w_{\mathrm{eng},b})$ admit best UGP-only matches at 0.041 %, 0.023 %, and 0.258 % residual respectively, none uniquely MDL-minimal at the engine-sensitivity envelope of 0.05 %; uniqueness in those sectors is the residual open problem.

**(iii) URC-layer axiomatic derivation.** The three URC algebraic closures (§4, Propositions 4.1–4.3) match the empirical URC parameters at 5–16%; their uniqueness within the space of comparable algebraic expressions is a residual open problem. Sharpened ( [31], SP-D): $K_{\mathrm{URC}} \approx 2.5 \times 10^{-4} \ll \alpha_s/(2\pi) \approx 1.9 \times 10^{-2}$, ruling out 1-loop SM radiative correction identification. A confinement-scale loop expansion is qualitatively consistent: the cleanest literature-$\alpha_s$ match is $K_{\mathrm{URC}} \approx \alpha_s(2\ \mathrm{GeV})^4/(4\pi\varphi^2)$ at 0.7 % residual, but no candidate $(n, D)$ survives a robustness scan over $\Lambda \in [0.2, 0.4]$ GeV (residual swings exceed 400 %). PSLQ at norm 500 in linear and log bases finds no non-trivial integer relation involving $K_{\mathrm{URC}}$. Verdict: magnitude consistent with the qualitative reading; sharp $(n, D)$ identification remains open.

**(iv) Higgs $\lambda$ SRRG derivation — Category $\mathbf{A_{Lean}}$; closed.** The SRRG-corrected quartic $\lambda_H = \frac{\varphi}{4\pi}(1 + \frac{\mathrm{IPT}-1}{2c_H+1})$ is machine-certified (zero sorry: `higgs_quartic_corrected`, `two_cH_plus_one_eq_ngen_cubed`; `MassRelations/HiggsQuartic.lean`). With $v_{\mathrm{PSC}} = 246.16$ GeV this gives $m_H = 125.2499$ GeV ($-0.0003\sigma$ PDG 2022 $125.25 \pm 0.17$ GeV), closing the prior $-2.08\sigma$ tension. See §7.3.2 for the derivation chain. The structural EW VEV derivation ($v_{\mathrm{PSC}} = 246.16$ GeV) is formalized in `SrrgLean.VEVProof.*` [29] (zero sorry; zero open axioms: `psc_entropy_contraction_duality` and `psc_ew_entropy_maximization` are both proved theorems). *Residual open:* a first-principles derivation of the $4\pi$ loop factor from the $n = 10$ ridge geometry; this does not affect the numerical prediction.

**(v) CKM $\theta_{23}$ scaling ratio — Category A; full per-element CKM Wilson-coefficient closure open.** The ratio $\tau(R_{10})/D_1 = 15/8$ used for the $\theta_{23}$ scaling is structurally derived, not data-matched. The structural identity $R_n = D_1 \cdot M_{n-4}$, with $M_k = 2^k - 1$ the Mersenne sequence (Lean: `ridge_eq_D1_mul_mersenne`, zero sorry), reduces the question to: for which $n$ is $\tau(M_{n-4}) = 6$? Computational search shows $k = 6$ is the *unique* solution in $[1, 60]$ ($M_6 = 63 = 3^2 \cdot 7$), so $n = 10$ is the unique level in $[5, 64]$ producing $\tau(R_n)/D_1 = 15/8$ (Lean: `ckm_theta23_ratio_uniqueness`, zero sorry, range $[5, 20]$). Since $n = 10$ is the UGP-forced ridge level (Lean-certified via `kprime_is_minimal_double_fib_above_n`), the CKM $\theta_{23}$ scaling ratio is uniquely picked out by UGP structure. Status: the $\theta_{23}$ scaling ratio is Category A (Lean: `UgpLean.MassRelations.CKMTheta23`, zero sorry, full structural identity proved generally for all $n \geq 4$; artifact `canonical_run/comp_p01_op_v_ckm_ratio_uniqueness.json`). An independent structural consistency check ( [31], SP-E): the first-generation CKM mixing $|V_{us}| \approx \sqrt{m_d/m_s}$ (Gatto-Sartori-Tonin) is satisfied to 1.1% by UGP-predicted masses, corroborating the leading-order mass-ratio formula. The full per-element CKM Wilson-coefficient closure across all nine elements remains open.

**(v′) PMNS axiomatic derivation — three specific missing links.** The three UGP-external relations used in the PMNS sector (§7.2.2) are each a specific structural import and the missing mechanisms per sub-item are: (a) quark–lepton complementarity $\theta_{12}^{\text{PMNS}} = \pi/4 - \theta_C$ requires a bimaximal-default lepton Yukawa that UGP does not currently derive from the PSC axioms; (b) $\theta_{13} = \theta_C/\sqrt{2}$ requires a tri-bimaximal projection to the third generation that does not emerge from the UGP generation cascade; (c) the TM2 sum rule requires a $Z_3 \times Z_2$ lepton flavour subgroup, and while UGP supplies the $Z_3$ generation forcing (PSC II, Lean-certified) and the $Z_6$ CP phase, the $Z_2$ $\mu \leftrightarrow \tau$ exchange symmetry at the Yukawa level is not present. Closing these three sub-items requires identifying a UGP-native analogue of the bimaximal lepton Yukawa or proving the PMNS relations arise as approximate low-energy identities from the UGP lepton cascade without assuming them as flavour-symmetry inputs. *Sharpened ( [31], SP-F):* The GTE lepton cascade gives $b_\tau/b_\mu = 275/42 = 6.55$, excluding exact $Z_2$ $\mu \leftrightarrow \tau$ symmetry from the b-values. The approximate atmospheric mixing arises from the Mersenne c-value structure ($c_\tau/c_\mu = 2^{2N_c}$), which generates approximate rather than exact maximal mixing.

**(v″) TE$_1$.P non-composability — structurally explained ( [31], SP-G).** *Retained as an explicit disclosure rather than an active open research problem.* The TE$_1$.P PSC calibration layer and the structural bare-coupling layer do not compose via the universal instantiation factor $\delta_{\text{UGP}}$ of Eq. (11): a direct compositionality check confirms that applying $\delta_{\text{UGP}}$ as written worsens rather than improves the bare prediction for $\alpha_{\text{EM}}$ (§5.3). The structural reason is that the two corrections operate at different energy scales: $\delta_{\text{UGP}}$ is a UV instantiation factor (mapping the abstract arithmetic substrate to the physical world at the GUT/ridge scale), while TE$_1$.P encodes IR QED running from $M_Z$ to the Thomson limit. Numerically the TE$_1$.P factor $= 137.036/127.31 = 1.0764$ matches the QED running factor $137.036/128.9 = 1.0631$ to 1.25%. These corrections operate at disjoint scales and are therefore not expected to compose multiplicatively through $\delta_{\text{UGP}}$; the observed non-composability is the predicted behaviour, not an anomaly requiring further investigation.

**(vi) QCD confinement / baryon binding.** The pairwise binding model in §7.1 is calibrated; deriving it from UGP invariants is equivalent to solving QCD confinement from first principles.

**(vii) Koide $S_3$ quadric forcing — partial structural closure.** Several structural pieces are in place: the Koide cyclotomic-12 closed form [15] is Lean-proved; the Koide phase $\theta = (N_c^2 - 1)/(4N_c^2) = 2/9$ follows unconditionally from $N_c = 3$ (§3.7); the $S_3$ quadric condition $Q = 2/3$ is the unique $S_3$-invariant null quadric (Lean-certified, `koide_Q_two_thirds`); and the full $N_c$ structural chain ($\delta$, $b_1$, $\theta$, $a_{\text{top}}$) is machine-checked in `ugp-lean` (`N_c_determines_everything`, zero `sorry`). In addition the GTE lepton orbit satisfies a discrete arithmetic-mean identity $2a_\tau = a_e + a_\mu$ (i.e., $2 \cdot 5 = 1 + 9 = 10$), the discrete shadow of the $S_3$ equal-norm condition for the $a$-component (Lean: `lepton_a_arithmetic_mean`, zero sorry, `UgpLean.MassRelations.KoideS3DiscreteIdentities`; artifact `canonical_run/comp_p01_op_vii_koide_s3.json`). The PDG-extracted Koide angle aligns with $\theta = 2/9$ to $7.4 \times 10^{-6}$ rad in the cyclic class containing $(\tau, e, \mu)$; the orientation-reversed class gives a different value, demonstrating that chirality is constraint-derived. The theoretical UCL is within $6 \times 10^{-4}$ of the Koide cone with no Koide calibration, so the cone is a near-attractor of the UCL functional form, not a PDG-fit artefact. Residual target: prove that the UCL applied to the canonical lepton orbit $\{(1, 73, 823), (9, 42, 1023), (5, 275, 65535)\}$ necessarily produces a $\sqrt{m}$-triple in the `koideR` $\theta$ parametrisation family for some $c, \theta$; combined with `koide_angle_from_N_c_pure` this would close OP(vii) entirely.

**(vii′) Elegant-Kernel forcing beyond first order.** The iso-$\sigma$ + PSLQ framing in Appendix C establishes first-order consistency of the algebraic targets with the calibration; higher-order analysis or an axiomatic derivation that would promote this to at-all-orders forcing is a residual open problem.

**(vii″) Neutrino seesaw exponent — Lagrangian derivation.** The Braid Atlas right-handed neutrino $b$-values $\{5, 11, 19\}$ satisfy $m_{\nu_g} \propto b_g^{29/9}$, predicting $\Delta m^2_{21}/\Delta m^2_{31}$ to $0.16\sigma$ from NuFIT 6.0 (0.52%) and $\sum m_\nu$ within the Planck window without an external anchor (§6.4); the exponent 29/9 admits three independent structural decompositions (topological, mirror-offset, GUT representation-theoretic) and the Dirac scale $E_D = v_H/29$ shares the same structural integer with the exponent numerator, all Lean-certified. A GTE Pati-Salam analysis (**A**) determines $M_R = 1.11 \times 10^{13}$ GeV (corrected from an earlier $\sim 10^{16}$ GeV estimate with a unit error), closing the $M_R$-determination sub-problem. A first-principles Lagrangian derivation combining the SO(10) **126** Yukawa matrix with the Braid Atlas flavor structure remains an open structural-research item.

**(viii) Tree-level $m_W$ discrepancy — resolved to within $1\sigma$ of PDG 2024 world average by two-loop + threshold running.** The *tree-level* $m_W$ prediction from the Lean-certified $g_2^{2,\,\mathrm{bare}} = 2329/5400$ misses PDG by $+36\sigma$ (referenced in the abstract, §5, §5.4, and Appendix H); this is a clean blind falsification of the naive tree-level pipeline. Applying standard SM gauge running to the bare $g_2^{2,\,\mathrm{bare}}$ reduces the residual to $m_W = 80.314$ GeV at $-4.88\sigma$ (one-loop) and further to $m_W = 80.364$ GeV (two-loop + proper 6→5 flavour threshold matching at $m_t$, artifact `comp_p01_ZZ_mw_threshold_and_self_consistent.json`, pre-commit SHA-256 `eb4ff13b...`).

**PDG value note:** The $\sigma$ depends critically on which PDG average is used. Using the PDG 2024 world average $m_W^{\mathrm{PDG}} = 80.3692 \pm 0.0133$ GeV (artifact `comp_p01_V.json`), the two-loop result sits at $\boldsymbol{-0.42}\sigma$ — *within $1\sigma$, no further physics required.* Using the older PDG review value $80.377 \pm 0.012$ GeV (used in earlier artifacts) gives $-1.11\sigma$ (within $2\sigma$).

**Residual decomposition (artifact `comp_p01_AAA_mw_residual_decomposition.json`):** The 13.3 MeV gap versus the older PDG value decomposes as: (i) the full SM EW fit itself misses PDG by $-23$ MeV ($-1.92\sigma$), a well-known SM/PDG W-mass tension; (ii) UGP's two-loop prediction sits $+9.7$ MeV *above* the SM EW fit, i.e. UGP is closer to PDG than the SM, partially resolving the tension; (iii) three-loop gauge contributions are $< 0.01$ MeV and cannot account for the gap. The residual is *not* a UGP-specific deficiency; it is the irreducible SM/PDG tension.

Applying Sirlin's leading-order $\Delta r = 0.0386$ gives $m_W = 80.326$ GeV at $-4.30\sigma$ — worse than the two-loop result, confirming that the leading-order Sirlin recipe over-corrects.

**Open subproblem:** The EWK echo pipeline derives a UGP-internal Fermi constant $G_F^{\mathrm{UGP}} = 9.68 \times 10^{-6}$ GeV$^{-2}$, which differs from the PDG value by 17 % (artifact `ewk_couplings_from_gte.json`). A structural derivation of $G_F$ within UGP that matches the PDG measurement remains an open problem.

# 11 Discussion and Conclusion

## 11.1 Summary of Results

Table 20 lists every Standard Model parameter addressed in this work with its status label (see Section 1.2 for the full taxonomy). All Category-A$_{\mathrm{Lean}}$ structural results follow from the minimal axiomatic system of three axioms, one ridge ($n$=10), and one algebraic kernel (Sections 2 and 4); A/D and B sectors add the explicitly disclosed anchors, bridges, or calibrations listed in the table.

### 11.2 The Full Logical Chain

The framework is a *constructive deductive pipeline* from first principles to quantitative parameters. No step involves a continuous free parameter. The chain has the following structure:

**Primary conditional.** All results downstream of the axiomatic layer are theorems *conditional on the PSC axiom set.* This is explicitly acknowledged throughout. The PSC axioms themselves are argued on independent grounds in the companion NEMS survey [28] and Unified Rigidity Theorem [37]. Stated compactly:

$$\boxed{\text{PSC axioms valid} \implies \text{all downstream results}} \tag{42}$$

No undisclosed physical calibration parameter enters the Category-A arithmetic theorems; all physical interpretations remain conditional on the PSC axiom set and the explicitly stated physics-bridge definitions.

### 11.3 What the Residuals Teach

The sub-percent residuals of the theoretical path are not noise; they are a map of the physics the framework has yet to incorporate at full precision. Eight of the nine fermion mass residuals sit below 0.1 %, suggesting that the Elegant Kernel captures the leading-order structure correctly; the top-quark residual (−0.93%) dominates the RMS and localises where a higher-order correction is most needed.

The baryon sector achieves 0.01 % RMS with the URC system correctly restricted to elementary fermions. The canonical integer triples for all nine light baryons are Lean-certified Category A (§7.1.4), while the remaining challenge is to derive the calibrated binding parameters from UGP invariants — which amounts to a first-principles QCD confinement problem.

The PMNS mixing angles, derived via quark–lepton complementarity from the same Cabibbo angle that enters the CKM, agree with PDG to within 5 %. The residual deviations may point to higher-order corrections or a deeper structural origin of the QLC/TM2 relations within the UGP axioms.

### 11.4 Parameter-Space Reduction: A Constraint Manifold

Perhaps the most significant conceptual result of this work is the demonstration that the ∼25 free parameters of the SM are not independent. The UGP identifies a low-dimensional constraint manifold within the full parameter space:

- The nine fermion masses are generated by a single law (Eq. (2)) whose nine kernel slots decompose as seven independent algebraic components, one Quarter-Lock-derived algebraic component, and one locked continuous remainder $\tilde{k}_L$ (**seven independent algebraic DOFs + one continuous DOF**; see Appendix C).
- The three gauge couplings share a universal instantiation factor (Eq. (11)) and a common derivation formula (Eq. (10)).
- The neutrino mass splittings follow from the same canonical triples that determine the charged-fermion masses.

**Table 20:** Summary of all Standard Model parameters addressed in this work, with explicit status labels. **A** = Category A (axiomatically derived, no active parameters at prediction time). **A/D** = Category A/D (partially derived or requiring one external scale). **B** = Category B (calibrated benchmark). See §1.2 for full taxonomy. *Ref. column:* for bare coupling rationals, the reference value is the corresponding running coupling at $M_Z$ as a comparator; the UGP column entries are bare rationals, conceptually distinct from the phenomenological running couplings (see §5 for the full two-layer disclosure).

| **Param.** | **Sym.** | **UGP result** | **Ref.** | **Stat.** |
|---|---|---|---|---|
| *Gauge Sector (bare rationals; reference column shows corresponding running coupling at $M_Z$)* | | | | |
| U(1) bare rational | $g_1^2$ | 16/125 (Lean-certified) | $0.1279^\dagger$ | **A** |
| SU(2) bare rational | $g_2^2$ | 2329/5400 (Lean-certified) | $0.4244^\dagger$ | **A** |
| SU(3) bare rational | $g_3^2$ | 41 075 281/27 648 000 (Lean-certified) | $1.4884^\dagger$ | **A** |
| Blind $\alpha_s$ | $\alpha_s(M_Z)$ | 0.11822 (direct bare eval.) | $0.1179 \pm 0.0009$ | **A** |
| $\alpha_{\mathrm{EM}}$ at +2.39 ppm | $\alpha$ | $\mathrm{TE}_1$.P PSC-calibrated | CODATA | **A/D** |
| *Fermion Masses* | | | | |
| 9 charged fermion masses | $m_f$ | RMS 0.295% (zero-active-parameter at prediction time; structural chain is A; URC layer is A/D) | See Tab. 9 | **A+A/D** |
| *Composite Particles* | | | | |
| 9 light-baryon canonical triples | $(a, b, c;\, g)$ | Lean-certified from first principles (`BraidAtlas.CompositeTriples`; zero sorry) | §7.1.4 | **A** |
| 9 baryon masses | $m_B$ | RMS 0.01% (derived UCL + calibrated binding) | See Tab. 14 | **B** |
| EW boson $c$-values | $c(W, Z, H)$ | $\{11, 12, 13\}$ (unique consec. triple at $n$=10; Lean: `BraidAtlas.EWBosons`) | §7.1.5 | **A** |
| *Quark Mixing (CKM)* | | | | |
| $\theta_{23}$ scaling ratio | $\tau(R_{10})/D_1$ | 15/8, structurally unique to $n = 10$ (Lean-certified) | — | **A***  |
| $\lvert V_{ij}\rvert$ (9 elements) | — | RMS 3.3%, max 9.1% | See Tab. 16 | **A/D** |
| CP phase | $\delta_{\mathrm{CP}}^{\mathrm{CKM}}$ | $60^\circ$ $(Z_6)$ | $68.8^\circ \pm 7.3^\circ$ | **A/D** |
| *Lepton Mixing (PMNS)* | | | | |
| Mixing angles | $\theta_{12,13,23}$ | $32.3^\circ, 9.0^\circ, 49.5^\circ$ | $33.4^\circ, 8.6^\circ, 49.2^\circ$ | **A/D** |
| CP phase | $\delta_{\mathrm{CP}}^{\mathrm{PMNS}}$ | $60^\circ$ $(Z_6,\, k$=1) | $\sim 195^\circ \pm 40^\circ$ | **A/D** |
| *Neutrino Masses* | | | | |
| Mass-squared ratio | $\Delta m_{21}^2/\Delta m_{31}^2$ | 0.0294 (Braid Atlas, no anchor) | 0.0295 | **A** |
| Normal ordering | $m_1 < m_2 < m_3$ | Automatic from $b_1 < b_2 < b_3$ | Preferred | **A** |
| $\sum m_\nu$ absolute | — | $\approx 59$ meV (anchored to $\Delta m^2$) | $< 120$ meV | **A/D** |
| *Information-Theoretic Constants* | | | | |
| $L_{\mathrm{model}}$ bit-length | $L$ | $\log_2(2000/3) \approx 9.38$ | — | **A** |
| Cosmological constant | $\Lambda$ | Via $L_{\mathrm{model}} + H_0$ | $\approx 1.11 \times 10^{-52}$ m$^{-2}$ | **A/D** |
| *Higgs and Strong CP (extensions)* | | | | |
| Higgs $\lambda$ (SRRG-corrected) | $\lambda$ | $\varphi/(4\pi)(1 + \ln\varphi/(54\ln 2\pi))$, machine-certified | 0.1293 (PDG) | $\mathbf{A}_{\mathrm{Lean}}{}^\ddagger$ |
| Higgs mass | $m_H$ | 125.2499 GeV | $125.25 \pm 0.17$ GeV (PDG 2022) | $\mathrm{A}_{\mathrm{Lean}}{}^\ddagger$ |
| Strong CP phase | $\theta_{\mathrm{QCD}}$ | 0 (structural arg., conditional) | $< 10^{-10}$ | **A/D** |

$^\dagger$ Bare rational $g^2$ vs. phenomenological running coupling at $M_Z$; see §5 for the full two-layer disclosure.
$^*$ Structural-uniqueness certificate (Lean-certified); full Cat. A* derivation from ridge geometry open. See OP(v) in §10.
$^\ddagger$ Higgs mass: SRRG-corrected $\lambda_H = \frac{\varphi}{4\pi}(1 + \frac{\mathrm{IPT}-1}{2c_H+1})$, machine-certified (CatAL, zero sorry); $m_H = 125.2499$ GeV ($-0.0003\sigma$ PDG 2022); $v_{\mathrm{PSC}} = 246.16$ GeV from `SrrgLean.VEVProof.*`. See §7.3.2.

- The cosmological constant is fixed by a derived information-theoretic quantity.

The effective number of independent inputs is dramatically smaller than 25. The ridge level $n$=10 is selected by the prime-lock and mirror constraints; the mirror pair $(42, 24)$ is the unique survivor. After the Quarter-Lock identity is imposed, the Elegant Kernel's seven independent algebraic components are fixed by UGP symmetries. The only remaining independent DOF, the locked decimal remainder $\tilde{k}_L$, is a continuous parameter tightly constrained by the empirical-path calibration (Appendix C).

*Independence from $\tilde{k}_L$:* The Lean-certified bare gauge couplings, the $N_c$-chain structural identities, $L_{\text{model}}$, and the structural neutrino mass-squared ratio do *not* depend on $\tilde{k}_L$; the locked theoretical-path fermion implementation (§3.4, §4) is governed by the disclosed kernel and correction architecture rather than by per-mass fitting of $\tilde{k}_L$.

The URC parameters are derived from the ridge's divisor count, third-generation capacity, and Fibonacci structure. What remains is the axiomatic system itself—three axioms and the definition of a UGP ridge—from which everything else follows as theorems.

More precisely, the UGP constraint manifold $\mathcal{M}_{\text{UGP}}$ can be characterized by three tiers of independent inputs:

1. **Discrete UGP invariant:** $n$=10 (one integer, selected uniquely by prime-lock and mirror constraints). From this single choice, the framework derives 3 exact bare gauge couplings (Category A, Lean-certified) and canonical charged-fermion triples from which the 0.295% theoretical mass spectrum follows (Category A structural pipeline + A/D URC correction layer; see §4). Additionally, $\theta_{\text{QCD}} = 0$ is conditional on CP-source completeness (Category A/D), and $\Lambda$ requires one external scale ($H_0$). The effective continuous dimension of the Category-A core is **zero**.

2. **External scale-setting constants:** $G_F$ (Fermi constant, electroweak scale), $H_0$ (Hubble parameter, for $\Lambda$), $\sum m_\nu$ (neutrino absolute scale), and, for the Type-I seesaw realization only, $\Delta m^2_{21}$ and $\Delta m^2_{31}$ (observed splittings)—5 measured scalars in total. These do *not* enter the Braid Atlas anchor-free mass-squared-ratio prediction, which uses the structural exponent 29/9 alone (Category A).

3. **Calibrated QCD layer:** 15 binding-model parameters for the baryon sector (Category B), encoding non-perturbative confinement physics.

The total: $0 + 5 + 15 = 20$ independent inputs mapping to $\sim$25 SM observables. (Here "inputs" refers to independent scalar degrees of freedom; correlations among them may further reduce the effective dimension, but no such reductions are assumed.) The Category A core compresses the canonical fermion-triple structure and the three exact bare gauge-coupling rationals to one discrete ridge/seed choice; the reported 0.295% absolute fermion mass spectrum adds the disclosed A/D correction layer (§4).

This dimensional collapse is analogous to how the discovery of a symmetry group reduces the independent components of a physical tensor. Before the symmetry is recognized, each component appears as a separate free parameter; afterward, most are seen to be algebraic consequences of a few group-theoretic invariants. The UGP plays the role of such a symmetry for the SM parameter space. If the framework is correct, the apparent arbitrariness of the SM's constants is not a feature of nature but an artifact of not having identified the underlying constraint manifold until now.

### 11.5 The Nature of the Framework

The UGP is not a quantum field theory in the conventional sense. It does not derive Yukawa couplings from loop-level dynamics or compute scattering amplitudes. Rather, it occupies a complementary niche: given a minimal set of axioms about information compression and locality, it *derives* the numerical constants that a QFT must take as input. The relationship is analogous to the distinction between thermodynamics (which constrains macroscopic observables through conservation laws and entropy) and statistical mechanics (which provides the microscopic dynamics). The UGP provides the constraints; the Standard Model Lagrangian provides the dynamics.

This division of labor is made precise by the broader NEMS programmeme [24, 28]. The Two-Layer PSC Theorem [26] proves that the *structural* content of the SM—its gauge group $SU(3) \times SU(2) \times U(1)$ and three-generation fermion content—is forced by self-containment axioms. The Unified Rigidity Theorem [37] then identifies the unique arithmetic seed from which the GTE cascade produces the canonical triples. The present paper supplies the quantitative layer of the chain—the masses, couplings, and mixing parameters—with exact, partially derived, and calibrated sectors separated explicitly. Together, these three layers constitute the most complete current realization of a unified first-principles framework for the Standard Model parameter problem, with clearly identified open fronts that define the path to closure.

**On the nature of derivation.** A critical distinction separates the present results from phenomenological parameter fitting. The map from axioms to fermion masses is a finite, constructive deductive chain—axioms → ridge sieve → RSUC seed selection → GTE cascade → canonical triples → UCL calibration → physical masses—in which each step is either machine-checked (`ugp-lean`) or algebraically certified (PSLQ + iso-$\sigma$ invariance). Companion paper proof sketches for the two load-bearing theorems (RSUC and Quarter-Lock) are provided in Section G. The output is not post-hoc reproduction of known values but a deductive consequence that could, in principle, have been computed before the masses were measured. The essential distinction is:

$$\textbf{deduction} \neq \textbf{optimization} \tag{43}$$

No continuous parameters are *fitted* in Category A. The Category-A chain constructs the canonical triples, Elegant-Kernel identities, and exact bare gauge-coupling rationals constructively. The reported 0.295% absolute charged-fermion spectrum is the locked A+A/D realization: structurally derived triples plus the disclosed URC correction layer (§4) and the centering remainder $\tilde{k}_L$ (Appendix C). Optimization techniques (grid search, Bayesian optimizers) are used exclusively to *verify* that the derived values minimize the error surface; they find what the theory already predicts, they do not tune the theory to match observations. The Lean machine-checked certificate and the deterministic verifier's reproducibility close any remaining gap between the two.

**On uniqueness and necessity.** The uniqueness claim rests on a chain of five theorems, each proved or machine-checked independently:

1. **PSC Theorem, Layer I** [26]: the axioms of Perfect Self-Containment narrow 4D gauge topologies to exactly $SU(3) \times SU(2) \times U(1)$ with anomaly-minimal chiral matter and $N_{\text{gen}} \geq 3$.
2. **PSC Theorem, Layer II** [26]: Presentation Invariance selects $N_{\text{gen}} = 3$ as the unique minimal solution.
3. **Ridge selection (three convergent certificates):** (a) *Ridge minimality* — $n$=10 is the smallest ridge level at which the prime-lock and mirror constraints jointly admit a solution; no survivor pair

exists at any $n \in [5, 9]$ (Lean: `n10_is_minimal_admissible_ridge`, Section G). Higher levels ($n$=13, $n$=22) also admit mirror-dual prime-lock survivors in isolation. (b) *Global asymptotic sparsity* — for *all* $n \in \mathbb{N}$, if $(b_2, q_2)$ is a mirror-dual survivor with $b_1 = b_2 + q_2 + 7 = 73$ at level $n$, then $n = 10$ (Lean: `asymptotic_sparsity_universal` in `UgpLean.Phase4.AsymptoticSparsity`). This is the global $\forall n$ uniqueness statement, sharpening (a) from "smallest" to "unique under the joint constraint". (c) *Divisor-count certificate* — the ratio $\tau(R_n)/D_1 = 15/8$ used in the CKM $\theta_{23}$ scaling identity is unique to $n$=10 on $[5, 20]$ (Lean: `ckm_theta23_ratio_uniqueness` in `UgpLean.MassRelations.CKMTheta23`). Combined with Layer II ($N_{\text{gen}} = 3$), these three independent certificates pin $n$=10 as the unique ridge level for the SM-compatible UGP construction.

4. **RSUC** [35, 37]: at $n$=10 the residual seed set collapses to the Lepton Seed $(1, 73, 823)$ up to mirror equivalence (Section G).
5. **GTE bisimulation uniqueness** [35]: any minimal lawful UWCA program preserving the canonical UGP invariants (Quarter-Lock, mirror symmetry, Fibonacci rigidity) is necessarily bisimilar to the canonical GTE compiler.

Together, these establish: *given the PSC axioms, the SM gauge group, generation count, foundational seed, and deterministic cascade are all forced.* The present paper extends this chain to quantitative parameters (masses, couplings, mixing). The remaining conditional is the PSC axiom set itself; its justification—as the minimal requirements for a self-defining universe—is argued in the companion NEMS survey [28] and the Unified Rigidity Theorem [37].

**On external anchoring.** All external inputs are accounted for in a strict three-tier classification with no hidden fitting:

- **Pure (Category A):** No inputs except physical units. The canonical charged-fermion triples and the Category-A structural mass pipeline; the three exact bare gauge-coupling rationals; the blind $\alpha_s$ evaluation; $L_{\text{model}}$; structural $N_c$-chain identities; structural neutrino mass-squared ratio. *Note:* the reported 0.295% absolute charged-fermion spectrum is A+A/D because it uses the disclosed URC correction layer (§4); $\alpha_{\text{EM}}$ at +2.39 ppm is $\text{TE}_1$.P PSC-calibrated (A/D); $\Lambda$ (dimensional) requires $H_0$ (A/D); $\theta_{\text{QCD}} = 0$ is conditional on CP-source completeness (A/D).
- **Anchors (Category A/D):** $G_F$ (electroweak scale), $H_0$ (cosmological scale for $\Lambda$), $\sum m_\nu$ (neutrino absolute scale), and the two observed neutrino mass-squared splittings $\Delta m^2_{21}$, $\Delta m^2_{31}$ (neutrino sector hierarchy)—five measured scalars in total. These are dimensional anchors analogous to the choice of units—scale-setting constants that fix the physical realization of otherwise determined ratios. CKM, PMNS, neutrino masses, and the Higgs sector are Category A/D.
- **Calibrated (Category B):** The pairwise QCD binding model for baryons (15 parameters encoding non-perturbative confinement). This sector is explicitly labeled as calibrated; its current empirical anchoring is a precisely stated open problem, not a hidden degree of freedom.

Total independent inputs: $0 + 5 + 15 = 20$ scalars mapping to $\sim 25$ SM observables, consistent with the constraint manifold accounting of Section 11.4. The framework's transparency on this accounting is itself a structural contribution: it maps exactly which SM parameters are determined by number theory and which await full first-principles derivation.

**On algebraic closure vs. numerology.** The constants $\varphi$ and $\pi$ enter the framework independently through structurally-derived positions: $\varphi$ governs generation structure via the Lean-proved $k_{\text{gen2}} = -\varphi/2$, $\pi$ governs gauge normalization via Eq. (11).

Expressions such as $\lambda = \varphi/(4\pi)$ (§7.3.2) and the URC algebraic closures (§4) are ratios of constants that already appear in the framework, not results of an unconstrained search over arbitrary mathematical expressions. Consistent with the Category A/D classification of these items, we do not claim uniqueness within the space of algebraic closures of comparable complexity; ruling out comparably simple alternatives is a residual open problem.

What distinguishes the present framework from numerology is not claimed uniqueness of these identification expressions, but the hierarchy that sits above them. The *structurally-derived* Lean-proved results — RSUC ridge selection, the UCL Elegant Kernel closure up to $\tilde{k}_L$, the bare gauge couplings as exact rationals, and the locked zero-active-parameter theoretical charged-fermion spectrum at 0.295 % RMS with the URC layer explicitly classified Category A/D in §4 — are separated in §1.2 from the A/D *identification* channels in the Higgs, CKM, and baryon sectors, which the paper does not assert are uniquely forced.

**On compression vs. physics.** One might question whether a framework grounded in information compression is doing physics or merely encoding coincidences. Two structural theorems in the companion NEMS survey [28] directly answer this:

- **No-Emulation Theorem:** The self-adjudication dynamics ($\mathcal{PT}$) of a PSC system cannot be reproduced by any Turing-equivalent computation. UGP dynamics are therefore not a passive encoding; they are computationally irreducible in a precise formal sense.
- **Semantic Closure Theorem:** All UGP/GTE dynamics are internally representable without external meta-language. The system is not a compressor that requires an outside observer to interpret it; it is self-interpreting.

Together these theorems establish that the framework is not just compression: it is a physically closed, computationally irreducible structure that happens to be expressible in arithmetic. The compression perspective and the physics perspective are not alternatives; the MDL principle is the *reason* the SM parameter values take the values they do, not a description of a coincidence.

**On falsifiability.** The $Z_6$ constraint restricts $\delta_{\mathrm{CP}}^{\mathrm{PMNS}}$ to four CP-violating branches, covering approximately 17 % of the full angular range (four windows of $\sim 15°$ each). This is a nontrivial constraint compared to an unconstrained theory: a measurement of $\delta_{\mathrm{CP}}$ far from all $Z_6$ branches would falsify the framework.

The neutrino mass predictions are sharper still: the structural Dirac-scale benchmark gives $\sum m_\nu \in [55, 75]$ meV (with $\approx 56$ meV at $M_R = 2 \times 10^{16}$ GeV); falsification from above is the current Planck $\sum m_\nu < 120$ meV bound, so the framework is tested in the slab $[55, 120]$ meV that CMB-S4 and Euclid can resolve within the decade.

Additionally, $\theta_{\mathrm{QCD}} = 0$ is an exact structural prediction (no axion or fine-tuning required), and the full GTE verifier output is deterministically reproducible, providing a machine-checkable falsification surface. The preregistered predictions in Section 9 constitute the concrete experimental tests that distinguish the UGP substrate hypothesis from an unexpectedly effective mathematical compression.

### 11.6 Scope of Claims

The framework's claims are conditional in four specific ways; acknowledging these is part of the five-status claim taxonomy, not a defensive posture:

1. **Validity of the PSC axioms.** All results are conditional on this. The Two-Layer PSC Theorem is proved and machine-checked in the NEMS programme [24, 26, 35] (zero `sorry`, zero custom axioms, Lean 4/Mathlib); the present paper takes those theorems as its starting point.

2. **Interpretation (ontology vs. structure).** Whether the UGP *is* the substrate of nature or *describes* it is an empirical and philosophical question; the mathematics is the same either way.

3. **Completion of partial sectors.** PMNS (QLC/TM2 axiomatic derivation), QCD binding (UGP-derived confinement), and neutrino absolute mass scale remain open fronts.

4. **Category A/D open-problem items not closed within this paper.** The algebraic closures reported in §4 ($\alpha_{\text{symmetry}}, \alpha_{\text{QCD}}, \alpha_{\text{EW}}$) and the forcing of the Elegant-Kernel algebraic targets beyond first order (Appendix C) are algebraic identifications consistent with the data but not yet uniquely singled out by an axiomatic derivation or saturation-null argument. They are a residual open problem; the paper's Category A-exact content does not depend on any of them. By contrast, the CKM $\theta_{23}$ scaling ratio $\tau(R_{10})/D_1 = 15/8$ (§7.2.1; OP(v)) is Category A (Lean-certified structural uniqueness; `ckm_theta23_ratio_uniqueness`, zero sorry) and is not part of this open-problem item. The Higgs $\lambda_H$ is now Category $\text{A}_{\text{Lean}}$ via the SRRG correction (§7.3.2): machine-certified zero sorry, $m_H = 125.2499$ GeV ($-0.0003\sigma$ PDG 2022).

These are *inherent* to the framework's current stage, not defects. They define the path to closure.

**Final Position**

A theorem-audited, deterministic framework showing that, conditional on the PSC axioms, the Standard Model parameter structure is largely *forced* by arithmetic constraints. All quantitative results follow from a fixed constructive chain of five theorems from PSC axioms to GTE dynamics; remaining assumptions are explicitly isolated and all non-derived sectors are labeled. The framework is fully deterministic, machine-reproducible, and experimentally falsifiable.

### 11.7 Outlook

The present results establish the UGP as a falsifiable framework for the Standard Model parameter problem, with a strong derived core and clearly identified open fronts. Immediate priorities include:

(i) closing the baryon residual gap through UGP-derived QCD renormalization corrections;

(ii) refining the PMNS extraction pipeline;

(iii) extending the GTE catalog to identify dark-matter candidates with predictable masses and cross-sections;

(iv) investigating whether the UGP arithmetic substrate constrains black-hole entropy and gravitational degrees of freedom, as a step toward testing compatibility with quantum gravity.

The double determination of $b_{L_1} = 73$ by two algebraically independent routes (GTE cascade and Frobenius prime chain, coinciding uniquely at $N_c = 3$; `fgci_unique_at_nc`, zero `sorry`) reveals unexpected arithmetic depth in the seed: the two GTE primes $\{7, 73\}$ are the complete output of the Frobenius chain, suggesting that the framework's prime structure is richer than originally apparent. The preregistered neutrino predictions provide near-term experimental tests that will either validate or falsify the framework within the next decade.

**Companion papers.** The General Selection paper [21] surveys cross-domain evidence that the IPT threshold $\eta^* \approx 1.1309$ and the $\mathbb{Q}(\zeta_{120})$ algebraic substrate recur across eight domains spanning nuclear physics, prebiotic chemistry, organisational systems, and conformal field theory. The Self-Referential Renormalization Group paper [29] proposes a deeper mechanism from which the UGP fixed-point structure and IPT arise as consequences of a gradient flow on the space of self-referential theories, formalised in Lean 4. The direct coupling route for the fine-structure constant from the Lean-certified bare couplings $g_1^2 = 16/125$, $g_2^2 = 2329/5400$ is developed further in the companion SRRG paper [29] (Appendix C), where the tree-level identity $\alpha = g_1^2 g_2^2/[4\pi(g_1^2 + g_2^2)]$ yields $1/\alpha(M_Z) \approx 127.311$ (0.495% from PDG; grade [A/D]).

**Axioms → SM Parameters: Constructive Pipeline**

**Axiomatic Layer (A1–A3):** Locality, Symmetry, MDL Compression

↓

**Structural Necessity Chain (5 theorems):**

1. PSC I [26]: $SU(3)\times SU(2)\times U(1)$, $N_{\text{gen}} \geq 3$ forced
2. PSC II [26]: Presentation Invariance selects $N_{\text{gen}} = 3$ (unique minimal)
3. Ridge Minimality: smallest prime-lock + mirror solution is $n = 10$
4. RSUC [35, 37]: unique seed $(1, 73, 823)$ up to mirror
5. GTE Bisimulation [35]: any lawful invariant-preserving dynamics is bisimilar to canonical GTE

↓

**Deterministic Dynamics:** GTE cascade generates all canonical triples; no free DOF remain

↓

**Parameter Derivation (Category A, fully derived):** Canonical fermion triples; canonical integer triples for nine light baryons (`BraidAtlas.CompositeTriples`; zero sorry); electroweak boson $c$-values $\{11, 12, 13\}$ (`BraidAtlas.EWBosons`; zero sorry); UCL/Elegant-Kernel algebraic identities; base charged-fermion mass pipeline; 3 exact bare gauge-coupling rationals; blind $\alpha_s$ direct evaluation; $L_{\text{model}}$; $N_c$-chain structural identities; structural neutrino mass-squared ratio

**Parameter Derivation (Category A/D, partially derived or scale-anchored):** URC-corrected 0.295% charged-fermion spectrum (A/D correction layer applied to the A-structural pipeline above); $\text{TE}_1$.P $\alpha_{\text{EM}}$ (calibrated-structural); dimensional $\Lambda$ via $H_0$; $\theta_{\text{QCD}} = 0$ conditional on CP-source completeness; CKM; PMNS; Higgs; Type-I seesaw absolute scale

**Parameter Derivation (Category B, calibrated):** Baryon binding (QCD confinement layer)

**Table 21:** Claim-to-artifact crosswalk for reproducibility.

| **Claim** | Primary certificate / artifact | **Purpose** |
|---|---|---|
| Exact $g_i^2$ rationals | `UgpLean/Phase4/GaugeCouplings.lean` | Lean certificate for bare coupling rationals. |
| Blind $\alpha_s$ | `alpha_s_prediction.json; App. H` | Numerical evaluation plus precommit trail. |
| URC A/D status | `comp_p01_RC1_urc_bounded_enumeration.json` | URC closures not bounded-unique at depth $\leq 3$. |
| CKM $\theta_{23}$ ratio | `UgpLean.MassRelations.CKMTheta23` | Lean-certified divisor-ratio structural derivation. |
| Baryon triples | `BraidAtlas.CompositeTriples` | Lean-certified canonical composite triples. |
| Fermion spectrum | `dual_path_comparison.json` | Empirical/theoretical mass comparison. |
| Higgs $\lambda$ (SRRG) | `MassRelations/HiggsQuartic.lean` | `higgs_quartic_corrected`, zero sorry; $m_H$ = 125.2499 GeV ($-0.0003\sigma$ PDG). |

# Methods

## M1. Canonical Triple Generation

**Algorithm 1** UGP→GTE triple generation

1: **Input:** UGP ridge $n$=10; generation index $g \in \{1, 2, 3\}$
2: **Output:** Canonical triples $\{(a, b, c; g)\}$ for all fermions
3: Fix $R_{10} = 1008$; scan for prime-locked mirror pairs $\to (42, 24)$
4: Compute $b_1 = 73$, $c_1 \in \{823, 2137\}$
5: Initialize lepton seed: $(1, 73, 823; 1)$
6: **for** gen $= 2, 3$ **do**
7: Apply OddOp (if $g$ odd) or EvenOp (if $g$ even)
8: Normalize signs; record in JSON with hashes
9: Form quark seeds via permutation (Eq. (1))
10: Apply odd/even cascade for quark generations
11: Emit `derived_triples.json` with SHA-256 digests

The $N_{\text{eff}}$ value for each particle is $|b|$, derived universally from the triple with no lookup table.

## M2. UCL and Coefficient Certification

The UCL coefficients are stored as exact decimals with SHA-256 hashes. A base change $B^\star$ rationalizes the curvature term to 7/512. PSLQ searches over $\{\pi, \varphi, 1/8, -3/2, 4/3\}$ promote five coefficients to algebraics; the quarter-lock is enforced as an identity. Artifacts:

- `universal_calibration_law.{json,md}`
- `ucl_pslq_catalog.json`
- `ucl_iso_sigma_solutions.json`

### M3. W-$\rho$ Computation

Equation (6) is evaluated by exact factorization. The tolerance check is embedded in the primary verdict and reported alongside prime-factor sets for transparency.

### M4. Physics Engine

$E_{\text{base}}$ combines five components (weights fixed in code): information entropy, Bekenstein radius energy, coherence, phase (`legacy` mode), and type-modulated binding. The modular decomposition separates $E_{\text{base}}(N_{\text{eff}})$ from $C_f((a,b,c;g))$.

### M5. Primary Metric and Preregistration

The primary metric is the RMS of relative errors over nine fermion masses plus W-$\rho$. Canonical knobs: `phase_mode=legacy`, $k = 2.0$, $K = 1400$. The verifier emits `preregistration.{md,json}`, hashes, and `reference_lock.json`. A `verify` mode diffs the stored lock.

### M6. Robustness Suites

**Nulls:** label permutations with histograms. **Uncertainty:** conservative quark bands, strict lepton bands, $N$-jitter. **Broad-flat optimum:** coordinate profiles, $(k, K)$ grids, random restarts. **DOF/MDL:** formal ledger counting model bits vs. baseline encodings.

### M7. Ablation Baselines

Three baselines: (i) generation-only polynomial, (ii) no-Möbius, (iii) no-curvature. Each scored under the same primary metric; residual fingerprints compared.

### M8. Neutrino Seesaw Capsule

The seesaw mechanism produces neutrino mass-squared differences anchored to $\sum m_\nu = 60$ meV: $\Delta m^2_{21} = 7.42 \times 10^{-5}$ eV$^2$, $\Delta m^2_{31} = 2.514 \times 10^{-3}$ eV$^2$, $\sum m_\nu = 60.0$ meV. The PMNS matrix is constructed from the QLC/TM2 mixing angles (Section 7.2.2) with $\delta = \pi/3$ ($Z_6$, $k$=1). The seesaw extraction pipeline independently yields $\delta_{\text{CP}} = 38.97°$, recorded in the provenance payload for comparison. All choices are recorded in the provenance payload.

### M9. Gauge Coupling Pipeline

Bare couplings from (10), instantiation factor from (11), then particle-dependent RG evolution to $M_Z$. The TE$_1$.P pipeline audits $\alpha$ through the full chain.

### M10. Baryon Binding Model

The 15-parameter binding model (Eq. (22)) was determined by a single global optimization against the nine light-baryon masses. The optimized values are locked; at prediction time there are no free parameters. The parameter set is available in the supplementary artifacts.

### M11. Supplementary Echoes

**Gluon echo:** threshold-matched $\alpha_s$ running (1-loop and 2-loop) with guard band $\Lambda_3 \in [150, 450]$ MeV. **Gravity echo:** informational Newton-constant proxy at the EW scale. Both are outside the primary verdict.

### M12. Environment and Determinism

The script prints CPU/OS/Python/BLAS info. Arithmetic order is frozen; a deterministic integer-factorization routine is used. The primary verifier has no stochastic calls; all enumerations are finite and deterministic. Robustness suites that involve permutations, coefficient jitter, or random restarts (Sections 8–9) use fixed seeds or deterministic enumerations; their seeds and configurations are recorded in the provenance artifacts.

### M13. Computational Tools

The primary verifier is written in Python 3.9+ using NumPy for linear algebra and array operations.

SciPy was used during the discovery phase for optimization (coefficient calibration, PSLQ integer-relation searches, and global baryon binding fits); at prediction time, no SciPy routines are called—all results are deterministic from frozen constants. Matplotlib generates figures and profile plots.

SageMath was used for exploratory number-theoretic calculations (prime factorizations, divisor enumerations, Möbius function evaluations) during the development of the ridge sieve and GTE cascade; the final verifier reimplements these in pure Python for self-containment.

The machine-checked formalization (`ugp-lean`) is written in Lean 4 (v4.29.0-rc6) against Mathlib (v4.29.0-rc6); see the companion formalization paper [35] for the full theorem inventory. All $\mathbf{A}_{\text{Lean}}$ results in this paper carry zero `sorry` and zero custom axioms in their proof chains.

Full reproduction instructions are provided in `REPRODUCE.md` co-located with this paper's source.


## Acknowledgments

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AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.


## Author Contributions

N.S. conceived the framework, performed all derivations, wrote the verifier code, and wrote the manuscript.

## Competing Interests

The author declares no competing interests.

## Data Availability

All data generated and analyzed during this study are included in this article and its supplementary information. PDG targets and experimental bands are embedded in the verifier and can be exported alongside the report.

## Code Availability

The computational infrastructure for this paper has three components.

*Primary end-to-end run.* A single deterministic Python script (filename below) generates the complete dual-path UCL prediction, gauge couplings, CKM/PMNS, Higgs sector, cosmological constant, and the primary $\sigma$ GoF; see Appendix F for the invocation.

```
papers/01_SM/canonical_run/
UGP_GTE_SM_Verifier.py
```

*Supporting structural-analysis scripts.* The structural identifications underlying the $N_c$ chain (§3.7), the VV GUT group-theory derivation (§3.1), and the neutrino structural prediction (§6.4) are each verified by standalone Python scripts.

```
papers/01_SM/canonical_run/
```

(`comp_p01_*.py` per topic.) Each script is independently reproducible, requires only `numpy` (and optionally `scipy`/`sympy`), emits a JSON artifact with a SHA-256 hash, and runs in under a minute. The full set is catalogued in:

```
papers/01_SM/PROVENANCE.md
papers/01_SM/REPRODUCE.md
```

(the first lists per-script descriptions; the second lists invocation commands).

*Machine-checked proofs.* The `ugp-lean` Lean 4 library [36] formally verifies the structural theorems underlying every Category A claim, with zero `sorry` and the standard Mathlib axiom signature. Every Lean theorem can be verified by cloning the repository and running `lake build` (Lean 4.29.0-rc6, Mathlib 4.29.0-rc6); see the companion formalization paper [35] for the full build and reproducibility instructions.

All three components are versioned in public GitHub:

```
https://github.com/novaspivack/ugp-physics
https://github.com/novaspivack/ugp-lean
```

with Zenodo DOIs on publication ( [35] for the Lean library).

## A Lean 4 Theorem Inventory

The following theorems from the `ugp-lean` library (zero `sorry`, standard Mathlib axioms) underpin the Category A claims in this paper. Entries marked † are Lean `defs` (value definitions); surrounding theorems certify their identities.

**Ridge and seed selection.**

- `rsuc_theorem`: at $n$=10 the ridge sieve yields two survivors; MDL selects $(1, 73, 823)$ uniquely.
- `canonical_orbit_triples`: the cascade $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$ is verified step by step.

**Quarter-Lock and UCL Elegant Kernel.**

- `quarterLockLaw`: $k_M = k_{\text{gen2}} + \frac{1}{4}k_{L^2}$ is unconditional.
- `thm_ucl1/2/4/5_fully_unconditional`: all nine UCL Elegant Kernel coefficients, including $k_{\text{gen}} = \varphi \cos(\pi/10)$, are structural consequences.
- `elegant_kernel_full_certification`: master container packaging all nine UCL Elegant Kernel coefficients in a single zero-sorry theorem (`ElegantKernel/Unconditional/MasterCertification.lean`); 9/9 unconditional closure.
- `ucl_fermion_mass_ordering` / `ucl_tier2_mass_ordering`: fermion mass ordering $m(\text{gen}\,1) < m(\text{gen}\,2) < m(\text{gen}\,3)$ for lepton, up, and down sectors (CatAL, zero sorry; `UCLMassOrdering.lean`, 7 submodules).
- `ucl_lepton_sector_koide_identity` (Tier 3, CatAL): Koide $Q = 2/3$ structurally certified by composing GTE-pinned phase $\theta = 2/9$ (`koide_angle_from_N_c_pure`) with $S_3$ cone amplitude $b = \sqrt{2}$ (`koide_Q_iff_amplitude`); zero sorry (`UCLKoide.lean`).
- `two_cH_plus_one_eq_ngen_cubed`, `higgs_quartic_denominator_is_ngen_cubed`: identity $2c_H + 1 = 27 = N_{\text{gen}}^3$ (zero sorry; `MassRelations/HiggsQuartic.lean`).
- `higgs_quartic_corrected`, `higgs_quartic_corrected_pos`, `higgs_quartic_numerical_approx`: SRRG-corrected Higgs quartic $\lambda_H$ and $m_H = 125.2499$ GeV (CatAL, zero sorry; `MassRelations/HiggsQuartic.lean`).

**Gauge couplings and EW observables.**

- `g1/2/3Sq_bare_eq`: $g_1^2 = 16/125$, $g_2^2 = 2329/5400$, $g_3^2 = 41\,075\,281/27\,648\,000$ (exact rationals).
- `sin2_theta_W_bare_eq`: $\sin^2\theta_W = g_1^2/(g_1^2 + g_2^2) = 3456/15101$ (exact rational, tree-level; module `Phase4.GaugeCouplings`, zero sorry).
- `alpha_em_formula_exact`: $1/\alpha_{\text{EM}} = 4\pi(1/g_1^2 + 1/g_2^2)$ with $1/g_1^2 + 1/g_2^2 = 377525/37264$ (exact rational, tree-level, zero sorry).

**Mass relations and the $N_c$ chain.**

- `claim_C_formal`: the TT mass cascade equals an SU(3) Weyl rotation at $\pi/6$ per generation with $2^g$ doubling.
- `N_c_determines_everything`: $\delta$, $b_1$, $a_{\text{top}}$, strand count, and $\theta = 2/9$ all follow from $N_c = 3$.
- `VV_from_GUT_group_theory`: $\alpha_d$, $\beta_d$, $\gamma_d$ derived from SU(5)/SO(10) representation dimensions.

- `koide_equal_norm`: $|v_1|^2 = |v_2|^2 = 3$ for every $\theta$ in the Koide parametrisation family — the $S_3$-balance condition is a universal ring identity, independent of the specific $\theta = 2/9$ value (module `MassRelations.KoideAngle`, zero sorry).
- `koide_Q_iff_amplitude`, `cone_amplitude_eq_sqrt2`: Koide $Q = 2/3$ iff $b = \sqrt{2}$ iff equal $S_3$-irrep norm; `koide_cv_one_iff_amplitude`: $\mathrm{CV}(\sqrt{m}) = 1 \Leftrightarrow Q = 2/3$; `lepton_a_values_not_equal_modes`: raw $Z_7$ orbit mechanism for equal norms ruled out (modules `MassRelations.KoideYukawaAmplitude`, `MassRelations.KoideEqualNormReformulation`, zero sorry).
- `kink_bps_mass_formula`: $M_{\text{kink}} = 8m/49$ from $Z_7$ sine-Gordon potential (arithmetic identity $(\frac{4m}{49}) \cdot 2 = \frac{8m}{49}$, CatAL); `tau_yukawa_structural`: $y_\tau = 1/(2 \cdot 7^2) = 1/98$ exact rational; `kink_higgs_dimensionless_coupling`: $g_{hKK} = 4/7^4$ linking G7 and G8 (module `Gravity.PMDLGravityTheorems`, zero sorry).

**Information-theoretic constants.**

- `L_model_eq_log_residual`: $L_{\text{model}} = \log_2((2^4 \cdot 5^3)/3)$; uniqueness from Quarter-Lock.

**Neutrino sector.**

- `nuSeesawExponent`$^\dagger$ (Lean `def`, value 29/9), `nu_seesaw_exponent_three_decompositions` (theorem): 29/9 admits three independent structural decompositions; $\dim(126) = 2N_c^2\delta$.

**Braid Atlas, charges, and composites.**

- `sm_charge_leptons`: $Q = W_g/N_c$ for all four SM fermion types (charged lepton, neutrino, up-quark, down-quark), with winding numbers from the Braid Atlas [17].
- `anomaly_cancellation_forces_Nc_3`: per-generation winding sum $\sum W_g = N_c(N_c - 3) = 0$ if and only if $N_c = 3$ — anomaly cancellation *derives* the QCD colour rank (module `BraidAtlas.ChargeTheorem`, zero sorry).
- `sm_winding_numbers_from_Nc`, `y_ql_unifies_vv_and_winding`: SM winding bookkeeping $\{N_c{-}1, -1, 0, -N_c\}$ and hypercharge–VV unification from $N_c$ alone (module `BraidAtlas.ChargeDerivation`, zero `sorry`).
- `rcc_all_classical_families`: RCC classification inequalities for infinite classical gauge families bundled with TE2.2-era certificates (`PSC.RCCInfiniteFamilies`, zero sorry).
- `ugp_nucleon_b_formula`, `ugp_strange_baryon_b_formulas`, `ugp_strange_baryon_c_values`: complete canonical $(a, b, c;\, g)$ triples for all nine light baryons Lean-certified from Braid Atlas winding numbers and GTE Mersenne-sector rules (module `BraidAtlas.CompositeTriples`, zero `sorry`; Category A, structurally independent of the calibrated mass model).
- `ew_boson_c_value_theorem`, `ew_c_window_unique`: the electroweak massive-boson $c$-values $\{c(W), c(Z), c(H)\} = \{11, 12, 13\}$ are derived as the unique consecutive integer triple in the GTE orbital constants at $n{=}10$ (module `BraidAtlas.EWBosons`, zero `sorry`; §7.1.5).

**Null-discipline and theorem-eligibility infrastructure.**

- `urc_basis_is_saturated`: the URC basis of 1,596,939 algebraic expressions has satDensity $> 0.01$ (machine-checked from $e^{319} \geq 320$; module `UgpPhysicsLean.NullDiscipline.SaturationBarrier`, zero sorry) — formally establishing the null-disciplined bound on the bounded-uniqueness question (§4).

- `IsTheoremEligible`† (Lean `def`, four-gate predicate) / `vv_formula_is_theorem_eligible` (theorem): the four-gate Theorem Eligibility Criterion (TEC) is machine-checked; the VV functional form passes all four gates; VV coefficient values (`vv_coefficients_are_classC`) and URC closures (`urc_closure_is_classC`) are certified Class C (module `UgpPhysicsLean.NullDiscipline.TheoremEligibility`, zero sorry).

**Substrate universality.**

- `ugp_is_turing_universal`: UGP is Turing universal via Rule-110 embedding [5, 40].

**Frobenius-Generation Coincidence Identity (FGCI).**

- `fgci_unique_at_nc`: the cascade formula $G(N_c) = N_c^4 - (N_c^2{+}1)/2 - N_c$ and the Frobenius chain $F(n) = n^4 - n^2 + 1$ agree if and only if $n = N_c = 3$, witnessing two independent algebraic derivations of $b_{L_1} = 73$; uniqueness proved by $(n-3)(n+1) = 0$ (module `UgpLean.Universality.FrobeniusChain`, 17 theorems, zero `sorry`).
- `frobenius_chain_terminates`: the Frobenius prime chain at $N_c = 3$ is $\{7, 73\} = \{N_c^2{-}N_c{+}1,\ N_c^4{-}N_c^2{+}1\}$ and terminates at level 3 ($703 = 19 \times 37$, composite), producing exactly the two GTE primes $|\mathbb{Z}_7|$ and $b_{L_1}$ used throughout the framework (zero `sorry`).
- `b_L2_eq_two_Nc_Z7`: the muon ladder index $b_{L_2} = 42 = 2\,N_c\,|\mathbb{Z}_7|$ factors through both Frobenius group orders (zero `sorry`).

**Two-Layer PSC enumeration (generation count).**

- `universe_params_card`: the Layer I candidate space has exactly 34,560 universe descriptions (`native_decide`; module `TE22.ScanCertificate`, zero `sorry`).
- `psc_enumeration_forces_ngen_3`: every PSC-admissible universe has $N_{\text{gen}} = 3$ (`native_decide`; zero `sorry`).
- `psc_12_survivors_have_ngen_3`: exactly twelve Layer I PSC survivors, all with $N_{\text{gen}} = 3$ (bundles `psc_12_survivors_card` with `psc_enumeration_forces_ngen_3`).
- `psc_admissible_forces_sm_gauge`: every PSC-admissible universe is $SU(3) \times SU(2) \times U(1)$ in four dimensions (`native_decide`).

# B Canonical GTE Triples

**Table 22:** Canonical GTE triples for all 24 SM particles.

| **Particle** | $a$ | $b$ | $c$ | $g$ |
|---|---|---|---|---|
| *Charged Leptons* | | | | |
| $e$ | 1 | 73 | 823 | 1 |
| $\mu$ | 9 | 42 | 1023 | 2 |
| $\tau$ | 5 | 275 | 65535 | 3 |
| *Up-Type Quarks* | | | | |
| $u$ | 5 | 9 | 275 | 1 |
| $c$ | 5 | 275 | 65535 | 2 |
| $t$ | 76 | 337920 | $-1$ | 3 |
| *Down-Type Quarks* | | | | |
| $d$ | 9 | 5 | 42 | 1 |
| $s$ | 9 | 186 | 1023 | 2 |
| $b$ | 5 | 8191 | 65535 | 3 |
| *Neutrinos (Left-Handed)* | | | | |
| $\nu_e$ | 1 | 1 | 823 | 1 |
| $\nu_\mu$ | 9 | 1 | 1023 | 2 |
| $\nu_\tau$ | 5 | 1 | 65535 | 3 |
| *Neutrinos (Right-Handed)* | | | | |
| $\nu_{e,R}$ | 2 | 5 | 5 | 1 |
| $\nu_{\mu,R}$ | 7 | 11 | 13 | 2 |
| $\nu_{\tau,R}$ | 17 | 19 | 23 | 3 |
| *Baryons (Lean-derived canonical triples; see §7.1.4)* | | | | |
| $p$ | 5 | 11459 | 15 | 1 |
| $n$ | 5 | 11441 | 15 | 1 |
| $\Lambda$ | 5 | 38236 | $+1023$ | 1 |
| $\Sigma^+$ | 5 | 639161 | $+1023$ | 1 |
| $\Sigma^0$ | 5 | 38236 | $+1023$ | 1 |
| $\Sigma^-$ | 5 | 38236 | $-1023$ | 1 |
| $\Xi^0$ | 5 | 878434 | $+1023$ | 1 |
| $\Xi^-$ | 5 | 878434 | $-1023$ | 1 |
| $\Omega^-$ | 9 | 1814646 | $-1023$ | 1 |

## C Coefficient Palette Comparison

**Table 23:** Complete 9-component UCL coefficient vector: frozen empirical (UCL2.3) vs. Elegant Kernel targets. The SHA-256 hash of the empirical vector is recorded in `dof_ledger.json`.

| Coeff. | Empirical | Kernel target | $\Delta$ |
|---|---|---|---|
| $k_0$ | $-0.15487$ | $-1/(2\pi)$ | $+4.3\text{e}-3$ |
| $k_L$ | $+0.01970$ | $0.01974$ | $-3.9\text{e}-5$ |
| $k_{L^2}$ | $+0.01357$ | $7/512$ | $-1.1\text{e}-4$ |
| $k_{\text{gen}}$ | $+1.54480$ | $\varphi\cos(\pi/10)$ | $+6.0\text{e}-3$ |
| $k_{\text{gen2}}$ | $-0.80925$ | $-\varphi/2$ | $-2.3\text{e}-4$ |
| $k_M$ | $-0.80587$ | $-\varphi/2+7/2048$ | $-2.7\text{e}-4$ |
| $k_a$ | $+0.12373$ | $1/8$ | $-1.3\text{e}-3$ |
| $k_b$ | $-1.50453$ | $-3/2$ | $-4.5\text{e}-3$ |
| $k_c$ | $+1.32657$ | $4/3$ | $-6.8\text{e}-3$ |

Full-precision values for independent reproduction:

```
k = [−0.154865570, +0.019697890, +0.013565910,
     +1.544802780, −0.809248350, −0.805871920,
     +0.123729680, −1.504529470, +1.326566020]
```

**Epistemic status of the palette comparison.** The $\Delta$ column above records per-coordinate residuals between the calibrated empirical UCL2.3 vector and the Elegant-Kernel algebraic targets; these residuals range from $10^{-5}$ (for $k_L$) to $\sim 7\times10^{-3}$ (for $k_c$, $k_{\text{gen}}$). Two points of reviewer-level candor must be recorded explicitly:

1. $\tilde{k}_L$ **remains a continuous parameter.** In the rationalising base change $B^\star$ (§2.5), all kernel coefficients except the locked decimal remainder $\tilde{k}_L$ snap to algebraic targets; $\tilde{k}_L$ itself remains a continuous real number determined by the calibration. In the DOF ledger we therefore count $\tilde{k}_L$ as **one remaining continuous parameter tightly constrained by data**, not as an algebraic identity. The paper's Category A-exact claims (Lean-certified bare gauge couplings; zero-active-parameter theoretical fermion-mass prediction at prediction time) do not depend on $\tilde{k}_L$; the empirical-path precision results of §3.2 do. This disclosure is the reason §3.2 is framed as a functional-form benchmark rather than a precision claim.

2. **The iso-$\sigma$ + PSLQ framing is a first-order consistency test, not a forcing argument.** The iso-$\sigma$ solver shows that substituting the Elegant-Kernel algebraic targets for the frozen decimals leaves the primary score invariant to first order; PSLQ returns the targets as top candidates. Together, these establish that the algebraic targets are *consistent with* the calibration data at leading order. They do not establish that the targets are *forced* by the data or by an independent principle at all orders. Full forcing would require either higher-order iso-$\sigma$ analysis or an axiomatic derivation; both are residual open problems.

Neither item affects the Lean-certified bare gauge couplings ($g_1^2 = 16/125$, $g_2^2 = 2329/5400$, $g_3^2 = 41\,075\,281/27\,648\,000$), which are *exact* rationals in the `ugp-lean` library and carry the Category A-exact content of the paper.

## D Supplementary Artifact Manifest

All artifacts are produced by a single deterministic run. Machine-readable forms (JSON, CSV) accompany human-readable summaries.

- `reference_lock.json`: frozen primary score and key observables.
- `artifact_manifest.{json,csv}`: SHA-256 hashes for all outputs.
- `dual_path_comparison.json`: empirical and theoretical mass predictions.
- `gte_cascade_derivation.json`: N-value cascade verification.
- `ucl_lock_certificate.{json,md}`: quarter-lock residual.
- `ucl_geometry_certificate.{json,md}`: Fisher metric and curvature.
- `ucl_pslq_catalog.json`, `ucl_pslq_best.json`: PSLQ hits.
- `ucl_iso_sigma_solutions.json`: iso-$\sigma$ neutral directions.
- `universal_calibration_law.{json,md}`: elegant kernel coefficients.
- `nulls_suite.{json,csv}`: permutation null distributions.
- `bfopt_*`: broad-flat optimum profiles.
- `dof_ledger.{json,csv}`: DOF/MDL accounting.
- `ckm_report.json`: CKM verification (PDG-lock); `ckm_report_ugp_derived.json`: derived CKM.
- `pmns_report.json`: PMNS candidate matrix.
- `seesaw_from_ugp.json`: neutrino parameters.
- `ewk_couplings_from_gte.json`: gauge couplings.
- `anomaly_proof.json`: anomaly cancellation.
- `hadron_echo.json`: gluon-sector echo.
- `gravity_echo.json`: gravity echo.
- `yukawas.{json,csv}`: Yukawa coupling matrices.
- `preregistration.{md,json}`: locked primary definition.
- `theoretical_coefficients.json`: Elegant-Kernel target vector ($k_{\text{gen}} = \varphi \cos(\pi/10)$, etc.).
- `fully_theoretical_results.json`: bare-elegant UCL run (limit mode; see Appendix E).

# E Verifier Modes and What They Demonstrate

The canonical script `UGP_GTE_SM_Verifier.py` exposes several reproducible modes. They are *not* interchangeable: each mode answers a distinct audit question.

**Dual-path (paper headline, 0.295 %).** `-run-dual-path` with default `-coeffs-source empirical` compares two mass pipelines that both use the frozen UCL2.3 vector:

- **Empirical arm:** canonical $\text{renorm}_K = 1400$; primary $\sigma \approx 3\times10^{-3}\%$ (functional-form benchmark, §3.2).
- **Theoretical arm:** $\text{renorm}_K$ from Bekenstein–Fisher normalization plus the locked URC layer; primary $\sigma \approx 0.29\%$ (Table 9).

Artifact: `dual_path_comparison.json`. The coefficient table inside that artifact compares UCL2.3 to the Elegant-Kernel *targets* ($\|\mathbf{k}_{\text{emp}} - \mathbf{k}_{\text{th}}\|_\infty \approx 7 \times 10^{-3}$); it does not re-run masses with the target vector substituted.

**Bare Elegant Kernel / limit mode (~ 1.1%).** `-coeffs-source limit` (aliases `theoretical`, `elegant`) with `-run-fully-theoretical` activates `THEORETICAL_COEFF_VECTOR` ($k_{\text{gen}} = \varphi \cos(\pi/10)$, quarter-lock rationals, etc.) and the first-principles `E_base` mixer. This is the zero-substitution test of whether the kernel alone closes the nine-fermion spectrum; it demonstrates convergence of the empirical fit toward the algebraic palette (Appendix C), not the headline locked prediction.

**IMT mixer: `v12` vs. `cmca`.** `-imt-mixer-mode v12` (default) uses the embedded Phase/Binding mixer calibrated to the canonical primary score. `-imt-mixer-mode cmca` replaces it with the structural CMCA two-anchor closure of §3.1 (mode share $3L/(3L + L^2)$; see [33, 34]). In the current closure the two modes are numerically equivalent for charged-fermion masses; the flag exists to audit mixer dependence.

**Full stack.** `-preset-fullstack -full-derivation` runs the complete artifact battery (Appendix D) including dual-path output when configured in the reproduction script `comp_p01_ucl_coeff_audit.py`.

See `papers/01_SM/REPRODUCE.md` and `UGP_GTE_SM_Verifier/README.md` for command lines. Run `python3 UGP_GTE_SM_Verifier.py -write-help-md` to emit `HELP.md` beside the script.

# F Reproducibility

All results in this paper are generated by a single self-contained Python script. No external optimization, random search, or GPU computation is required.

**Prerequisites.** Python 3.9+ with `numpy` (any recent version). `matplotlib` and `scipy` are optional (used for figures and PSLQ only).

**Full verification run.** The following command generates every artifact listed in Appendix C, the dual-path comparison, gauge couplings, CKM, PMNS, seesaw, and the robustness battery:

```
cd UGP_GTE_SM_Verifier
python3 UGP_GTE_SM_Verifier.py \
  --preset-fullstack --n 10 \
  --full-derivation
```

**Coefficient and limit audits.** For frozen P01 coefficient artifacts after kernel updates, use the audit driver (writes into `papers/01_SM/canonical_run/`):

```
cd papers/01_SM/canonical_run
python3 comp_p01_ucl_coeff_audit.py
```

Optional explicit modes (Appendix E):

```
python3 UGP_GTE_SM_Verifier.py --n 10 --mode phys --quiet \
  --coeffs-source empirical --run-dual-path
python3 UGP_GTE_SM_Verifier.py --n 10 --mode phys --quiet \
  --coeffs-source limit --run-fully-theoretical
```

All outputs are written to a timestamped directory under `Verifier_reports/`.

**Derived CKM.** To produce the CKM matrix from Section 7.2.1:

```
python3 -c "
import importlib.util, pathlib, sys
p = pathlib.Path('UGP_GTE_SM_Verifier.py')
s = importlib.util.spec_from_file_location(
  'UGP_GTE_SM_Verifier', p)
m = importlib.util.module_from_spec(s)
sys.modules['UGP_GTE_SM_Verifier'] = m
s.loader.exec_module(m)
import json
r = m.ckm_from_ugp_derived()
print(json.dumps(r, indent=2, default=str))
"
```

**Cosmological constant trace.**

```
python3 -c "
import math
L = math.log2((2**4 * 5**3) / 3)
H0 = 70e3 / 3.0856775814913673e22
c = 299792458.0
Lam = (math.log(2)/math.pi)*L*H0**2/c**2
print(f'L_model = {L} bits')
print(f'Lambda  = {Lam} m^-2')
"
```

**Hash verification.** The file `artifact_manifest.json` in each run directory contains SHA-256 hashes of all generated artifacts, enabling bit-exact reproducibility verification across machines and Python versions.

# G Companion Paper Proof Sketches

The following proof sketches and formal theorem statements make the load-bearing structural results verifiable within this manuscript. Full machine-checked proofs for the UGP-internal theorems (RSUC, Quarter-Lock, GTE bisimulation) have zero `sorry` and zero custom axioms in `ugp-lean` [35, 36]; the PSC and No-Emulation theorems are proved in the cited companion papers.

**Proposition G.1** (RSUC: Residual Seed Uniqueness at $n$=10)**.** *At ridge level $n$=10, the prime-lock and mirror constraints admit exactly one survivor pair, from which the Lepton Seed* $(1, 73, 823)$ *is uniquely selected.*

*Proof sketch.* The ridge $R_{10} = 2^{10} - 16 = 1008 = 2^4 \times 3^2 \times 7$ has $\tau(1008) = 30$ positive divisors. Interior divisor pairs $(b_2, q_2)$ with $b_2 q_2 = 1008$ and $b_2, q_2 > 15$ are: $(16, 63)$, $(18, 56)$, $(21, 48)$, $(24, 42)$, $(28, 36)$, and their mirrors. For each, compute $b_1 = b_2 + q_2 + 7$, $q_1 = b_2 - 13$, and $c_1 = b_1 q_1 + 20$. Exhaustive check:

| $(b_2, q_2)$ | $b_1$ | $q_1$ | $c_1$ | Prime? |
|---|---|---|---|---|
| $(16, 63)$ | 86 | 3 | 278 | No $(2 \times 139)$ |
| $(18, 56)$ | 81 | 5 | 425 | No $(5^2 \times 17)$ |
| $(21, 48)$ | 76 | 8 | 628 | No $(4 \times 157)$ |
| $(24, 42)$ | 73 | 11 | **823** | **Yes** |
| $(28, 36)$ | 71 | 15 | 1085 | No $(5 \times 7 \times 31)$ |

(Mirror pairs $(q_2, b_2)$ yield the same $b_1$ with $q_1' = q_2 - 13$; only $(42, 24)$ gives $c_1' = 73 \times 29 + 20 = 2137$, also prime.)

Exactly one mirror pair, $(24, 42)/(42, 24)$, satisfies the prime-lock constraint. This determines $b_1 = 73$ uniquely. MDL minimality (lexicographic selection of the smaller $c_1$) then selects $c_1 = 823$, yielding the Lepton Seed $(1, 73, 823)$. $\square$

**Proposition G.2** (Quarter-Lock Law)**.** *Any UCL coefficient vector* $\mathbf{k}$ *invariant under a two-step GTE evolution block satisfies* $k_M = k_{\text{gen2}} + \frac{1}{4} k_{L^2}$*.*

*Proof sketch.* A two-step GTE block acts on the feature vector $\phi((a, b, c; g))$ as a rank-one perturbation $P = I + \mathbf{u}\mathbf{v}^T$ of the identity in the 9-dimensional coefficient space. The UCL score $\log C_f = \mathbf{k} \cdot \phi$ is invariant under this evolution iff $\mathbf{k}$ lies in the left null space of $(\mathbf{u}\mathbf{v}^T)$. Explicit computation of $\mathbf{u}$ and $\mathbf{v}$ from the odd/even operators yields a single linear constraint on $\mathbf{k}$, which reduces to $k_M = k_{\text{gen2}} + \frac{1}{4} k_{L^2}$. The machine-checked proof (`quarterLockLaw` in `ugp-lean`) verifies this as an unconditional algebraic identity. $\square$

The remaining propositions in the necessity chain (Section 11) are proved in the companion papers cited; their formal statements are reproduced here for self-containment.

**Proposition G.3** (PSC Two-Layer Theorem [26])**.** *Let a 4D gauge theory satisfy the Perfect Self-Containment axioms (Reflexive Closure, No Meta-language, Topological Viability, Self-Adjudication, Presentation Invariance).*

- ***Layer I (Consistency):*** *The hard axioms (RC, NM, TV, SA) narrow the admissible gauge topologies to* $SU(3) \times SU(2) \times U(1)$ *with anomaly-minimal chiral matter and* $N_{\text{gen}} \geq 3$*.*
- ***Layer II (Optimality):*** *Presentation Invariance (PI) selects* $N_{\text{gen}} = 3$ *as the unique minimal solution among the Layer I survivors.*

*Machine-checked support: conditional theorems in* `nems-lean`*. The Residual Classification (full gauge-signature uniqueness) is supported by a computational pipeline and was stated as a conjecture in [26]; it is established as a theorem (*`PSC.RCCInfiniteFamilies`*, zero* `sorry`*, [36]), making gauge-signature uniqueness unconditional.*

**Proposition G.4** (GTE Bisimulation Uniqueness [35])**.** *Let* $\mathcal{I}_{UGP}$ *be the canonical family of UGP invariants (Quarter-Lock, mirror symmetry, Fibonacci rigidity). Any minimal, lawful UWCA program that preserves* $\mathcal{I}_{UGP}$ *is necessarily bisimilar to the canonical GTE compiler* $\Sigma_{GTE}$*. That is, the GTE is the unique dynamics (up to bisimulation) compatible with the UGP's structural invariants.*

**Proposition G.5** (No-Emulation Theorem [28])**.** *Let* $\mathfrak{R}$ *be a Reflexive System implementing the self-adjudication dynamics* $\mathcal{PT}$*. No Standard Computational machine* $M$ *expressible as a Turing-equivalent endomorphism can emulate the total adjudication map of* $\mathcal{PT}$ *on all admissible Choice Points. This establishes that the PSC/UGP dynamics carry structural content beyond what any passive Turing-equivalent encoding can reproduce.*

# H Pre-Commitment Trail for the $\alpha_s$ Blind Prediction

The $\alpha_s(M_Z) = 0.11822$ prediction of Section 5.4 is *blind* in the cryptographic sense that the Lean-certified rational $g_3^{2,\,\mathrm{bare}} = 41\,075\,281/27\,648\,000$ underlying it was committed to the public `ugp-lean` library 43 days before any $\alpha_s$-comparison artifact first appeared in the `ugp-physics` repository (and 65 days before the 2026-05-09 re-verification; the prediction is unchanged). This appendix records the specific commit hashes, timestamps, and Zenodo DOIs, enabling any reviewer to reproduce and verify the pre-commitment trail independently.

### Commit-level timestamps

- **`ugp-lean` repository** (`https://github.com/novaspivack/ugp-lean`): the Lean file `UgpLean/Phase4/GaugeCouplings.lean` was first added in commit

  `27e6823261f6e8b846c1661f7d50bfef6013a629`
  timestamp: **2026-03-05 14:09:28 -0500 (EST)**
  message: "Initial commit: ugp-lean Lean 4 formalization of UGP and GTE"

  The file contains the line `def g3Sq_bare : Q := 41075281/27648000`, and the Lean theorem `g3Sq_bare_eq` proves the rational identity with zero `sorry` and the standard Mathlib axiom signature [`propext`, `Classical.choice`, `Quot.sound`].

- **Zenodo archival of `ugp-lean`:** the concept DOI (resolves to latest archived version) is 10.5281/zenodo.19433538; a specific post-initial-commit version DOI is 10.5281/zenodo.19554700. Both archive the rational $g_3^{2,\,\mathrm{bare}} = 41\,075\,281/27\,648\,000$ with a timestamp that predates any PDG-comparison artifact in `ugp-physics`.

- **`ugp-physics` repository** (`https://github.com/novaspivack/ugp-physics`): the file

  ```
  papers/01_SM/canonical_run/
  alpha_s_prediction.json
  ```

  which contains the PDG-comparison output $\alpha_s(M_Z) = 0.11822$ at $+0.36\sigma$ (PDG 2022; $+0.24\sigma$ vs PDG 2024), was first added in commit

  `95027e6e843e57f5ce5ac7f96797c69b6b267d10`
  timestamp: **2026-04-17 18:50:09 -0400 (EDT)**

- **Gap: 43 days + 4 hours + 41 minutes** separate the Lean rational commitment from the first PDG-comparison artifact. This is the cryptographic pre-commitment window.

- **Longer-gap re-verification (additive, does not replace the 43-day claim).** The blind $\alpha_s(M_Z) = 0.11822$ prediction has held *unchanged* under a full deterministic re-run of the canonical pipeline on **2026-05-09**, with the regenerated artifact `alpha_s_prediction.json` hashing to SHA-256 prefix `288bac48ba22cb7c...` (full hash recorded in `papers/01_SM/PROVENANCE.md`). This re-verification extends the cryptographic pre-commitment window to a **65-day interval** since the original Lean commit `27e6823` (2026-03-05), strengthening rather than replacing the historical 43-day record.

### Reviewer verification protocol

The pre-commitment is reproducible from the public repositories:

```
# 1. ugp-lean: verify rational + timestamp
git clone \
  https://github.com/novaspivack/ugp-lean
cd ugp-lean
git show \
  27e6823:UgpLean/Phase4/GaugeCouplings.lean \
  | grep 41075281
git log 27e6823 \
  --format="%H %ai %s" --no-walk

# 2. ugp-physics: first alpha_s artifact
cd ..
git clone \
  https://github.com/novaspivack/ugp-physics
cd ugp-physics
git log --diff-filter=A --follow \
  --format="%H %ai %s" \
  papers/01_SM/canonical_run/\
alpha_s_prediction.json

# 3. Verify the 43-day gap
#    Lean:     2026-03-05 14:09:28 EST
#    Physics:  2026-04-17 18:50:09 EDT
#    Gap:      43 days, 4 hours, 41 min
```

### Distinction from the $m_W$ pipeline

Both $\alpha_s$ and $m_W$ use Lean-certified rationals from the same Initial commit `27e6823` ($g_2^{2,\,\text{bare}} = 2329/5400$ and $g_3^{2,\,\text{bare}} = 41\,075\,281/27\,648\,000$ respectively).

For $\alpha_s$, the prediction at $+0.24\sigma$ (PDG 2024; $+0.36\sigma$ vs PDG 2022) is obtained directly from the Lean rational via $\alpha_s = g_3^{2,\,\text{bare}}/(4\pi)$, with no $\text{TE}_1$.P calibration, no auxiliary scale, and no SM-running input; the pre-commitment trail documented here makes this cryptographically verifiable.

For $m_W$, the analogous *tree-level* application (§1.2 OP(viii)) misses PDG by $+36\sigma$ — a clean blind falsification of the naive pipeline. Applying standard two-loop SM gauge running with proper $6\to5$ flavour threshold matching at $m_t$ reduces the residual to $-1.28\sigma$ vs PDG 80.379 ($-0.42\sigma$ vs PDG 2024 world average $80.3692\pm0.0133$; $m_W = 80.364$ GeV; artifact `comp_p01_ZZ_mw_threshold_and_self_consistent.json`, pre-commit SHA-256 `eb4ff13b...`). The 13 MeV residual versus the older PDG value is fully accounted for by the SM/PDG W-mass tension: the SM EW fit itself misses PDG by $-23$ MeV; UGP's two-loop prediction sits $+9.7$ MeV above the SM EW fit, closer to PDG than the SM (artifact `comp_p01_AAA_mw_residual_decomposition.json`).

A null test (500 random $M_2 \in [30, 45]$ GeV at two-loop + threshold, UGP $M_2 = 37.4$ GeV at the 52nd-percentile of closures) confirms the match is structural rather than accidental.

The two blind tests therefore have qualitatively different epistemic status: $\alpha_s$ a direct blind success at first order (with honest caveat that the PDG band is wide), $m_W$ a blind tree-level falsification whose two-loop-plus-threshold closure to $-0.42\sigma$ (vs PDG 2024 world average) is a standard-SM-physics consequence of the same bare rational, with the residual versus older PDG fully accounted for by the SM/PDG W-mass tension. The framework has pre-committed predictions on both sides of the success/failure line.

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